

Letters To
A Young Architect



Christopher Benninger

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Christopher Benninger (b.1942) is one of India's most highly decorated architects. His well known award winning projects are the Mahindra United World College of India, the Samundra Institute of Maritime Studies, the "Suzlon One Earth" world headquarters, the National Ceremonial Plaza at Thimphu, Bhutan, and his own studio and residence "India House" in Pune. Each project addresses the complex issues of design, context, climate, materials, sustainability, and technology, amidst the client's programmatic needs. Benninger studied urban planning at MIT and architecture at Harvard, where he later taught. He settled in India in 1971, founding the School of Planning at Ahmedabad as a Ford Foundation Advisor, where he continues as a Distinguished Professor. His name appears alongside Geoffrey Bawa, Charles Correa and Balkrishna Doshi as one of the six recipients of the Great Master Architect Award in India, presented every three years.

His firm, Christopher Charles Benninger Architects, has recently been awarded India's most sought after commissions: the Azim Premji University at Bengaluru; and the Indian Institute of Technology at Hyderabad, along with the Indian Institute of Management at Kolkata. His city and regional planning works range from Sri Lanka, across India to Bhutan. Benninger's expansive campuses reveal an understanding of Indian "place making", reflecting the great temple complexes and the Mogul campuses. His narrative presents a language that lies between American ideals embedded in wooded Arcadian landscapes and sacred notions enshrined within Indian courtyards, generating a unique approach to architecture and place making.

Letters To A Young Architect

Letters to a Young Architect is a sensitive memoir of Christopher Benninger's life in India and his personal concerns about architectural theory and contemporary urban issues. Through the medium of articles and lectures presented over the past decade, a lucid collection of essays emerge that testify the commonality of mankind's condition. This is a collection of autobiographical narratives and ideas reflecting a man's journey of the spirit from America to India and the philosophical considerations that matured from his experiences. His travels are not only stories of the dusty roads he traveled on, but also of the passions and emotions of those he met along the way.

Letters to a Young Architect reflects on the role and direction of architecture in framing a new man and a new society in the new millennium. Benninger notes his encounters with gurus like Jose Luis Sert, Walter Gropius, Arnold Toynbee and Buckminster Fuller and the manner in which their personal passion for humanity shaped the lives of others. Benninger is a strong believer in tradition, in gurus and in students and in a lineage of values, ideals, principles and practices which have been matured from generation to generation. He is concerned with the education of architects; the nature of architecture itself; and the role of urbanism and planning in the creation of a new society. The role of Indian masters like Balkrishna Doshi, who guided him in his search, is a touching tribute to the Indian "Guru-Shishya" tradition.

Letters to a Young Architect

Christopher Benninger

Foreword by Balkrishna Doshi

August 2013

INDIA HOUSE
Pune, India

Createspace

USA

“Architecture is a curious craft!

*One structure may follow all the laws of design
yet be worthless, while still another may break all
the principles and be profound!*

A building may be bad without doing anything wrong,

*while yet another may have to sin against
architecture to reach perfection!”*

Christopher Benninger

Letters to a Young Architect

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*To
Ramprasad*

A Dedication

These ‘Letters’ are a compilation of many interactions with students and young professionals across the Indian subcontinent, at venues ranging from Chandigarh to Chennai and from Ahmedabad to Kolkata, and abroad in North America, Europe and Australia. The collection includes material written over the years.

I am a believer in tradition, lineage and gurus. I believe there are competing ‘schools of thought’ and that there are, to borrow an idiom from Hindustani Classical Music, *gharanas*, or chains of ideas that are passed down from teacher to student. Each *gharana* has its explicit and implicit values and knowledge. I believe that the teacher re-manifests into a student, continuing the tradition. The core of this tradition is composed of values, ideas, concepts and attitudes that are shared, analyzed, adapted, changed and made relevant to new contexts and problems.

Good teachers inspire their students. Inspiration is that magical moment when one discovers something deep within oneself that has lain dormant – something that one can aspire to be; a good

characteristic one can assume; or the latent talent one did not know existed, yet defines one's future. Inspiration is the spark that ignites a flame. It also catalyzes rebellion against stereotypes and biases. It makes one question the very tradition that formed us. Good teachers spark a sense of doubt, 'cause' and struggle.

I want to acknowledge the teachers who touched my heart and made me think. They made me know 'who I am'. When I awoke to their call I began a journey that has never ended. They were surely all part of the same humanist movement. They were clear in their thoughts and passionate about their work. They were artists, architects, planners, economists, sociologists, historians and philosophers. They were deeply concerned with the contexts and societies they lived in. They were actively trying to make this world a better place to live in, and through direct contact and interaction with them I became a better person and a better architect.

My journey began with *The Natural House*, a book by Frank Lloyd Wright, gifted to me on Christmas day in 1956. I was inspired. When I turned the last page, I knew I was destined to become an architect! Then I found a clear path through a young teacher called Harry Merritt who inspired me further through his wonderful designs and thoughtful questions. He advised me to leave Florida and go to Harvard. From then on good fortune introduced me to a chain of true gurus whom I would like to mention:

Robert Tucker
Norman Jensen
Blair Reeves
Turpin C. Bannister
Charles and Ray Eames
Paolo Soleri
Buckminster Fuller
Walter Gropius
Jose Lluís Sert
Jerzy Soltan
Jaqueline Tyrwhitt
Joseph Zalewski
Mirko Basaldella
John F. C. Turner
Barbara Ward
Shadrach Woods
Dolf Schnebli
Fumihiko Maki
Alexander Tzonis
Liane Lefaivre
Yona Friedman
Arnold Toynbee
Margaret Mead
Constantinos Doxiadis
Panayis Psomopoulos
Roger Montgomery
Gerhard Kallmann
Jane Drew
Maxwell Fry
Kevin Lynch
Lloyd Rodwin
Herbert Gans
Lisa Peattie
John Kenneth Galbraith
Charles Correa
Piraji Sagara
Anant Raje
Louis Kahn
Vikram Sarabhai
Kurula Varkey
Otto Koenigsberger
Yoginder Alagh
Hasmukh C. Patel
Achyut Kanvinde
Preston Andrade
Arthur Row
Kamla Chowdhry
Dasho Lam Penjor
Giri Deshingkar
J.P. Naik
M.V. Namjoshi
V.M. Sirsikar
V.M. Dandekar
Charles Boyce
Shiv Datt Sharma
Dattatreya Dhanagare

Most of all I owe a debt of love, knowledge and poetry to Balkrishna Doshi who has been a source of inspiration for me to be here in India, to place my footprint on this soil and to call this land my home. Were it not for him, I would not be here. He is my true guru.

Finally, all of my fellow architects at India House who sustain me and carry forward our tradition, are part and parcel of my journey. Rahul Sathe and Daraius Choksi have been my guides, soulmates and studio leaders. Without them, my studio does not exist. My associates Deepak Kaw, Shivaji Karekar, Harsh Manrao, Madhav Joshi, Jagadeesh Taluri, Navin Ghorecha and Shashi Mohandas have each spared me valuable years of their creative lives. I must also thank Tsutomu Sato for the first layout and the continual help from my support staff, Geeta and Shantaram.

This book would not have been possible without the persistent encouragement of my research associate, Naveen Bharathi. Vivek Khadpekar favored me by accepting the commission to edit this disparate collection. He has been a friend and source of ideas since our first meeting in 1968.

My companion and sharer in this journey, over the past seventeen years, has been my life partner Akkiseti Ramprasad Naidu. It was he who conceptualized the idea of our successful studio and carried it forward as its Managing Director. Perhaps he alone has the energy and vision that makes us 'be' as a functioning entity. He created a vessel in which I could sail smoothly through stormy waters.

Christopher Benninger

India House, April 2011

Foreword

When I first visited America on a Graham Foundation scholarship in 1959, I turned to Le Corbusier to introduce me to a guide in that complex land. He wrote to Jose Lluís Sert, Dean at Harvard, who was kind enough to advise me. Several years later Sert introduced me to a young man in his mid-twenties who was coming to India on a Fulbright Fellowship. I suggested that he be with me in Ahmedabad at the School of Architecture. It was in the spirit of our ‘global family’ that without any questions Christopher Benninger came to work with me as a teacher in 1968, at the age of twenty-five.

After his fellowship ended I asked him to stay on with an Indian salary. He stayed! While in Ahmedabad he worked with me on my idea to start a new school of planning. When he left to teach architecture at Harvard, I assumed his tryst with India was over. A year later, while with Kahn in Philadelphia, I got a call from Christopher to come up to Harvard and give a public lecture. While in Cambridge we talked of utopian dreams and about the future of India. I obtained his promise that if I could ever initiate a school of planning he would come back to Ahmedabad and help me start it.

As fate would have it that project materialized sooner than I ever imagined. But hopes for Christopher’s support evaporated as he had just been made a tenured Assistant Professor at Harvard, and I knew he would not leave that coveted post easily. In any case I wrote to him that he must come. I was surprised after a month to receive a letter that he would join me. From then on we have been on a long journey. Like the *gharanas* of music, we share a common school of thought and a lineage of great teachers. From Walter Gropius, Le Corbusier, Louis Kahn and Jose Lluís Sert, the *gharana* baton was taken over by people like Oscar Niemeyer, Fumihiko Maki, Dolf Schnebli and me. It is our good fortune that we could pass on part of our burden and inherited obligation to the next generation

and that they are doing the same. I am heartened to have a person like Christopher call himself my protégé and my *shishya*.

The ‘letters’ that follow are in fact a collection of interviews with the media, public lectures and articles in journals from 1967 onward. They cover a wide range of topics but are integrated through a number of concerns, ideas and concepts. Some of these are elemental to the ideas that energized Le Corbusier and concepts around which I built institutions in India. These are the ‘connectors’ that have linked generations of architects. They are not ‘styles’ or visual attributes, but concerns for the human condition. These are a way of looking at the world. These letters are Christopher’s very personal revelations about his life, which should catalyze young architects to think out their own careers. There are ‘letters’ about modernism and his concerns that architecture has become isolated from its social, historical and contextual base. There is a concern that architecture has turned into eclectic academic postmodern styles employed for greed and mercantile purposes. Some of the letters focus on Christopher’s interest in social and economic change, appropriate technologies and their impact on urbanization, poverty, equality and the environment. Often he brings these concerns together through his discussions on urbanism and city planning. Some letters are tributes to people he knew well who he feels should be the role models for young architects.

Our journey has been one of friendship, learning, sharing and love. Our decades of work together never seemed like work. It was always an exploration, accentuated with discoveries and full of fun. We each brought out the child in the other and catalyzed each other’s creative instincts. I could say that our friendship has been a celebration of life. This collection of talks and articles, now presented as letters, is familiar to me. It embodies our shared values, life’s work and hopes for the future.

Balkrishna V. Doshi
Ahmedabad, India, April 2011

Prologue

A wise sage I once met in his cave retreat, somewhere on the rocky slopes of Mount Abu, offered to read my fate from my palm. As a young student of empirical method, I recoiled from his outrageous suggestion. What would my teachers think of a protégé who curried the favors of godmen to know his fate? But he charmed me with his charisma, and questioned my logical abilities to reject his conclusions, should I find them so whimsical. I suppose his piercing eyes and the lyrical landscape, perched high above the Rajasthan desert, swayed me like some magic potion.

He told me that I was a person of little wealth, but of great fortune. He declared that luck was my life's companion.

Tempted further, I coaxed him, 'But what do you mean by good luck?'

With an incredulous sneer on his face, he informed me that there is only one kind of good luck, and that is to have great teachers.

I felt a chill run down my spine and across my skin, momentarily leaving goose pimples all over. He had unraveled a truth within me that he never could have made out from my appearance, or from his imagination. It was true that I had been blessed with wonderful teachers.

From then on, what had been a youth's good fortune became my life's endless search. To meet wise people became my passion.

That passion, and my fated trajectory of good luck, brought me decades ago to India. My good fortune led me to the likes of Balkrishna Doshi, Achyut Kanvinde, Anant Raje, Kurula Varkey, Vikram Sarabhai, Piraji Sagara, Yoginder Alagh, Hasmukh Patel, Dasho Lam Penjor, Lyonchen Jigme Thinley, and to many youngsters who are now practicing architects and teachers. There have always been values, ideals and principles that have bound us together and made us one. I often imagine all of my gurus as 'one being' and 'one soul' who have embodied many avatars, worn many masks and appeared in many manifestations. But the path we walked on has been bound on either side by values. The destination has been our shared ideals and utopian ambitions. Through the medium of the 'letters' which follow I am trying to share this legacy with future generations.

What I am presenting to you is the idea that we are all one, but also that we are many.

We are all one because we share binding values. We are many because we each have our own personality and our sense of destiny. But, again, we share optimistic hopes. Asked to paint a picture of paradise, we may all paint the same image, though in different tones and hues.

Values, idealism and anything that hints at an ideology is unpopular and even suspect today. It is suspect because *effeteness* and mercantile mindsets have become our gods. But we are not fools or idiots; we are professionals. To be professionals we must 'profess' values.

What I am presenting to you is the idea that we are all one, but also that we are many.

What are these values?

I wish to propose a few generic beliefs, themes and causes that have taken us all on an exciting journey. I am doing this with the explicit hope that students and young architects may see sense in what I say and join our mission and carry the journey forward.

In the ‘letters’ that follow I would like to explore this journey, and then look into the future. I believe that in the past and in the future, we are all there. Like a flame from a candle that lights the next candle, the body changes but the spirit remains the same, and the journey continues. What is important is that our mission becomes more explicit and clear from generation to generation. It is important that we all speak with sure voices. It is important that we have an impact on the society around us. These ‘letters’ are passed on with that one purpose in mind. Let me introduce them by laying down the values I learned from my teachers, which are elements in what I have to say.

They include attitudes toward truth, objectivity and equality; toward justice, a worldview and respect for regional contexts; love of liberty, fraternity and modern ideas; trusteeship and planning; embracing knowledge and openness to new ideas.

Truth and Objective Reality

Underlying all our work and fundamental to our personalities is our commitment to TRUTH. By truth I don’t mean a middle-class sense of right and wrong, but an intellectual commitment that ‘what we see to be true is true’. This commitment extends through a faith in the empirical method and a disdain for superstition, nostalgia and romanticism. It means going to the field and using life as a laboratory for learning. In the end it means acting on OBJECTIVE REALITY. This is what is called intellectual honesty and it first involves a dialogue with ourselves, assuring us that what we speak is factual and rational. If there is one value we all share, it is a commitment to truth. It demands introspection. It makes us ask, ‘How can architecture be dishonest, or honest?’ Do we believe what we speak? Are we true to ourselves? Can we see what we know?

Equality

The EQUALITY OF ALL LIVING THINGS, particularly human beings, is also fundamental. This can be distilled into a slogan (paraphrasing Marx): *From each according to their abilities and to each according to their needs*. I feel we all believe in some kind of social net that does not allow anyone to fall below a minimal level of dignified human existence. This means access to medical care, a minimum sustaining diet and ‘humane’ shelter. It also means access to opportunities to develop ourselves to do things in life that bring us joy. It means that everyone has equal right and access to opportunities that exist within society. This makes us ask: is there a ‘social contract’ between an architect and society that transcends the client-architect relationship? Can architecture and urban planning further the cause of equality amongst all people? Can our design skills act as channels and

paths of access and opportunity for others?

Justice

These values evoke a need for JUSTICE and FAIRNESS. A rule applied to one must be applied to all and there must be no exceptions to a rule meant for everyone. We have questioned whether it is 'just' to destroy the ecosystem in our own lifetime at the cost of future generations. We have asked if it is fair for one city to consume twenty times the energy of another. We have pondered over the excesses of a minority at the cost of a majority. These are concerns we all share. Do such values impinge on how we design humane settlements? Are they a part of our personal plan of things?

World View

The times we live in require us to have a WORLD VIEW and understand that we cannot survive in isolation. Be it climate change, sustainability, human rights, nuclear war, terrorism or trade, they affect all our shared global space. A sense of geography, history and culture must temper our passions and grant us love and understanding. We must respect diverse cultures and co-exist with many beliefs. Can we validate our own ideas without their being vetted by minds from many lands? Can we design habitats for others without an articulate world view? Can we understand our own reality without understanding the people around us?

A sense of geography, history and culture must temper our passions and grant us love and understanding.

Regional Context

At the same time we have to build and to design contextually. We have to draw our inspiration from styles evolved from the past that have addressed climate, local materials, craftsmanship and culture. The design of any artifact begins with an understanding of 'place'. How does context become the starting point of design? How do the functions reflect the culture of the users? Do the footpaths and public domains we create gift conviviality to their users? What lessons do we learn from the history of the site and of the people who will occupy our structures? How are we contributing to, or detracting from, the regional culture?

Liberty

Finally, I think each one of us demands our personal freedom and the LIBERTY to pursue our own fate, explore our own talents, and be ourselves in every creative manner possible. Many of my teachers fled oppression in Europe to be where they had a voice. Their diaspora in fear of oppression became my legacy of freedom. Liberty is also an attitude about the involvement and participation of the users of our works. How do we make our design processes more interactive and more participatory? How do our plans gift humanity more opportunities and choices? Do we see the trampling on the rights of one minority as a theft of freedom from us all? Do we see land grabbing and evictions as affronts to human dignity? Can we allow such things to happen in the name of city and urban design?

Fraternity

We have undergone a number of personal reincarnations, explored manifestations and lived different avatars in life's search. What binds us together is a sense of FRATERNITY – sisterhood and brotherhood that attracts us and makes us want to come together. When I would meet my gurus these were, and are, the kinds of concepts that molded our dialogues and guided our thoughts. The strength of shared values gave us a sense of meaning and assurance. It kept us on a path. By sharing these 'letters', I want to expand our fraternal circle. I want each student to think of their life and their values. I want each student to become a teacher casting our *gharana's* net much wider. To be a part of our fellowship, of our fraternity and our *gharana*, all one must do is to imbibe and promote the simple values I am outlining.

Modern Ideas

All of us on this path have been deeply concerned about and personally confronted with The Modern Era. This has been a time that took away dignity from the craftsman; a time that turned self-sufficient homesteads into mono-cropped commercial fields; a time that pushed ill-prepared, illiterate people into chaotic cities. It is a time when a disproportionate share of the earth's resources shifted into the hands of a few persons and agencies that lacked what Gandhiji called TRUSTEESHIP. Right from John Ruskin, Charles Rennie Mackintosh, through the Deutscher Werkbund, the Weimar School of Art, the Bauhaus, the CIAM, the Team 10 and the Delos Symposia, up until today, our mission has been to bring a better life to humanity. Do we see modern architecture as a mere style, or as a social commitment? Do we see architecture as an agent to enhance the human condition?

Trusteeship

We live in an era of greed and self-aggrandizement. It is an era in which a person's respect and social position is determined by their wealth, no matter what the methods of its acquisition. Good deeds are thrown to the wind. Perhaps the fundamental error of socialism was that it did not recognize greed as a generic human quality, but construed it as a symptom of freedom, or thought it drew its source from capitalism. The fundamental error of capitalism was that it has not accepted an 'equal playing field' as the basis of true competition and the rise of a meritocracy. Greed is in all of us. It is the darkness struggling with the light. Greed is in my nature and in yours.

Perhaps the following 'letters' are my reaction to greed. They may help young architects take a stance on their own values during their difficult careers. We must understand that our talents and our wealth are only loaned to us, in trusteeship, to use toward the greater good.

Planning

The issue of the accumulation and distribution of wealth has been one that called for our attention and drew us to the planning of common resources and assets. To me planning is the study of stresses faced by different segments of society, and the rational distribution of investments, creation of incentives and enforcement of rules that alleviate society's stresses. It is the logical allotment of available resources toward the equitable reduction of human stress. This may result in the creation of infrastructure, services, facilities and institutional modalities. It must result in a humane society.

The word PLANNING has now taken on a new dimension. In 1971, when I started the School of

Planning at Ahmedabad, it was all of these concepts and values that fired the new institution. We were no longer doing colored maps that designated land use zones. We were looking at minimum needs, availability of goods and services and links between human needs and their fulfillment. The word ‘design’ took on a new dimension.

Architecture and urban planning took on the character of a social tool. All of us saw these disciplines as vehicles of social change. We saw these professional callings as paths to the discovery of truth; for the analysis of objective reality; for the realization of human equality; as harbingers of justice; and as voices for individual and community freedom. We saw not just problems, but their resolution.

We must understand that our talents and our wealth are only loaned to us, in trusteeship, to use toward the greater good.

Knowledge

Ignorance breeds ugliness. Good intentions not informed by knowledge can lead to counterintuitive results. Values lack context unless informed by knowledge. The collection of facts helps us form ideas. Analyzing and understanding the relationships between ideas allows us to form concepts. Without concepts we cannot state problems. Concepts are the seeds of good design.

Complex systems can be modeled on the understanding of interrelations between functionally interactive concepts. To design a good chair an architect must know the nature of the materials being employed; the weight to be applied at different angles and points of impact; the resulting bending moments, shear loads at connecting points; and, the continually changing dynamic distribution of loads; and the responses from the materials and connectors. Then the chair must feel good, be comfortable and be ergonomically designed. It must be beautiful to look at. All of these facts and ideas lead to the concept: CHAIR. The chair is then put in a house.

The house must respect its site, fulfill its functions and use appropriate materials. It must be durable and cost effective, sustainable and beautiful. Likewise the house is part of a designed street, which is part of a neighborhood, which is part of an urban district. The urban district is part of a city; and that is part of an interlinked metropolitan region. All of these can be designed if we can gather the relevant facts and their interrelationships; if we can understand the functioning of the resulting concepts in systemic models; and then realize them as actual physical fabrics. We must be analytical to isolate the performance standards of our solutions, creative to generate options to evaluate, and capable of selecting amongst them. This is where the word DESIGN comes in as an elemental process of the use of knowledge and its analysis to create objects and artifacts.

The preceding discussion raises questions about the theory of knowledge itself and the ‘meaning systems’ that buttress or destroy these ideas. Architects must have a keen interest in and an understanding of technology, and the way it impacts social structure and societal change. They must know the history of technology to understand their place in history and how their own work detracts from or contributes to society.

It would be arrogant to assume that any architect will change history. But surely every architect, for better or for worse, tempers the cultural fabric of the society they work in. Anyone who builds makes an impact on nature, society and the quality of life offered by the environment. No one is on the sideline. No one who claims to be an architect is an outsider.

Having a rough idea of the timeline of history is important to ‘fit’ oneself into the scheme of things. Is repeating the Eiffel Tower a contribution to the world, or is it an affront to our understanding of things? Is it progress in technology for the human good, or is it a cheap money-making stunt? Is making the tallest building in the world anything new, or is it gross exhibition of one’s insecurity?

Architects need to know something of the social structure of their surroundings; something of the stresses felt daily by, and experienced in, their societies; something of the materiality of their art; something of the resources they consume; and they must know how their acts that employ these materials and technologies influence the creation of a humane habitat. They must know how greed and mercantilism distract attention and resources from dealing with the human condition. Under the veil of nostalgia and romanticism, objective reality is disguised and great sins against humanity, realized in built form, are justified.

Architects must be continual learners. We are all teachers and we are all students. Knowledge makes us wiser as it helps us see the conditions and the points of view of others and helps us to understand their feelings and needs. As designers, we must continually place ourselves in the users’ shoes and walk in them, at least in our imaginations. This could even lead us to wisdom. To be wise merely means to be able to see all sides of issues dispassionately and to reach just conclusions based on that knowledge. Every good design is a just conclusion.

Lack of knowledge leads to lack of personal identity. It makes people play roles of fictitious characters instead of being themselves and feeling comfortable with what they are.

To young architects I say, ‘Don’t seem, BE.’ Understand your limitations and your potentials. Speak from where you are and not from where you would like to be. Be humble in your self-appraisal and be bold in doing well what you know you can do. Flaunt your pride and your ego through your work.

Each young architect must set their own standards and define their own character through the values they adhere to.

Each of us evolves a unique persona based on our values, attitudes, wisdom and compassion. Do not adopt your persona from people who are mean to you, making you reactive. If you are mean to a mean person you slowly begin to collect all that meanness into your own personality. Instead of mirroring all that is bad in the world, slowly, piece by piece, build upon all the good you see in little slivers in each person. All that good can become your model of yourself. It can be the persona that you project like an aura radiating out to others. That is true knowledge and true wisdom.

In the ultimate analysis we architects are not just creators. We are builders. We cannot live in the world of mere words and thought; we have to plant our foot firmly on earth and leave our footprint. We are ‘thinker-doers’ whose ideas must mesh with the materials around us and the technology available. We must address the society we live in and build for.

Don’t seem, BE.

I often hear architects blaming what they are doing on their context, their clients, or the lack of resources. These are just self-delusions constructed to rationalize defeat. Our role here on earth is to DO GOOD. If we have a bad client we can walk away. That is unless we are driven by greed. It is only through KNOWLEDGE, tempered by values and compassion that true architecture raises its head.

Openess

Our debates, dialogues, presentations and even arguments have made us all more open to constructive criticism and self-evaluation. Teaching was never a one-way flow of ideas; it was a back-and-forth flow of sharing and questioning. Most of my teachers taught through the Socratic method of asking questions. They maintained that it was more important to ask the right question than to know the correct answer. Dialogue and questions are more important than monologues and answers.

Legacy

These values were passed on to me by my teachers. I learned from being with them, seeing how they dealt with issues and understanding how they resolved problems. These values were woven into the fabric of everything they said, and underpinned their worldview.

Now it is my legacy to hold these ideas as my only wealth, and it is my duty to pass on this wealth. I have chosen to call these messages ‘letters’ as they are like post cards sent to family and friends while on a long journey. They are not chapters in a story, but rather serendipitous discoveries along life’s path.

(Based on the First Professor Akhildutt Dadkar Memorial Lecture delivered on the 30th of

January 2010 at the Twelfth International Conference on Humane Habitat, Mumbai, India.)

*... it was more important to ask the right question than to know the
correct answer.*

Omens of a Magic Gift

As a child I spent my days drifting in confusion. Nothing inspired me. Neither my teachers nor my studies inspired me to seek knowledge. My parents thought by going to school I'd be educated; by taking part in sports I'd become athletic; and by going to church I'd be in touch with the Ultimate Truth! They confused religion with spirituality; sports with health; and qualifications with knowledge. Most of what transpired in these institutions seemed like a dull black cloud hovering over me.

What did move me were the autumn leaves in reds, yellows and oranges. The nude winter fingers of trees reaching into moody skies depressed me as the beautiful leaves fell away. Come snowfall, the black branches and twigs would morph, covered with a white powder of snowflakes, melting momentarily, and then freezing the black stick trees into gleaming ice-glass candelabras, glittering upside-down in the bright sun. Amazing beauty emerged from angry darkness. These were the things that grabbed me and fascinated me.

My personal life was composed of all things natural and my friends were the squirrels in the trees and the rabbits in the fields. All these were portents of an organic truth to be revealed! Springtime full of growth, flowers, transformation; summer full of form, shape, insects, fruits; autumn with its withering, death and destruction; and the long winter's sleep: all these made a full cycle of reincarnations, the one manifested in the many. From disaster emerges realization, renewal and transcendence. The flocks of birds flew in formations: south in winter, north in spring. These were 'signs'.



Thus, I was composed of two different parts, each amplifying the meaning and the meaninglessness of the other. Like the **yin** and the **yang**, a white and a black force intertwined within me, chasing each other. The black made the white more pure and beautiful, and the white made the black more foreboding and ominous.

One Christmas morning my eyes were drawn to the one gift I had not stumbled upon in my parents' usual hiding places. I knew all of the others from looking under beds, crawling up in the attic, or peering over the high shelf above my father's cupboards where he hid his condoms and porn magazines. So I reached for this first, as my family members all gasped with hypocritical surprise, opening boxes and small parcels they'd secretly discovered only days before. Like all children, on that fateful morning I reached out for the most intriguing gift first; for the unknown. But unlike the others this gift turned out to be a talisman of my future. It was a magic book that would change my life forever.

As I read the first words, sentences, paragraphs and pages of Frank Lloyd Wright's *The Natural House*, I discovered who I was, and what I wanted to be. I gained my first insight into the nature of my life's meaning and search. Reading the pages I felt like a reincarnated person discovering who he

had been in previous lives, and what he would be in the future.

It was not just that I liked the designs, the drawings and the photographs, or that I gleaned meanings from the words; it was a testament that unfolded a truth within me: a truth that in fact had dwelt deep inside. Something that had always been there, slumbering, concealed from my consciousness, was now unfolding. I suppose this is called inspiration, or even self-discovery. From the moment I opened *The Natural House* I did not put it down until I had finished reading the last page. In a sense I have never put it down. I am still reading it in my soul, discovering and searching for what inspired me on that Christmas Day over fifty years ago.

When I closed the book just past midnight I had been transported into a different world. I walked out of my house into the freezing night air with thousands of stars glistening in the vast nocturnal heaven. The air was amazingly fresh and the sky was wonderfully transparent. Everything I saw looked different, fresh and new. It was not only nature that was singing a song in my heart, but my soul had switched on and my mind had started to think.

The black made the white more pure and beautiful, and the white made the black more foreboding and ominous.

In the days that followed I ‘saw’ things I had never comprehended before. Finely carved balustrades caught my fancy. Sculptured stone gargoyles made me smile. I noticed that one wood was different from another in its color, grain, density and use. I was drawn to ‘feel wood’ and to slide my fingers across it, appreciating its inner soul. I noted that a wood floor was warm in the winter and cozy to look at, while a marble floor was cool in the summer and soothing to sit upon. Stained glass windows, fine brass handles, and well thought-out paving patterns were my companions. Grasping materials with my hands I could hear their inner voices and I spoke to them. I began to argue with sloppy workmanship and clumsy details. Unnatural, synthetic and artificial finishes fired a sense of anger in me. I developed a self-righteous sense of the right and wrong uses of materials; good and bad expressions of functions, and revulsion toward exaggerated applications of expensive finishes. Monumentalism annoyed me. Motifs crafted in Plaster of Paris to look like marble carvings repulsed me. I divided the world of artifacts into those of honest expression and those of lies. There were the master architects, and there were hacks who churned out ugliness for money. I realized that my ‘holier-than-thou’ attitudes were verging on fundamentalism. But I loved the order, the devotion and the balance these thoughts brought me. A new passion had entered my soul and fired my spirit.

Wright taught me that the human mind is a huge analogue for all things beautiful and all things ugly. He taught me that a human being is both a monster and a saint all rolled up into one; capable of creating incredible beauty, or of inflicting deplorable destruction and ugliness. It is the human mind that separates humans from other animals, which makes us the monsters of terror and the creators of poetry, art and architecture. We alone can know the exhilaration of transcendence. Humans are the only species that creates values which temper and modulate its behavior. They employ values to design and create their habitats.

After I had read *The Natural House* the **yin** and the **yang** within me merged, no longer playing out against each other, exhausting me. The black force empowered the white beauty. I was now driven by a passion in whatever I did. I gave up on education and embarked on an inner search. Something magical had gripped me. I stopped attending church and I found spiritual moments in fits of creative discovery.

Such a moment of self discovery is what I call INSPIRATION. It is a flash of wisdom that calls out to us, telling us what we want to be and forces us to yearn to be that. It catalyzes life's search; it embeds an urge; it creates a desperate need to seek what we do not possess; it beckons us to know our inner soul; it leads us on a path from which we cannot deviate.

Wright taught me in that simple book to seek the generic order in things and to see beauty in truth. I understood that buildings are merely mirrors of the people who live in them. They reflect how people behave, how people think, what their aspirations are and how they deal with materiality. Built form illustrates how evolved people and societies are in their spiritual realizations, whether they live for material things or employ material artifacts to reach transcendence. Buildings place people and societies somewhere on a scale between beasts grabbing at survival and sages blessed with transcendental awareness. Architecture distinguishes people who only 'take' from patrons who nurture and 'give.' Buildings indicate the extent to which people are in touch with the environment in which they live; the extent to which they are part of the places in which they build, and are in harmony with the social traditions and modalities which bring bliss and peace.

But life is not a fairy tale. It is a maze of choices and we have to learn as we go. We make some good decisions and some bad ones. But I believe we are driven by our GENERIC INSPIRATION to learn from our mistakes and move on. We are guided to recognize lessons when they come our way and to learn from them. With the fire of inspiration inside us, life itself becomes a great university of learning. We are learning all the time.

With the fire of inspiration inside us, life itself becomes a great university of learning. We are learning all the time.

So today, almost half century later, those portents of a magic gift still inspire and guide me. I pray that all young architects have had, or will have, such an inspiring moment that gifts them a sense of meaning and direction in life.

What I learnt from Wright was very simple: find the generic order in things, seek out the truth and see beauty in the truth. What he meant by 'the natural house' was the natural self and the natural life. He used the word 'organic' as an idea with many meanings. He meant that everything has its own putative nature; that architecture is part of nature; that each material has its essence, its nature and its unique potentials and limitations; that there is an innate truth to each thing; that each individual has an organic personality, a generic originality and unique character; that one must be true to the nature of things and to one's own nature; that there is a lie in misusing materials, ideas and facts; that there is hypocrisy in misrepresenting oneself for what one is not; that everything has its organic opportunity and potential to contribute, and that is nature's journey and purpose. He meant that each and every person has a unique opportunity and potential to contribute, and that is nature's journey and purpose.

Wright ended his book with a *Credo* (Latin for 'I Believe). I state Wright's simple credo as follows:

'I believe a house is more a home by being a work of art.'

'I believe a person is more a human by being an individual, rather than a committee meeting.'

‘For these two reasons I believe Democracy (though difficult) is the highest known form of society.’

‘I believe Democracy is the new innate aristocracy our humanity needs.’

‘I believe success in any form consists in making these truths a reality according to ability.’

‘I believe all agencies tending to confuse and frustrate these truths are now continuous and expedient – therefore to be exposed and rejected.’

‘I believe truth to be our organic divinity.’

Wright was for beauty, for democracy and for the honest individual. He was strongly against fake, mercantile and feeble expressions, like the commercial glass-clad buildings we see today. About this effete architecture he commented, ‘Our architecture is the significance of insignificance only. We no longer have architecture; at least no buildings with integrity. We only have economic crimes in its name. Now, our greatest buildings are not qualified as art.’

In revolting against the ugly Greco-Roman Plaster of Paris fakery of his time, Wright provided me with an example of the struggle against false art and aesthetic lies. I realize that Wright’s struggle a century ago is our struggle today. Wright’s voice arches across the decades, calling out to us today. In revolting against the status quo, Wright talked of five great integrities. These were influenced by the 18th-century Chinese philosopher Ong Giao Ki who insisted ‘Poetry is the sound of heart.’ In other words beauty is in the essence of true feelings.

Wright’s five integrities were:

First, ‘The room must be seen as architecture, or we have no architecture.’ What Wright means is that architecture is not just a decorated box, but a group of interrelated, integrated spaces that interlock, one within the other, creating a relevant form. He echoed the dictum of his guru Louis Sullivan: ‘Form follows function.’

Second, he continued that line of thought saying, ‘We no longer have an outside and an inside as two separate things. They are **o f** each other. Form and function thus become one in design and execution, **if** the natures of materials are all in unison.’

Third, he proposed the concept that design grows from the site. The ‘space’ and the ‘interior’ and the ‘form’ all rise from the ground into light. The fact is that contemporary culture is in bits and pieces, not organic, integrated or natural. Cities were, as they are today, growing against nature. Wright believed that light-architecture-culture-happiness-work-faith all lie in the ‘natural’. He revolted against conspicuous waste and insulting artificiality as being oppressive to our own intelligence.

Fourth, Wright proposed that the idea of ‘organic architecture’ is that the reality of a building lies in the space within it, to be used and to be lived in. Wright rejected the idea, so popular today, that a building is an ‘envelope’ or a decorated package.

Fifth, Wright felt this organic idea was the essence of democracy too. He believed that democratic societies unlock the essence of the individual, to flow forth in an honest manner. He

believed each of us has an organic reality that is the integrity of the individual. Honesty, comes from within, not in what one wears or looks like. Integrity is not something to be put on and taken off like a garment. Integrity is a quality from **within** and **of** the person himself. This internal integrity cannot change from external pressure, or from outward circumstances; it cannot change except from within because **it is ‘that in you’ which is you** and due to which you will try to live your life in the best way possible.

Wright began to rebel and protest against the idea of ‘façade’ architecture and ‘façade’ people. He reversed the decorative idea of façade architecture, employing a new paradigm that form emerges from the inside. He took inspiration from the Japanese philosopher Okakura Kakuzo who, in his *Book of Tea*, proposed that the reality of a room was to be found in the space enclosed by the roof and the walls, not in the roof and the walls themselves.

Wright was a collector of oriental art and an admirer of Lao Tse and Taoist thoughts. His work on the Imperial Hotel in Tokyo brought him closer to this culture. His concept of organic architecture sprang from the ancient concepts of the essence of things, rather than the superficial, often dishonest appearances. He insisted, ‘what is needed most in architecture today is the very thing that is most needed in life – Integrity.’

What is true for people, integrity, is also the deepest quality in a building. Wright felt that integrity is a natural human characteristic, but with the advent of consumerism a craving for possessions emerged. The urge to become a ‘success’ has destroyed this precious quality, because success has been overshadowed by greed and materialism.

To build a man or a building from within is always difficult, because ‘deeper is not so easy as shallow’. Naturally you would want to live in a manner and in a place that is true to this deeper thing in you, which you honor. That place you live in must also be integral in every sense; integral to site; integral to purpose; and integral to yourself. This concept seems to have been forgotten. Houses have become a series of anonymous boxes that fit into row upon row of bigger boxes in mass nuisance. But a house in an ‘interior’, or a deeper organic sense, may come alive as organic architecture, just as human personality comes alive as honest character.

Our sketches and our designs are paths to self-realization. Buildings reflect how we behave, how we think, what our aspirations are, and how we deal with materiality. Scholars like Liane Lefaivre and Alex Tzonis have reinforced this credo, through their work on what they call Critical Regionalism, in which new functions and technologies are honestly integrated with cultural contexts, climates and cultures. This concept was embedded in Wright’s ideas of ‘organic architecture’.

Young architects, I implore you, please understand and imbibe this great philosophy. It will send you on a wonderful journey full of joy and discovery.

Legacy and Endowment: An Architect's Self Discovery

An Awakening

After *The Natural House*, I read *A Testament*, *An American Architecture* and anything else by Wright that I could lay my hands on. *Broadacre City*, *An Autobiography* and *The Story of a Tower* were consumed in rapid succession. Wright ignited an energy within me that burns until this day. Decades later, his designs and his ideas follow me. It was not just the brilliance of structures embracing the vast, undulating American landscapes, but the rationalist's principles of 'honest expression' that grabbed my soul. The truth of Wright's thoughts became my **credo** – a beacon showing me the direction to my future.

As I read other books by Wright I learned that there are gurus whose wisdom helps to navigate the ocean of facts and knowledge, and they could guide me on a meaningful path in life's journey. This inspired me both figuratively and literally to be an adventurer of the spirit and a traveler of the mind. It catalyzed me to search out wise men and women who could pass on to me their understanding of the meaning of life. To find these people I had to wander and explore, stumbling serendipitously across wonderful people. As I traveled, I learned that there are 'schools of thought' that live on over decades and even centuries, with their spirit passing from teachers to their students, who then become teachers. Like a flame from one candle lighting another in a row, one guru passes away, but the flame of his spirit – his vision – passes on. In Indian music these schools are called *gharanas* and through gurus like Balkrishna Doshi I came to know that such traditions exist in architecture also.



Travels and Discoveries

My physical travels took me on a bicycle from Boston to Montreal through undulating green mountains; from Berkeley to Los Angeles down the San Joaquin Valley, through the Santa Lucia Range, past San Simeon through Cambria to the Mojave Desert on to Los Angeles; from Paris to Athens through the Vosges Mountains and two thousand kilometers across Europe through Tito's Yugoslavia. By boat I traveled around the Aegean Sea, the Baltic and the shores of Florida. As a youth I went to South America, crisscrossed North America and explored Europe. As a young man I visited Russia, Japan, Southeast Asia and in 1971 made an epic overland journey from London to Mumbai.

With no plan, program, or knowledge of the means of travel, I set off from Victoria Station by train, crossing the English Channel by boat. I rested at Dolf Schnebli's house in Switzerland and at Jaqueline Tyrwhitt's in Greece. In Turkey I entered a new society, a new religion, a new culture and an ancient civilization. The cuisine, the dress and the people were all new to me. As I traveled east, vast arid lands and immense mountains greeted me. A harsher territory dared me to enter. The rail line came to an end at Erzurum, and one could only thumb a ride on a mini-bus to the Iranian border. Along the way I met people who shared happy and sad stories, echoing their love of life. In Turkey I met a young engineering student, Prem Chand Jain, headed from Germany to Delhi, carrying a television set as his luggage. We became traveling companions on to Delhi. We met hippies and

back-packers, unshaven and dirty.

In Tehran we found an oasis of modernity, urbanity and bustling streets; then on to Mashhad, where boys at the city mosque threw pebbles at Western girls in shorts. "Serves you right!" I said to them. I travelled by mini-bus across the border, into the mud city of Herat, where I slept in a walled serai garden on a platform under a vast sky. A simple meal and a good massage under the shining stars put me to sleep. Several days by buses, like desert schooners floating over the arid sands, got me to Kandahar and Kabul. The bazaar was exotic and complex. The people were affectionate, full of laughter and bound by brotherhood. That is now an antique land that lives only in my memories. Harsh new realities and wars have destroyed its past. Coming down the Khyber Pass I discovered the Indian subcontinent. The people, the greenery, the increasing humidity, and the smell of *rotis* roasted on coal fires were all sub-continental!

All along my journey I made friends. Our physical destinations were common, but our life's searches were not. One could tag along with a local traveler, from whom one learnt the ropes of each segment of the journey and the segments overlapped. The flame passed from one candle to the next. One never knew with whom one would be on the next day, or where one would be going.

I think life is like that, based in people sharing joys and sorrows, seeking the meaning of where we are and where we are going. The photographer Cartier-Bresson, a great traveler, once said, "To be a great photographer one only needs one eye, one finger and two legs!" While he chronicled the meaning of life through his lens, I knew I would have to capture the essence of life in brick and mortar. But before I started putting lines to paper, I wanted to see what the masters had done across the world and through the ages, creating my present, in order that I could contemplate the future. For that I would need to move; I would have to travel on two legs, meeting people along the way, enriching one another's lives listening, telling stories and sharing the small belongings we carried with us on life's journey.

As a boy traveling in America, I would pick up the phone in New Canaan and call Philip Johnson, and he'd say, "Come over tomorrow!" In Los Angeles Charles and Ray Eames said, "Come see our house and then we'll go to see a new film we've made." In Phoenix Paolo Soleri said, "Come see my bells." In Miami Buckminster Fuller said, "Come have coffee at my hotel," and lectured me till three in the morning. And all along the way I was meeting people and listening to them: Fumihiko Maki in Tokyo; Otto Koenigsberger, Barbara Ward, Jane Drew and Maxwell Fry in London; Shadrach Woods and Yona Friedman in Paris; Dolf Schnebli in Switzerland, and Jackie Tyrwhitt, Panayis Psomopolous and Constantinos Doxiadis in Greece.

I particularly remember a Sunday afternoon with Louis Kahn and Anant Raje in Philadelphia. It was springtime in 1970. We listened and Kahn lectured the two of us for several hours. At one point he crushed a sheet of plain typing paper into a tight ball, tossed it on the table, and asked me to sketch it. As I struggled, he laughed. Then he drew four lines making the original simple flat sheet. We all laughed. Behind every conundrum lies the joke of life.

In India I met Achyut Kanvinde, Habib Rahman, Balkrishna Doshi, Charles Correa, Kamal Mangaldas, Hasmukh Patel, Mrinaliniben and Vikram Sarabhai, Anuradhaben and Sanat Mehta all within the first month of my arrival in 1968. Those were heady days full of discovery, hope, dreams and sharing.

Just as I marveled at the character of mountainous landscapes and great rivers, I tried to grasp

people's unique characters, their great spirits and their world views. I never bothered much about what they said of facts and figures. That I could learn on my own. I was interested in comprehending how they looked at ideas and what they felt about life. I wanted to understand their subtle mental organization of things and their nuanced reactions to the world. How did they extract poetry from the banal? How did they filter the profound from the mundane? How did these subjective traits temper their designs for the future? Deep down, each of them was asking, "What is the meaning of the universe? Why am I here? Are my actions directed towards a mission or a purpose?" All of these people emitted positive energy and were humanists. Many of them are no more. And there were hundreds of unnamed wise people I met along the way who shared their lives with me, and from whom I imbibed the nourishment of life's truths. I had to absorb their wisdom; I had to inculcate their spirit and to digest all the subtle knowledge that I could absorb. This was my search and the catalyst of my life's journey. This was the stimulant of my imagination.

Wisdom and Enlightenment

In retrospect I see that it is only through emotions, through subjective twists in things, through the poetry of life and through the love of people that profundity emerges. We can formulate ideas and concepts and nurture attitudes and 'constructs', but without a welling up of emotions these are merely dull academia. Truth dwells in the emotions of the heart. That is where poetry originates.

Behind every conundrum lies the joke of life.

Yes, we have to draw, measure and make buildings and create a civilization, but with what elemental material do we do this? It is from the emotive energy of the people we meet – their positive energy – that we gain the wisdom and the power to do things. Methods and techniques can be sub-contracted, but without passion and care there can never be anything really new. I think this is how I came to love the arts. This is how I came to love poetry and architecture. This love and passion is my legacy. It is *our* legacy.

So I realized that I did not travel to document or to photograph; nor did I read to learn facts and figures. I did all of these things to discover a larger emotive aura around myself and within me; to gather into myself sharper feelings about the nature of the universe, to seek fundamental answers to life: Who am I? Why am I here? Why do I build? For whom do I create? What is the meaning of the universe? Can one ever have a lien on eternity?

The Good Life

In the end there is something called 'the good life,' which underpins everything. I do not mean wealth, fame or conspicuous consumption. I do not mean that one has to drink the best Scotch, eat the best cuts, or know the names of the most famous wines. Yes, one should freely enjoy things to know what life is all about! But there has to be a balance.

One should eat well, but not become obese; one should drink good wine, but not become a drunkard; one should discuss ideas, but not argue incessantly; and one should be a good lover, but not let it envelop one's entire being. There is a kind of art in life and a measure in how one does things. Maybe one should be open to experiment just to know them, but then retreat to a middle path between boredom and hyper-excitement? That is where the good life lies, in the in-between zone. It is just a step outside of materiality and it is the essence of our work to find that realm and to recreate it for others. The Greeks called this optimality the Golden Mean, manifested by balance, proportion,

harmony and scale.

Truth dwells in the emotions of the heart. That is where poetry originates.

While Wright gave me a legacy of Truth, my travels in Asia taught me about the search for ‘the Good.’ I have often said it is better to know the Good than to search for Truth. The Good is lived; it is walked on under the clouds; it is smelled in gardens; it is there in bed at night and in our arms in the morning. The Good is known through smiles and laughter. The Truth can be sought in books, on the internet and gleaned from journals. It lies waiting in our study bookshelves; it is debated over by politicians, clerics and academics. They are all wondering and seeming; they are often pretending, while we are experiencing and being.

‘Be, not SEEM’, could be a simple way of expressing our legacy, because we will never really know what the truth is, though we must also search it.

Through my travels such questions and propositions began to haunt me. While I travel a great deal (and indeed am on a journey in Australia even as I write this piece) I have always settled into different kinds of *ashrams*, or retreats. Perhaps they mirror my stages of life from that of a *Brahmachari* to that of a *Rishi*?

There have been four figurative *ashrams* in my life that I would like to mention, as each had presiding *gharanas*, or schools of thought, nurturing them. They all had gurus and clear credos. There were my years in Cambridge, Massachusetts; my years in Ahmedabad, India; two decades at the Centre for Development Studies and Activities (CDSA) in Pune; and my present life at India House in Pune. In my previous *ashrams* I was an object of the *gharanas*, and in the present one I am the subject and the verb; that is to say that I have become more formative and deterministic as I grew older. That, in fact, is the anomaly of being an architect; as one is embattled by age, one becomes stronger; as one retreats, one becomes more engaged, making a greater impact on one’s context.

The Ashram of the Student: The Life of a *Brahmachari*

As I have already mentioned, my interest in architecture was ignited by Wright. Through Wright the windows opened onto Mies van der Rohe, and from there to the entire European school. Great buildings were emerging all around me: Lever House; Dulles International Airport; the TWA terminal; La Tourette, Ronchamp and the Unité d’Habitation in Marseilles; Baker House at MIT; the Carpenter Center at Harvard; the United Nations Building; the Ford Foundation Building; the Guggenheim Museum in New York; the Yale School of Architecture; Marin County Civic Center and many others.

Le Corbusier was building Chandigarh, Paolo Soleri was building Arcosanti, and Wright’s Taliesin West was my Mecca. Houses by Wright, Oscar Niemeyer, Charles and Ray Eames, Philip Johnson, Richard Neutra, Mies van der Rohe, Mary Otis Stevens and Thomas F. McNulty, Harry C. Merritt and Paul Rudolph all amazed me. In those days it was actual constructed art that inspired us. Now it is what people say and write about architecture that intrigues us. Today, without a theory to explain them, the queer shapes have no meaning. After reading and re-reading, and visiting construction sites, I got a chance to study real architecture. My father was a professor at the University of Florida where there was a large school of architecture. There were two hundred and

thirty students in the first lecture series by Professor Blaire Reeves and only sixteen students graduated from that batch. Survival was a struggle. The campus was a vast spread of lawns, trees and ivy-clad brick structures dating from the mid-nineteenth century. We studied architecture in Grove Hall, a temporary wooden barrack put up in the Second World War for officers' training. There was a pond next to it where alligators lived. Nature and new ideas surrounded us.

The amazing thing was that we had a core group of great teachers. Blair Reeves, Robert Tucker, Turpin C. Bannister, Norman Jensen and Harry Merritt grabbed our attention. They challenged us with ideas and puzzles. They opened windows on ideas, concepts and issues, creating competition between us youngsters while we studied. Tucker started an evening discussion group of motivated students from all the classes. Hiram Williams, the painter, Jerry Uelsmann the famous photographer and many others hosted us for dinner symposia. We worked, slept and ate in Grove Hall. Blair Reeves and his gracious wife were caring guides, opening our eyes to all aspects of modern art and design, walking the talk in their garden home; Robert Tucker was the Socrates. He was our philosopher and the teacher always posing questions. Norman Jensen, the painter, changed my perspective, asking me if I were a bird and admonishing me to sketch from eye level only. Harry Merritt was the embodiment of a 'great man.' He was a builder-teacher of great talent and confidence, and we learned through his beautiful works of construction, his studios and his discussions. It was Professor Merritt who guided me to leave the small university town of Gainesville and to attend Harvard. He pushed me off the edge of a cliff, seeing if I could free-fall, open my wings and fly. He pushed and I flew!

In Cambridge I studied urban planning at MIT and architecture at Harvard, where I later taught. I also worked in Jose Lluís Sert's studio. He was my mentor and my teacher at the Graduate School of Design. That was a time when we had only twelve students left in the Master's Class, after the four or five poor performers were asked to leave in the first month of studies. Times were hard. Sert was a 'no-funny-business', down-to-earth rationalist who loved life, and the people around him. But he wasted no love on laggards; out they would go. He found humor in our immaturity and exposed the 'joke of life' in letting us know our follies. From his chuckles we learned to think. I lived in Perkins Hall for a year and then shifted to Irving Street, into a large wooden house built by William James. They say he was the inventor of behavioral sciences. Julia Child lived two doors down the street across from E. E. Cummings and Elliot Norton.

The campus motto VERITAS, or 'Truth', was the object of our search then. Harvard and MIT were steeped in the empiricist school of thought, where "what is true is only that which can be seen to be true." Cambridge was also a bastion of the European way of thinking and doing things. While Wright employed natural materials, integrating buildings into the landscape, Le Corbusier and Gropius poised their buildings off the ground on crystalline white pillars. Civilization for them seemed to float above nature. Europe worshipped sky gods, abstractions and ideas. Americans loved nature, wildlife and the turn of seasons. Gropius, though retired, would visit our studios and pose questions. Jerzy Soltan, who developed *Le Modulor* for Le Corbusier, pushed us to 'be better than the best'. Joseph Zalewski, who survived the Jewish Ghetto in Warsaw, and worked at 35 Rue de Sèvres, would often just stare at a drawing and mutter, *rien* ('nothing' in French). Albert Szabo, a deep thinking Hungarian, treated us like his own sons.

At MIT, Kevin Lynch awakened our understanding of cities and urban experiences, while with Herbert Gans we explored the social ecology of cities and how people used them.

At Sert's personal studio on Church Street, I worked on the Harvard Science Center and on the details of his new studio coming up on Brattle Street. These were very much 'to the point,' rationalist buildings that inspire me even today. The word *Brahmachari* hardly describes the personal lives of young scholars living along the Charles River! But it surely reflects an innocence, a persistent quest for the truth, and working from 'tired to tired' to reach perfection. This was an *ashram* where modern meant 'progress'. We sincerely believed that history was a continuous path of improvement and problem solving. All diseases would be vanquished; poverty would be eradicated, borders between countries would dissolve, and the world would become one fellowship. We were 'thinker-doers' charged by the urge to create a better world.

Even though I was an architect I studied economics under John Kenneth Galbraith and became Barbara Ward's protégé, traveling with her to Greece to attend the Delos Symposium on Doxiadis' yacht. There a few of us lucky youngsters came to know Edmund Bacon, Margaret Mead, Arnold Toynbee and Buckminster Fuller. Later, over the years, Greece became a rest stop on my way between America and India. *Sparoza*, Jaqueline Tyrwhitt's house in Attica, was my true retreat. Panayis Psomopoulos became a life-long 'elder brother' sharing stories, good wine and long nights in the cafes of Kolonaki Square. After summers in Europe I would return to Cambridge, where the green leaves were turning bright red, yellow and orange. And at my true *ashram* I taught studios with Gerhard Kallmann, Jane Drew and Roger Montgomery. Dolf Schnebli became a friend. Fumihiko Maki became a lifelong mentor. During my many retreats at Gloucester Place in London, where Jane Drew and Maxwell Fry lived, I was exposed to their wide circle of philosophers, authors and artists. There were lunches with Freddie Ayer and Stephen Spender, evenings at the Albert and Victoria Museum and functions at the Architectural Association of which Jane was the President. I learned of Chandigarh, heard Jane's stories of love affairs between great people and heard first-hand stories of modern architecture in the making.

My fellow students were my teachers also. Thomas Cooper, Bruce Creager, Marc Trieb and Louis Kiszonak who also came up from Florida to Cambridge, remain sources of inspiration and shoulders to lean on even today! Michael Pyatok and Urs Gauchat were my classmates and competitors in Sert's studio. When I was twenty-six years old I was promoted as a tenured Assistant Professor at Harvard, and became a Member of the Faculty Senate. That's when my mentors cautioned me to move on. "Leave now," Joseph Zalewski warned me, "or you will rot here". Jaqueline Tyrwhitt scolded me, "You know too many famous people. You are becoming Harvard's official greeter!"

The Cambridge *ashram* was the center of the modernist *gharana* and had, as its gurus-in-residence, Gropius, Sert, Mirko Basaldella and many others. The intellectual atmosphere was full of ferment. We gleaned enough ideas and concepts to carry us through a lifetime. We learned that by opening one window of knowledge, we would be presented with ten more windows to open. But the real learning was not through the multiple windows, it was through the human examples around us who lived amazing lives. Their grasp of eternity and their place in history was astounding. Their vision, their missions and their sense of purpose infused passion in us. It set us off on a long journey and search. So when my mentor Balkrishna Doshi called me to Ahmedabad, I resigned from Harvard and embarked upon a great adventure.

Making an Institute: The Life of a Householder

In Ahmedabad, it was Doshi who inspired and encouraged me. I had been there on a Fulbright Fellowship in 1968-69 and thereafter returned to teach in the summers. He called me back to start the School of Planning at CEPT in late 1971 and encouraged me to run my own architectural studio. There I got my first design commissions: the Alliance Française in Ahmedabad, Dr. Bhanuben Parekh's House in Bhavnagar, low cost housing for hundreds of families in Jamnagar, and the SOS Children's Village outside Delhi and another in Kolkata. For the World Bank I designed a program called Site and Services Shelter in Chennai, which created habitats for five thousand families in our first experiment at Arambakkam. More sites followed sheltering another ten thousand households at Villivakkam and other sites where habitats were built by the people with their own hands.

In Ahmedabad we were limited to a few basic materials – concrete, brick, mild steel window and door frames and local glass. We made floors out of concrete and Kota stone. With these simply expressed materials I tried to carve out interlocking visual spaces, and to shape my sites into convivial social spaces, using the settings to guide my designs. A simple concrete column centered in a two-storied space, a cantilevered balcony, a bridge, skylights and plug-in modular toilets were the parts that made a building 'come alive'. These were the components from which I tried to make art.

At the other end of the scale was the massive low income shelter scheme at Chennai from 1973 onwards. This was one of the first projects I did with Kenneth Bohr and Donald Strombolm of the World Bank. Here, self-help construction and the most efficient layout of plots set the theme. The narrower and deeper the plot configuration, the shorter was the length of all of the infrastructure networks, reducing capital and running expenditures. This made the entire scheme cheaper and more economically accessible to the urban poor. On these small plots people built their own shelters, confident that with land ownership, their sweat and savings would not be lost. This typology became a model for the World Bank around the world, to use and misuse! Another challenging project for the World Bank was the Busti Improvement Scheme at Kolkata. Here we accepted what we found, improving health and hygiene, safety and comfort levels. We improved the hutments as we found them. Paved pathways and corner street lights were added. Communal bathing places and separate toilets for men and women were carefully placed. Potable water from common water taps was introduced. Storm drainage was added. Thousands of families had a better life due to this intervention. In Kolkata I came into contact with Sivaramakrishnan, a brilliant urbanist, and Arthur Row, a hands-on urban development man.

In Ahmedabad for the first time I earned my own way and lived as a householder and teacher. The Ahmedabad *ashram* integrated the modernist movement of the West into a search for an 'Indian tradition.' Balkrishna Doshi, Hasmukh Patel and Anant Raje were my gurus. Mentors like Charles Correa, Achyut Kanvinde and Laurie Baker guided me.

But in Ahmedabad I had to try being a Guru too! Doshi had asked me to start a School of Planning. He just said, "Do it! Pick up on your Harvard teaching experience and run!" The Ford Foundation supported our efforts through a grant of books, visiting professors and office equipment. Suddenly I had to write a curriculum, hire professors and scout all over India to find good students. I had to plan teaching schedules, make rules and try to bring a sense of order into natural chaos. My office attendant Baldevji put it very aptly. He would say, "This is not the School of Planning; it is the School of Problems!"

The School of Planning was an important experiment. It drew students from architecture,

engineering, social work, the liberal arts and technology, upsetting the accreditation committees of civil engineers, geographers and architects from New Delhi. Our pedagogy was through 'laboratories' rather than classrooms and studios. Students and teachers lived in villages, rural towns and slums 'learning from the people' with whom we made plans for the future. Theoretical subjects were cross-disciplinary involving economists, managers, social scientists, architects, community leaders and technicians. With Jaswant Krishnayya we introduced the first computer link in India between social science research, public policy and decision making. With Yoginder Alagh we prepared India's first district plan. That was way back in 1972. We all worked in multidisciplinary teams, where teachers were students and students were teachers. After all I was only twenty-eight years old when I initiated the School of Planning. My life in Cambridge, Massachusetts was in the *ashram* of truth, empiricism and progress. In Ahmedabad it was the ashram of devotion, social change and passionate service to community. 'From each according to his abilities and to each according to his need' was our battle cry. In Ahmedabad the word 'modern' meant transformation, not progress. We wanted to create a 'new man' and design a 'new culture'. Maybe we even wanted a revolution!

The CDSA Ashram: The Life of a Hermit

An eccentric administrator named Vasant Bawa unexpectedly entered to catalyze a new ashram. He had taken my advice on the new legislation creating the Hyderabad Urban Development Authority (HUDA). Now he wanted me to design the HUDA's very first project. It was the new township for two thousand Class IV employees of the Government of Andhra Pradesh. Twelve labor unions had formed a housing federation and within their limited means they wanted shelter. Working closely with these people, who were securely employed and had access to housing finance, we built a township of two thousand houses and amenities at Yousufguda near Hyderabad. The scheme provided one hundred square meter plots on a grid of serviced access roads. Each plot had a water tap, a separate toilet, a bathing cubicle and one room. These incremental houses had a secret agenda: each owner could add more rooms and rent them out to less fortunate relatives and fellow villagers. Thus, an 'organized informal sector' rental housing market was catalyzed. Five families could use one set of wet cores designed for one family. A lower income user group was addressed who would never have access to land or institutional finance. The owners became low cost housing managers. A rental market was created. This project drew lessons from the housing township for economically weaker groups that I had done for the Gujarat Housing Board in the early 1970s and the site and services schemes in Chennai. The fees from this project translated into my opportunity to jump out of Doshi's nest and fly! I wanted to start my own institute and this project provided the finances for that. Thus, the Centre for Development Studies and Activities (CDSA) in Pune was born.

When I shifted to Pune to start CDSA in 1976 I was thirty-three years old. This was my second institution. This was a place of intellectual research and social action. The design of my new campus followed the lead of the Alliance Française in Ahmedabad. Honestly expressed concrete beams and window boxes; natural stone from the area; clay tile roofs; and the then newly introduced powder-coated aluminum sliding doors. Again, large waterspouts from Le Corbusier; naturally expressed materials from Wright and Kahn. Then I was on my own to create interior and exterior spaces, find human scale, interlock spaces, and play with the landscape. At the peak of CDSA's ascendance more than eighty young professionals worked with me at offices in Pune, Bhutan, Goa, Almora in (then)

Uttar Pradesh, and in Jaffna and Galle in Sri Lanka. We prepared town plans, slum improvement programs, village, district and regional plans. We pioneered micro-watershed planning, micro-level social services planning, decentralized planning and participatory planning. It was a retreat from the real world, from where we wanted to change it. The ‘thinker-doer’ was our model.

My twenty years at CDSA were spent inventing, enabling and facilitating programs, all to assist households to climb out of poverty. Most of our ideas were put into practice. Again we worked in teams; again we used villages and slums as our research laboratories, classrooms and teachers; again we drew from all disciplines. The Centre was part of an active network including the Indian Planning Commission, the National Housing Bank, the Housing and Urban Development Corporation and the Ministry of Urban Affairs. Internationally we worked with the UNHCS (Habitat) in Nairobi, the Asian Development Bank in Manila and the World Bank in Washington.

The new School of Planning at CDSA was founded on didactic techniques initiated in Ahmedabad. The real world became the learning laboratory. Students lived in slums and villages learning from those they would plan for. Poverty, sustainability and inequality were the main concerns and issues to be debated. Methods of ‘stress identification’, problem identification, program design and monitoring the results of development inputs were evolved. At CDSA ‘modern’ came to mean planned amelioration and facilitated change. It involved the civil society, NGOs, governments and people.

Toward the end of my stint at CDSA my interest in design and architecture was awakened again. A commission to design the Mahindra United World College of India allowed me to employ my architectural language, while discovering a new sense of poetry. The stone walls reached up to the sky like the massive surrounding mountains; the little streets focused views toward lakes and valleys; small spaces, large spaces, low ceilings and high ceilings were all interlinked into an articulated experiential built fabric.

It was this commission from Harish Mahindra that lit a flame in me to confine myself to the *ashram* of a simple studio, yet re-engage with society as a maker of artifacts. Thus, my academic life came to an end and I left my life as a teacher and householder, entering a more inward and meditative stage. At CDSA the *ashram* was both one of retreat and one of engagement; it was the place of thinker-doers who explored more relevant ways of doing things with optimism to change the world.

The India House *Ashram*: The Life of a *Rishi*

Perhaps India House is more of a real *ashram* than either CDSA or CEPT were. It is even more of a retreat than Harvard Yard, or the Endless Corridor at MIT. It is a self contained residence, guesthouse, art gallery, office, studio, and public space for cultural events. One can even swim in the lap pool and thus pass weeks without ever leaving its limited compound walls. Surely this is my true forest retreat.

At India House about fifty creative people work on a range of design and design management activities. Major new projects have been initiated here and older ones completed: the Capital City Plan of Bhutan; the Indian Institute of Management at Kolkata; the Samundra Institute of Maritime Studies near Mumbai; the YMCA Retreat at Nilshi; the Suzlon One Earth at Pune; works at the College of Engineering, Pune; the National Ceremonial Plaza in Thimphu; the Bajaj Science Centre at Wardha; the Supreme Court of Bhutan; the Aamby International School near Mumbai; the Super

Computer Laboratory for the Tatas; the Center for Life Sciences, Health and Medicine at Pune; the Life Care Hospital at Udgir and many more. It is an *ashram* where young people gain their confidence and work from 'tired to tired.' It is my retreat into my secret world of ideas, sketches, design concepts and putting things into 'buildable' technology. It is a centre of art and architecture; a place of self discovery and transcendence.

At **India House** I have found the solace to write this book. I have found the peace to focus on a series of articles on urbanism published by the *Times of India* group. New design 'starts' have been initiated almost every month. There is a team of devoted architects contributing to a fellowship of creativity. India House is a place of incessant creative activity. The clock never stops.

Legacy

If you are an architect your true legacy is the love and passion of those who went before you. It is the smile in your guru's eyes when you excel and the annoyance in his voice when you fail. Your legacy is the stream of works built over the years, decades and centuries that were honestly created and are integral to their places. Your legacy is a challenge to be honest in your works and build in yourself the integrity that is expressed in your work. When you reach this plane true art will be your gift, and your endowment to the future.

My good fortune was in having great teachers, mentors and gurus. My great luck was having wonderful students, friends and a great lover. I searched for them and found them. I stumbled upon some and some found me. From them I inherited an endowment of passion for architecture and a love for life. I was taught life is a shared journey. I have thrived on the brotherhood and camaraderie of my 'fellows in arms' on the great battlefield called Architecture. That battlefield spreads into urban and regional planning, cultural conservation and environmental management.

Endowment

In a small, isolated Mizo village on the border with Burma I came across a hand painted sign that read, "You have traveled so far to come here, leave something behind, for good or for bad!" I suppose as architects and designers we leave a large footprint. For better or worse, it is a kind of 'endowment for the future.' We plan cities, build campuses, create stone and concrete reliefs, and we shape the way people teach and learn. We are there when people die and when they cry. We are there years after we die when they laugh and make love. While we can 'go on' to the next project and our clients often change jobs, the inheritor of our 'built wealth' (or poverty) is the user community of our buildings and spaces. It is humanity. Let us hope that our humble legacy becomes a considered endowment for the future. The buildings we make are our lien on eternity. From living and working in our spaces and experiencing our choreographies of place, let us hope we are able to pass on the passions, the feelings and the humanity that becomes a legacy for the future.

(Article in Insite Magazine published in June 2010 - Volume 3, Issue 5-6)

Five Lessons Life Has Taught Me

LESSON ONE

To gain something beautiful, one may have to give up something beautiful

One day, while I sat in my garden campus near Pune, surrounded by fifteen acres of fruit trees, flowering plants and lawns, a young architecture student came unannounced to meet me, insisting on having our picture taken together. Like many students who visited my campus at CDSA he was studying my designs and my campus layout. At that moment I was completing the fiftieth policy paper I had written on ‘Development’ and it struck me that no student had ever approached me for a photo session after reading one of my hefty policy papers!

At about the moment we said ‘cheese’ I decided to quit my post as Founder-Director of the institute and to devote my remaining life’s efforts to architecture. Among other things, I had to give up the sprawling campus I had created and move into a tiny apartment studio with modest equipment. The years since that impulsive decision have never allowed me time for regrets, or even to look back with nostalgia. But I had to give up my very own little dream world, created over twenty years of toil, to seek transcendence through my art. By giving up something beautiful, I found something even more beautiful.



LESSON TWO

It is better to BE what you are than to SEEM what you are not

In October 2001 I made a presentation of my new capital plan for Bhutan at the European Biennale along with some of the greatest painters, cinematographers and architects of our times. I noticed something very interesting. To seem a ‘creative artist’ in Europe you must wear the black uniform of an artist! To be a creative youth in Europe you must attend rock concerts waving your arms high in the air just like several thousand other conforming youth, pretending to be ‘free’. To be different, unique, free and an individual, you must wear the ‘uniform of the different’! You must wear a uniform – dress totally in black; wear black shoes; black socks; black pants; black belt; black shirt; black tie and black jacket. Perhaps even the underwear must be black! I realized that for these people, in fact for most people in the world, being creative is not a form of liberation, but is living a lie. There are people who never design anything, never write, never draw, and never search, never question, but who dress in the black uniform of creators. They are not ‘being’, they are ‘seeming’. If I have any lesson to share with young students, it is to BE, not SEEM.

LESSON THREE

Don’t be euphoric when people praise you, or depressed when people criticize you

In Buddhist thinking there are axioms called the *Sixteen Emptinesses* from which I have learned to keep my emotions ‘empty’. I became euphoric when my design won the American Institute of Architects Award: 2000, but having reached the final list for the Aga Khan Award, I lost. I realized that my happiness should come from the process of design and from my own understanding of the

inherent beauty of my efforts. About the time I settled with myself in this philosophy of emptiness, I learned that the project which won over us was disqualified as a fraud; the authors had misrepresented it as a design created by the village people. But that did not make me happy either. I have learned that creation is a patient search, not some kind of competition. To be true to one's art one must be empty to both praise and criticism and know oneself.

LESSON FOUR

Truth is the ultimate search of all artists. Even then I feel it is better to search for the good, than to know the truth

I suppose it took me too long in life to distinguish between *Ethics* and *Aesthetics*; Morals and Artistic Balance. Ethics is a rather exact science of rules; of right and of wrong; and there must be some generic truth within them. However, this world is not black and white, but rather grey and fuzzy.

On the other hand, aesthetics is the search for pleasure, which I call 'The Good.' Aesthetics is a question of balance, or what the Buddhists call the Middle Path. Beauty is a search for that Golden Mean; that harmony which brings all forms of visual, sensual and intellectual pleasure into balance. Harmony is the search.

If you are a lover of food, don't eat too much; don't overdo this or that spice; don't cook too long or too less. If you love wine, don't drink too much or never at all. In your love life don't be too passionate or too neglectful. The Good Life, or the Sweet Life, is all about pleasure and the pleasure principle. I realize that most of us are trapped in our Victorian fear of pleasure and have no aesthetics.

We are on an endless trip seeking the truth. We judge others, meting out what is right and what is wrong; dying as empty drums that never knew happiness and never spread that happiness to others. Art and architecture are but spiritual paths to 'The Good!' They stimulate enjoyment, delight and balance...*la dolce vita*...the sweet life. It is better to search in this life than to think one can know the truth.

Yes, it is better to search for the good, than to know the truth.

LESSON FIVE

There is only one form of good luck, which is having good teachers

Years ago Adi Bhathena, the founder of Wanson Industries that morphed over the years into the giant Thermax, introduced me to his ninety-year-old teacher. Adi himself was nearing eighty! We were sitting on the lawn of the Turf Club and Adi went into a long story about how he quit his comfortable job at age forty to risk all in a new venture here in Pune. He explained to me his middle class roots and that it was not within him to embark upon such a big financial adventure. Smiling at his teacher, he noted that without his encouragement, guidance and assurance he would have continued marketing Godrej products as a salesman. Then he turned to me and said, 'Christopher, in this world there is only one kind of good luck, and that is to have good teachers.' In what he said I felt I could hear the voice of an ancient sage somewhere on a mountain. I have never been able to forget that truth over the years, and I realize that all my teachers in India and America have been my 'good luck.'

(Interview by Sunanda Mehta for the Maharashtra Herald, published on 5th March, 2005)

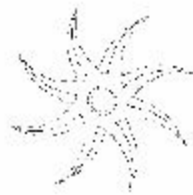
An Uncertain Journey: The Education of an Architect

From the moment young people decide to study architecture they begin a journey. At first it is an uncertain journey guided by glamor, images and hopes.

Their experiences in the classroom, in the studio, interactions with fellow students and chance meetings with practicing architects begin to set an agenda. They begin to hear stories of architects; they see architects' houses and they walk into interesting studios. Slowly they take upon themselves the images seen through the mirrors of others. Some have heroes, some are silent and some are the heroes of their own lives. Gradually these students start drawing realistic pictures of themselves and embark on the journey of meaningful self discovery. This should be the most beautiful time of their lives.

We as teachers play a pivotal role in shaping their drama of self discovery. We guide them onto the paths of their journey. Most important, we as teachers can inspire students. We can give them an insight which makes them realize something about themselves that they never knew. As I often say, there is only one kind of good luck in life, and that is a good teacher.

The experience for many students is exhilarating and transcendental. As mature and wise persons we can see them in their totality from a distance. With objectivity we can guide them on the path they need to follow. We can see their weaknesses and their strengths and help them ameliorate the former and reinforce the latter.



There is a critical point in their education when they either imbibe the concept of 'being a professional' or they drift off into fashionable, glamorous or celebrity paths. This is the first moment when our education fails young architects.

As mentors and guides we have to ask, 'Why do young people enter architecture?' Let me pose a few possible reasons:

Journey One:

'My father is an architect and I am planning to enter the family business.'

Journey Two:

'I saw an architect's picture in the newspaper and, by chance, the next day I saw him get out of a Mercedes. I want to be rich and famous.'

Journey Three:

'I would like a calm, artistic life, sitting in a serene studio surrounded by plants and paintings, contemplating and letting art flow.'

Journey Four:

'I wanted medicine, but my school-leaving scores were too low; then I tried electrical engineering, but I failed the entrance exam; so I paid a capitation fee and entered architecture.'

Journey Five:

'My parents want me to get married to a good professional as soon as possible; they just want me to graduate so that I can find a good partner.'

Journey Six:

‘I want to migrate to America and I think architecture is the best way. As soon as I graduate I will apply for a Masters’ program in South Dakota, get my visa and leave.’

Journey Seven:

‘My art teacher introduced me to the subject of architecture. He showed me a book on Frank Lloyd Wright’s houses and I was amazed. I want to be this person.’

Journey Eight:

‘I want to serve society and make a decent living while doing it. If I hone my skills, study technical systems, learn about materials and learn professional ethics, I can be a serious professional.’

These multiple possibilities continue. But we as teachers need to know where our paths merge with those of our students and where they diverge. Where do we touch their lives and what are our limitations to change them? What small gifts can we give them along the way? At some juncture of their journey can we make a small impact? Can we do this without becoming involved with students, as their friends and as their confidants? Can we leave personalities and campus politics out of all of this? Can we see the strong points of even our weakest colleagues and help them to be better teachers, instead of ridiculing them? Can we keep the distance of a wise guide and still pass on values, inspirations, sensitivities and understanding?

On a larger canvas the course curriculum is the highway, or even expressway, down which all of our students are racing. It has many lanes, many entries and many exits. This road can be made monotonous or exciting. It is the quality of teaching that makes it either a smooth and scenic ride, or a bumpy and tortuous one.

I feel we have two kinds of gifts as teachers that we can bestow on our students:

One is to help youngsters **see an image of themselves**. We give them images of what they can be and how their own inner strengths and values can transform into a ‘life’ and a meaningful role in society. That is a very personal gift from one person to another. It is called inspiration. But our collective gift, our group goal, must reside within the course content, the required reading, the meaning of projects and **the experiences we create** for them and into which they immerse themselves. We have to be good at teaching this curriculum and skilled in making it real and lively to the students.

We as teachers need to know where our paths merge with those of our students and where they diverge.

In brief we have to provide an excellent grounding in essential knowledge; in necessary skills and in underlying values. As a group we have to decide what are those skills, knowledge and sensitivities.

I can compare the first year of medical education and that of architecture and I know that the young doctors have mastered *Gray’s Anatomy*, embracing the nervous system, the skeletal system, the circulation system, cells and their nourishment and all of the organs which control, monitor and fix this complex system. I am sure that at the end of the first nine months of architecture our students will not have a clue of the electrical, plumbing, air conditioning, structural and functional systems which are elements of every building. Even upon graduation we send ill-prepared people out to solve the problems of society. The history of architecture is not made up of the sum of all of the buildings constructed, but of the structures in which a new insight, a new material, a new technique or a new way of looking at space is revealed. I wonder how many young architects are equipped with a complete knowledge of this stream of history and know where they can make a contribution, or how they can employ what has already been discovered.

I even wonder how good our new graduates are at drawing and sketching free-hand so that they can quickly study options and

conceptualize solutions. Most of our youngsters visualize 3D images on a 2D computer screen. Are we making a marriage between the real and the virtual world?

- Do students know the values and design logic of harmony, proportion, scale, and balance?
- Do they know that architects can become the touts of builders who only care for municipal drawings and how much FSI can be harvested? Are we exposing them to the processes of urbanization and the role of architects, builders and planners in creating a vessel in which the multitudes can live a beautiful and poetic life?
- Do our graduates know what phases one goes through to make a real building?
- Do they know that there are numerous roles they can play within this maze of procedures and expected outcomes?

Within this conundrum it is very important for young architects to know that being ‘creative’ comes far down in the list of logical, rational and responsible things they will have to do to be good professionals.

We spend far too much time trying to teach what we cannot teach, which is creativity, and very little time teaching what we can teach, which is knowledge, skills and sensitivities. The result is that our graduates have a very wrong impression of what we actually do in a studio and they lack the real skills to do those things. Many imagine that after a two- or three- year stint of work they can open their own offices and do large projects. They are not ready to suffer the low salaries and long working hours that chartered accountants, young doctors and lawyers put in during their apprenticeships. We have not properly grounded them on the path they must endure.

The greatest gift we can endow students with is the knowledge that they will always be students. We must teach them how to be continuous seekers and learners. We must show them how opening one window of knowledge shows us the way to more windows and still more.

We have to bring our students and young architects back to basics. We have to make them into responsible, capable and sensitive young professionals. I believe this is **doable**. I believe we are here to think this through. I believe that working together we can indeed do this.

(Lecture delivered at Rachana Sansad Academy of Architecture, Mumbai on January 16, 2009)

The greatest gift we can endow students with is the knowledge that they will always be students.

In Search of Architecture

Architecture has always been a part of the mentalities which criticize, question and ultimately rule society. Architects have always left lasting images of the societies which patronized them. As societies fade away it is the architect's footprints which remain. They leave the final images by which each era is remembered – made into a legend.

Contemporary Architecture

Today the dominating role of the architect is fading in the fashion-driven market that gives form to most of our environment. Modern civilization seems to be enchanted by the realm of the image, by projecting the values of 'packaging' to the detriment of architectural contexts. Contemporary architecture is seen as a permanent surface decoration – the wrapping of materials around functional interiors. Urban deterioration and ugliness reside in the interaction between economy and politics in a manner which determines the role of architects in society. Mostly, contemporary architecture is a kind of escape from the vapid world of materialism into a shallow amusement, or at best into self-indulgent deception. It is aiming at a new 'creative' statement which actually betrays a widespread poverty of ideas. The architect is driven by tasteless clients to provide forms, colors and textures which are understood in the media of fashion as 'being there,' when in fact the resultant buildings are nowhere. The architect has merely put his stamp of approval on a client who craves social recognition. Beyond the graphic frenzy, somewhere past the Babel of symbols and signs, the abiding force of architecture demands a commitment to human dignity, an honest expression of materials and technology and a search for meaning, as opposed to the frantic stimulation promoted by current design trends.



On the Role of Patrons

One must add that goodness lies deep in the human soul and there are patrons (as opposed to clients) who call forth the good in the architect, and from such a relationship beauty can emerge. Whatever good we can achieve lies deep in our patrons' faith in architecture and the free hand they give to *create something beautiful, which would be a lasting gift to the world*. Inundated by the desensitized environment, the thrust of architecture constantly struggles to create precincts of peace and meditation. In such a sanctuary, liberated from the bondage of time, architecture exchanges its mechanical form and its crass packaging for a spirit of poetry and mystique. The immediacy of contexts and their poetic ambience reveals what is most real and fundamental. The architect must aspire to construct a sustaining spatial domain that goes beyond the allure of mere packaging.

Architecture and Fashion

Architecture re-impassions a world whose values have been destroyed. In an era when civilization has deployed its most devastating forces against humanity and our environment, architecture must maintain faith in a transcendental future – a future that can mend a wounded world, crippled by the onslaught of signs, symbols, images, tricks and flippant styles; a future that challenges a society immersed in self-indulgent visions. Architecture ceases to be a mere wrapping when it ceases to conjure fashion and begins to unfold its unique pursuit. It is in the bliss of its presence that architecture deducts from all the chaos in life, affirming reality.

Architecture and Context

My own pursuit for architecture was rekindled in the vast Sahyadri Mountains; in nature; where trees meet the sky; a place of unencumbered horizons, where nature dominates each possible view. There is a compelling beauty in the profusely barren hills of this dispersed environment, haunting in its solitude – not a solitude filled irreverently with the urban glamor of disposability. The ever-present mountains tenaciously project fantastic architectures of shade and form. During the hot season they offer no shade from the relentless sun. During the monsoon they offer no protection from the storm unleashed. In such a setting one cannot hide in fashions.

Beyond the graphic frenzy, somewhere past the Babel of symbols and signs, the abiding force of architecture demands a commitment to human dignity, an honest expression of materials and technology.

The mountains cast cool shadows over villages, over lakes, across rivers and vast territories. Each shadow points to another; freed from contrived economic impetuses, generating a transcending landscape of immense power. This is not a setting for the fabricated urban packaging, all wrapped in yesterday's new idea. Architecture in such a setting must take a stance resisting alien, urban conditions, rather than an attitude serving to endorse and perpetuate them. In such an empowering setting, I built my own institute, CDSA and later the Mahindra United World College of India.

Principles that Guide Design

The struggle with this awesome landscape, trying to find a meaningful way to build, drew me toward some abiding principles. Guided by these, I felt prepared to address the mountains; to work with nature and to reject fashion. Let me spell out these principles – the values that I feel should rule architecture:

Context

A building should be part of its context. It should reflect and extend the scale, proportions, textures and colors of its surroundings. It should integrate into the existing movement system, into the contours, and into the visual backdrop.

Scale

Buildings should engender a human scale. An inhabitant, or a visitor, should be greeted by a subdued landscape and portal, enter through low spaces or a small foyer, and then be introduced to larger spaces which emphasize the human scale through counterpoint. There should be motifs like windows and doors which scale down massive walls; or like waterspouts – which are almost

anthropomorphic – which throw dramatic shadows over strong stone walls.

Proportion

Buildings are assemblies of elements and motifs. These must all relate to one another. The sizes, measurers, placement of things, and locations of elements, must all fit into a system. As in a human body, everything has its place, its proper size, its relationship to all the other parts. What appears to be fanciful must have some deeper logic.

Simplicity

A quote often attributed to Einstein is, “Genius is making the complex simple, not the simple complex”. In architecture this quest for simplicity means defining a language. For each element (support/span/enclosure) of a building, or a campus, one must define the simple terms one wants to use and stick to them. For ‘support’ one could say ‘stone bearing wall’; for ‘enclosure’ one could say ‘glass sliding walls’; for ‘span’ one could say ‘sloped tile roof.’ Whatever the terms, choose them carefully and stick to them.

Nature

We should use natural materials, expressing their inherent beauty. Climate, budget and context may temper this; we may have to dress a brick wall in plaster clothes and color the plaster with paint. But we should seek out natural colors – earthen hues. Our buildings should not appear like over-decorated and painted harlots. Their natural beauty should be highlighted. This aspect can be enhanced by merging landscape with built form – bringing the outside into the building. Courtyards, quadrangles, verandahs and porches all work toward this end.

Function

Buildings have specific functions and also include more important generic functional systems. They demand to be divided into long spans and short spans; into noisy areas and quiet areas; into public areas and private areas. The ‘zones’ must be connected by an appropriate circulation system, separating pedestrians from vehicles and service areas from user areas.

Motifs/Decoration

Buildings are not mere machines to live in. They transcend mechanical necessity. But the spirit of transcendence must not be confused with the glitter of costume jewelry, with gaudy make-up – a kind of interior decoration turned inside out. A more relevant search may be for ‘motifs’ or ‘objects’ which solve little problems, and in doing so add an element of delight to our work. These could be waterspouts, columns, steps, seating platforms, little windows, doors, statues, reliefs and lintels. These could be incidental, yet powerful ‘adjectives’ and ‘adverbs’ which describe and embellish our architectural language. These details must be used with constraint and consistency. They must counterbalance the strong ‘nouns’ and ‘verbs’ of the architectural language (support, enclosure and span).

My struggle to design the Mahindra College was resolved largely through these axioms, or guidelines

by which I could check myself. Perhaps every young architect needs to create his own landmarks against which they can check their work.

(Written in early April 1998 about the design for the Mahindra United World College of India, which later won the Designer of the Year Award in India [1998] and the American Institute of Architects/ Business Week/ Architectural Record Award 2000)

A Timeless Way of Living

In architecture we are passing through a period in which the baby who cries the loudest gets the milk. What I mean is that architects are screaming and yelling like babies to grab attention. Façade architecture – the packaging of buildings in trendy wrappings – is popular. Fashionable western architects are ‘selling styles,’ not making architecture. Each building they make looks like a copy of the one before it.

These architects are playing on only one sense, the visual, leaving touch and textures, smell and sound, volume and proportion to the winds. Common sense, context and integrating with nature have become passé. In other words architecture is at one of its low historical points where most of its practitioners are chasing style and crude popularity. This bad taste is media-driven from cities, outward to the smaller towns. It works on the center-periphery phenomenon where more and more energy builds up at a central point until the system explodes. While this is happening at the center, there is more calm, thought and reflection out on the periphery. Often more creative works can be expected from either Ahmedabad or Pune than from Mumbai, and from Mumbai more than from New York. But young architects always look into the distance to find local truths. When they seek truth at the center they will only find opaque theories, not relevant processes and methods to create true art.

Over the past decade young architects have grown up in a digital world. Their experience of architecture has been in virtual reality: 3D on a 2D computer screen. While this has helped to push the limits of the visual world, it has suppressed experiential architecture that finds its true measure not only in vision and sight but also in touch, smell, sound, sequence and movement. In the resulting cacophony we even find young architects wondering: What is Architecture? They want to know what the reality of architecture is.



Continuously Educating Ourselves

Education in architecture is a search for the reality of architecture. There are several ‘givens’ about this reality that form the basis of education and practice. I list a few of them:

1. Architecture is built; it is construction; it is technology.
2. Architecture is a response to functional needs; it is a product with performance standards.
3. Architecture is social action. Every building either gives to or takes from the social milieu. At the most basic level the exploitation of the maximum Floor Space Index – a commercial factor – becomes an indicator of the architect’s ‘social commitment.’ Architects can also create new public domains. They can make schools into places that stimulate learning. They can bring nature into people’s everyday lives. They can create social housing.
4. Architecture is an exercise in economic analysis. Every client has a budget that is an estimate of the value of the economic operation of the building in producing something – at least happiness in a home or inspiration in a school.

5. Architecture is history as it is a part of a behavioral pattern which persists over time. It is a process in the present, which draws on the past and creates the future.
6. Architecture is poetry, because in the end it must go beyond the programmatic. It must say something about the human condition that is not explicit. It must raise people's spirits and spark their curiosities.

Architecture exists just one step outside materiality. It dwells in our sense of experience, our immediate memories and in the identity of a place.

Critical Regionalism

I feel each country in the world, and each region in each country, has its own unique expression of architecture. There are elemental concerns (confused as global concerns) that attract the efforts of all sincere architects of good intention. All true architects seek the honest expression of materials, the employment of human scale and proportion, integration with nature, a belonging within context, a gift of meaning and sense of place for the inhabitants. All of these characteristics can be discovered through the study of traditional buildings and neighborhoods in their own contextual settings. One need not go to London to find architecture. Every regional context holds the secret of good architecture. Bangalore, Kochi, Aurangabad, Ahmedabad, Chennai, Pune, Kolkata, Delhi and many other such places are regional centers with strong local contexts to draw architectural lessons from. Why not begin one's search for architecture by studying the vernacular structures in one's own vicinity? Maybe all of the secrets lie within a ten-minute walk? In vernacular cultures there is a lot of variety and self-expression that comes from within people and projects out. In global culture there is a homogeneous mono-culture oppressing in on the individual and stifling creativity.

Vernaculars: Attitudes/Components/Elements

Thus, when a young architect is searching for their unique identity as a designer, that search for a language can begin within a hundred-kilometer radius from their home! Maybe it begins right at home!

Every architect must have a language, and in fact I believe each region enshrines an architectural language, or its own dialect. This is part of a cumulative heritage that can be studied, imbibed, and used as a vehicle to catalyze new and relevant design ideas. Such an approach generates a fellowship between local competitors, as a common and mutually shared approach emerges in their way of doing things. I am proposing that regional architectural languages can be defined by seeking out regional attitudes toward space and place making; by searching for and defining basic components; and by understanding the elemental characteristics that persist through and gift meaning to local structures.

Let me further elaborate on these aspects of regional styles and languages.

I Attitudes

Attitudes toward spatial arrangements and place making are embedded in the experiential use of architecture. What is sacred in a place and what is profane depends on local culture. Is there a difference between a mundane door that provides security and a profound 'portal' that is a transition into a sacred space? Does every place have its own sacredness that must be celebrated? Does a central courtyard define movement from the public realm of the street to the private, more personal

realm of a dwelling? How does a door 'proclaim' the transition from an impersonal street to a special place? Does the courtyard catalyze conviviality? Does it demarcate the domestic sphere from the occupational sphere? Does it tell outsiders to temper their behavior to a more respectful and considerate mode of interaction? Do murals, statues, paintings and artifacts begin to speak about the particular likes, nature and concerns of inhabitants? Are they cues telling visitors who they are meeting and what behavior would be appropriate? Different regions use and modulate attitudes toward 'space and place making' in different ways. There can also be significant shared ideas and concepts between regions. Attitudes are not exclusive, they just exist. Think of the various doors you have seen in your life. Imagine their sizes, materials, colors and shapes. Some are set back in shaded niches or are introduced by a cozy porch. Some have an alcove inside where one can sit and play chess on an *otta* built within the niche. Some are so large that they have a small door within a larger door. Some focus a view on a statue in a courtyard to draw you within. Some hide what follows, creating a sense of mystery, and then surprise. Various regions express their own attitude toward a door, an entry, a portal and the 'in-between space' separating the aura of one from that of another.

There are major themes or attitudes toward built form from which regional languages can be understood:

Attitudes towards Nature:

'Falling water' by Frank Lloyd Wright is striking in the way it integrates with its natural setting. It exhibits an attitude of being a part of nature. On the other hand, Le Corbusier's 'Villa Savoye' floats in solitude, detached from nature. It is an attitude of abstract reality. A Rajasthani mud house rises from the earth, while a Rajput palace towers magisterially over its domain.

Attitude towards Proportion and Scale:

If you stand in front of the High Court at Chandigarh you feel dwarfed by the immensity of its portal. If you enter 'India House' there is a scale that is intermediate. It shares the intimate experience with a hint of a larger realm of things. It startles with its combination of roles as a home, a studio and a public institution. Entering Dr. Oswal's Center for Health, Life Sciences and Medicine, one is made to feel at home, intimate and secure. Scale and proportion are used in different ways in different regions, be it the Chola temple complexes of South India or in the great Mughal tomb gardens of North India.

Attitude towards Materials:

Each region has its own local building resources which can be used and expressed in different ways. In Karnataka one finds wonderful granite fit for columns, beams and roof slabs. In some areas there is abundant clay for bricks, hollow tiles and roofing tiles. In Himachal Pradesh the abundance of sturdy wood generates its own attitude to how slate, wood and stone can be harmoniously employed.

These are not building technologies or techniques. They are commonly held concepts of how one crafts spaces and forms places. The work of Shankar and Navnath Kanade, Jaisim and Shashi Bhooshan emanate from their distinct regional attitude toward materiality.

Attitudes towards the Sacred

In New York City there are churches, synagogues, mosques, temples and meditation places. But these highly revered and sacred places are small events in the larger functioning of the city. In Paris the historical and heritage sites take on a character of sacredness. These are very special and cherished common properties that form an important part of the French psyche. In Indian cities like Pune, Old Delhi or Varanasi one finds thousands of small, medium and large temples that texture the urban fabric. Each shopkeeper has created a small temple within a shelf over his cash register, in every wall there is a niche with a deity, or a symbolic hint of one. Sacredness is kind of omnipresent and omnipotent. Even a house in India is made up of sacred space and place elements forming a sacred whole. The direction of the entry, the location of the kitchen, the alignment of the temple in the kitchen and the sacredness of the kitchen itself, are all part of local attitudes that must not be disrespected. The work of Girish Doshi, Sanjay Patil, Madhav Joshi and Deepak Guggari in Maharashtra speak of this attitude.

II Components and Connections

At a very simple level architectural language is made up of basic functional parts. These are nouns, or components which are things. Fundamentally, these are supports, spans and enclosing envelopes. A bearing wall of stone may be both a support and an envelope. A span can be in the form of a beam, vault, shell or a flat slab. Supports can be stone or concrete bearing walls and steel or concrete columns. The possibilities are limited, but we must conceptualize each of these three kinds of components separately. A language is also made of verbs, or connections, ‘hinges’ or stems to move through. Connectors can be arcades, courtyards, promenades, water pools, visual axes, passages, portals, bridges, stairs or ramps. These create experiences as people have to move through them. As people move, all the walls, columns, windows and objects in their line of sight move in relative terms and architecture becomes kinetic. To me, identifying these components is the easy part of making an architectural language. What are the local support, spacing and enveloping components?

Identify ten components and use them. What are the typical roofs, shading elements, types of stairs, supports, spans, envelope devices and their connections? How can we draw these components from history and from our contextual surroundings? Can we use them in a manner that employs respect toward and amplifies local attitudes? Architects in Ahmedabad are careful to articulate these components as a regional attitude to architecture.

III Elements

More difficult is the understanding of the **elements** of architecture:

Elements persist through systems. They are everywhere. A glance at them tells you where a building is located. You see light blue plaster walls – almost white in the bright sun – and you know you are in Jodhpur; pink, and you are in Jaipur; yellow floors and earth-colored stone walls, and you are in Jaisalmer.

The characteristics of buildings in different regions and contexts will be different. In Bhutan the red band around a building declares it as sacred. The pagoda roof over roof in Kathmandu valley creates a local identity. The community tanks of Kolkata are the focus of neighborhoods. The tanks of Tamil Nadu are more formal and defined.

Perhaps the attitudes, the components and the elements are all intermingled. But a young architect must take their sketchbook and document their environment. They must see how light is employed;

how shades of colors are used; how textures in floors are laid out; how echoes and reverberations of sounds are handled; how alignments, landmarks and axes generate sequences of modulated experiences.

Each school of architecture should run a workshop, sending students into towns, villages and hamlets to document attitudes, components and elements which define a regional architectural language. These should be looked at critically to assess their validity and relevance to contemporary needs, functions and lifestyles. Analogies between traditional and contemporary materials and technologies can be conjectured. Possibilities and potentials for applying local concepts to new designs must be debated. Findings should be listed. New prototypes should be attempted, based on old plan concepts and spatial arrangements.

Gharanas of Architecture

We have in Hindustani Music the concept of *gharanas*, and in philosophy we have ‘schools of thought.’ We need ‘schools of thought’ and *gharanas* in architecture too. In ancient regions one can see unique schools of thought emerging. The history of architecture in a given place is a narrative of the culture of the people who have lived there. The changes in style mirror the society which has evolved in that place. The buildings are landmarks along the journey of history that give meaning to a context. We can have clear attitudes toward nature, materials and proportion. We can have unique components to create support, span and enclosure. We can have special motifs for shading stairs, floors, seats and connections. We can have our own elements and unique ways to employ them. These will evolve as the functions, technology and the culture of a place evolve.

A wonderful challenge calls upon young architects to expose the meaning and reality of architecture in their local contexts. They will discover the identity of their own culture and begin to enrich and mature that identity. They will find meaning in their work and purpose in their lives. I challenge you, young architects of India. Make your own language and your own style.

(Keynote address to the Karnataka State Convention of the Indian Institute of Architects in February 2006)

Architecture as a Social Tool

Hypothesis

Historically architecture has addressed itself to the problems of a very small class of citizens. Though the social and economic context in which we work has changed, we are still burdened with an ethos of the past. Architecture has, by and large, been the expression of the interests of a small class of clients – an expression of enlightenment, progress and power. Even the contention that architecture should be functional has been a point of debate in this rarefied atmosphere.

It is important for the next generation of architects to consider my hypothesis that architecture can be a social tool, a tool for change, a tool which expresses new social relationships.

By focusing on the common man we not only resolve his needs but we express his new importance and motivate a reconsideration of priorities. By focusing on communities rather than individuals, architects express the meaning of mutual cooperation and collective social security. Architecture can revive respect for local resources, techniques and capacities. It can reflect a dignity in the culture of small and poor communities who have been humiliated through mass media, by high technology, large systems and a feeling that the important events are controlled in another context. Architecture represents one of the most visible, expressive tools for invention and change.

If architecture as a profession is to play its essential role as a social tool we must be clear about:



- our GOALS;
- our APPROACH;
- our STYLE;
- what is NEW;
- our STRATEGIES; and
- our relations with PEOPLE and POLITICS.

The following paper attempts to discuss these issues.

Goals for the Profession

In international circles today we hear of basic human needs. Habitat and environment institutions and leading thinkers have outlined a list of goods and services which are needed by each and every human being. What are these? Food, clothing, shelter, skills, awareness of health and hygiene, access to social and economic services, and participation in local governance. In India we have talked about a Minimum Needs Programme which focused on rural housing, clean drinking water, access roads, village electrification and the like. While this emphasis initiated a discussion on the tangible benefits of development that can be realized by the common man, it was only a beginning. The emphasis of architecture and planning did not change. Town planning continued its quest for the Garden City and macro-planners projected the needs of industry for basic inputs. Architects designed their urban monuments. But where were the individual, the rural laborer, the hutment dweller, the poor, in this

highly segmented, sectionalized planning process? It is clear that the common man was neglected and in fact there is evidence that this position, vis-à-vis basic human needs, deteriorated over the period of India's early development strategies.

Our planning has failed because it has not grasped and focused on the issues correctly. On the one hand we have in town planning tried to create principles out of middle class western ideals. Neatness, cleanliness and orderly façades have been the mirage which physical planners have chased. Our economists have looked at the structure of western industrial economies, at the inputs and products needed to build an industrial state, and have tried to use these models as a 'cookbook' for filling in the missing parts. Realizing the failure of the dream of importing heavy technology, the scientific community has tried to pull their fingers out of the fire with terms like 'appropriate technology' and 'intermediate technology,' hoping to keep their place in the questionable system of development they proposed in the 1950s and -60s. Why have we failed? The question haunts us! As we approach our goal as a 'high technology nation', we realize that technology does not reach those who need it the most.

We have failed because we have not taken the bull by the horns. We must declare that basic human needs are in fact **the basic human rights**. We must be clear that these basic human rights are our top priority and that all professions must be held responsible for delivering the goods. Let us make it clear at this point that when we talk of human rights we do not refer to elusive concepts of freedom, or a philosophy of justice. Let us be clear that we refer to nutrition, clothing and shelter; to the right to be trained to participate in the economy and to subsequently participate in society; that there is a specific 'basket of goods and services' to which everyone has a right, of which shelter, sanitation and hygiene are crucial. These, then, are the goals which we in a new profession must support.

Approach

The above immediately changes the emphasis of architecture away from the urban, the monumental and the stylish, calling for an entirely new way of looking at architectural education. It re-focuses on:

- An architecture that *works at the micro level*, that is the level at which communities or interests aggregate themselves. It could take the traditional village as a social unit or the watershed as a geo-economic unit. But it must be a small enough unit in which target populations and coverage can be analyzed and specified. It must also be small enough for a need-based 'identity of common interests' to emerge, yet large enough to support a basic management function. This unit, perhaps an entire village, should be considered the HOUSE of the community. As a community sharing a HOUSE a common set of services must be designed and shared.
- An architecture that is *cross-sectional and integrated*. Architecture must be holistic in embracing the total environment. The use of the word 'habitat' in this broader sense is important. It means that activities must be aggregated around the function and the user, not disaggregated into the technical specialties of the 'suppliers'. Processes should be designed which enhance the community. The community should not be approached piecemeal through roads, shelter, water, health education (i.e., 'departmentally' through the PWD, Housing Board, Irrigation).
- An architecture that *directly involves the people*. This means the creation of institutions at the micro level that have decision making and management capacities. It means that ideas must emerge from the people and that the paradigm 'government decides, people support' must be reversed to 'people decide, government supports.' Giveaway planning, which emerges before elections and then quietly dies away to be replaced by paternalistic voluntary work, must end. Architects should provide the 'idea tools' by which each village can prepare its own environmental action plans.
- An architecture that *builds on local resources and capabilities* and fulfills basic needs for shelter, amenities and services.

- Above all, architecture and planning that creates a base line of guaranteed consumption on the basis ‘to each according to his need; from each according to his abilities’.

Style of Working

Accordingly, as architects we must accept a new style of working. The age of the individualistic master builder is over. The age of the commercial architect sitting in his office is over. The age of the specialists doling out advice on forays into the field is over. The new style must include:

- Working in teams to discuss problems and find solutions, building on each other’s strengths and cancel out each other’s weaknesses.
- Bringing forth multi-disciplinary skills and abilities and using the social sciences as an analytic base.
- Keeping in contact with and learning from the people through field work. Moving in the villages, moving with the people and learning from them is an essential technique of the new professional style that must emerge.
- Taking a scientific stance with regard to understanding the structure of the areas within which we work.
- Developing, on the one hand, tools to document consumption levels, the impact of different inputs, production systems, local materials and techniques, and distribution systems; and on the other hand avoiding the pseudo-science of hypothesis formation, survey design, survey test, sample and control group – all neatly juggled from the comfort of an urban office and administered by inexperienced graduates in the field. We will only understand the nature and the character of problems through case studies, group interviews and area histories. These are the tools through which we can identify and define the stresses that threaten people and their contexts. Problems can only then be stated as questions to be answered. Sample surveys can then follow to tell us the size and extent of the problems, but let us be clear that surveys cannot identify problems or the issues of actors in the system. The field is our great laboratory and we must use it and learn from it.

I would take a similar stance on

- highly quantitative approaches which draw broad conclusions from faulty secondary data.
- the design and testing of new architectural processes which result in ALTERNATIVES for society.

It is our role and duty to show in concrete terms the alternatives and the choice system. We must state clearly, step by step, methods which incorporate the elements noted above in the new profession. Finally, the new style demands a selfless immersion into the problems of the poor. A discipline of doing, seeking results and re-learning must emerge.

The New Architecture

Such a new style of work and a new approach must define the **New Architecture**. And the shift from the old architecture to this New Architecture is a cause of conflict and tension because:

- it indicates a restructuring of the gains of development; it means redistribution of the benefits of

production;

- it undercuts the role of commercial architects and technocrats who plan in a top-down manner;
- it negates the importance of the old school academics, their abstract techniques and the institutions they have built around these effete methods; it also negates the importance of their internally ‘consistent’ models;
- it directly confronts the hold of feudal forces over the masses of the rural population because it initiates self sufficiency;
- it confronts the comfortable lifestyle of the urban elite crypto-leftists who favor the starvation of the poor as a catalyst of their intellectualized, self-centered, utopian future;
- it challenges the established system of production relations, placing decision making in the hands of the people, and guarantees them the right to consume at specified standards.

The six factors indicated above collectively represent vested interests opposed to the New Architecture and, given their individual and collective power, it should not surprise us that the New Architecture faces opposition from many quarters.

But the New Architecture **will** succeed. Why? It will succeed because:

- it is in the interest of the mass of the population and it is a majority interest;
- the collective vested interests noted above are in mutual conflict and they will work at odds with each other;
- there are progressive elements that persist through society in all walks of life and as individuals they will do their duty to ensure that the New Architecture endures;
- it is essential.

Strategies for the New Architecture

What are the strategies for the emergence of this New Architecture?

Institutions must be built around it. These will not receive funding from traditional sources, so they will have to support themselves and depend on progressive individuals for support. They will have to play a frontline role in facing the attack of vested interests.

A new profession of architects must emerge in which the basic goals and techniques of the New Architecture take priority over specialized disciplines and in which individual ambitions are curbed.

The New Architecture must grow from within existing systems and architects must work to create issue-oriented alternatives around which constructive political debate can emerge. By staying aloof from the system one becomes an uninformed and isolated critic. Architects should not become a burden on society and should not become a privileged class of dependents as academics have done. Only by acting within the system will architects learn and play a role in change. This does not necessarily imply that they will work only in government jobs, though that is clearly an option. Additionally they may:

- Work as ‘barefoot architects’ for communities through voluntary agencies. This would be the ideal way for young architects to begin their careers. Agencies like the Ahmedabad Study Action Group (ASAG); VIKAS (Ahmedabad) and UNNAYAN (Kolkata) present us examples of such initiatives.
- Work in an action oriented institution where their contact in the field would be intact and where

they can work on ‘live projects.’ The Centre for Development Studies and Activities (CDSA), Pune, is an example of such an institution.

- Teach other young architects.
- Work as independent authors, consultants or freelance researchers.
- Work with a relevant development agency, such as a financial institution or a development authority.

Whichever of these paths an architect chooses to take, he must be aware of his own careerist tendencies. He must put his goals and ethics above mere opportunities and position. He must convince his fellow workers of the validity of his approach and style.

The new architect must act as an agent of change. He must study problems, design new programs and promote these tools for change. This will involve ceaseless effort.

People, Politics and Architecture

Needs and resources emerge from the people, solutions emerge with the professionals’ assistance, and the power to change emerges from the politicians. These three essential actors in any change model should have a common set of interests or goals if that model is to deliver on its promise. The present crisis in architecture and planning – in fact in our nation’s development – emerges from the conflict of interests between these three groups. The traditional architect is isolated from the people and aligned with an array of interests of special groups. The goal of town planning will coincide with the needs of a very privileged group of house owners and land investors, elite groups concerned about the appearance of the city, or the needs of industrial houses for whom an ‘efficient’ city is important. Our macro-planners are aligned with the needs of basic industries for the supply of essential inputs such as diesel, cement, steel, electricity and railway wagons. In turn, these interest groups have their own political connections in the exercise of power in decision making towards common ends.

It is necessary that the New Architecture be clear about its goal and that it builds its ties around a common, systematic presentation of interests with politicians who share the same goals and objectives. As a new profession we must profess specific goals and our ethics must be built around an adherence to those goals. We must understand that our profession has two basic components:

- **tools**, which we share in common with all architectural professionals; and
- **interests**, which we profess and which distinguish us from all other professionals.

It is out of the latter, or out of the shared interest with the common man and our commitment to basic human rights that we will form our political ties. This distinguishes us from the technocrat who, committed only to his tools and techniques, can theoretically work for any master. Within the New Architecture we must also beware of individual power wielded at the expense of the interests or unanimity of the larger group. The New Architecture should not be hijacked and held to ransom by individuals whose style, approach and goals do not conform to the majority interests of the profession as a whole, or who use the language of revolution as a tool for acquiring personal dominance.

(Article published in 'Akshara', October 1980)

The Design of Design

We are all gathered here today by a common devotion to something called design. For each of us it may have a different meaning, but for all of us it is a ‘process through which something is achieved.’ We may think of ‘a design’ as an object like the Coca Cola bottle, or the Sony Walkman, or as a beautiful interior space. But the iconic designs which come to our minds are the outcomes of a **design process**. We are all involved in this process.

All of my friends sitting here are ‘designers’, be they industrial designers, architects, interior designers or artists. Each, in their own way, is a master of a unique design process. Design to me is a **method to achieve an end result**, whether it is creating the Tata logo, conceiving of a reading lamp, or rolling out a new automobile. The process starts with a vague image of what is needed and desired. It involves defining performance criteria and applying legal standards, creating optional solutions, and evaluating options against performance criteria. Then we create and refine prototypes before rolling out the final product. Reasoning, criticism, logic, questioning, simplicity and analysis are all fundamental to the design process. Limited resources – whether in the form of finances, human effort, or time-temper the process. The best designs often emerge when the defining resources are constrained; hence our attraction for tribal art, handicrafts, and the rustic architecture of villages.



Designs may range from the plan of a city, the design of a neighborhood, the layout of a public space, the design of a building to designs for lighting buildings and open spaces, designs of household goods and of small artifacts. They include the lighting of an entire road network or that of a small apartment...

As designers, we work as the catalysts of complex interest groups and stakeholders who will manufacture, use and judge these designed artifacts, whether large or small. Designers of entire cities, or an aspect like the systems that light them, work on timelines, following sequences of planned events, design criteria, performance standards, testing options, evaluation, and defined outputs, and they employ modulated processes to achieve results that match specifications.

Design has emerged as a necessity. Thirty years ago designers were viewed as frivolous artists, churning out fanciful ideas. Indian products were poor imitations of foreign designs from another thirty years before. The term ‘lighting design’ would then have sounded exotic if not weird.

Today a product will not sell – it will in fact flop – unless it is well designed. A city will be ugly and will not function unless it is carefully designed. The lives of its inhabitants will be miserable, frustrating and empty in the absence of design. This is the challenge we face in India. Our role has to expand from fanciful, lyrical stunts to the epic stage of social and economic transformation.

Industrialization has made it possible to bring thousands of daily use items within the reach of the average citizen. Things which were unaffordable when made by hand dropped in price when churned out in the thousands. Rustic oil lamps were difficult to maintain, awkward to operate and unsafe to

handle, while modern lighting is inexpensive, safe and accessible to all. We have moved from the design of crafted objects to the creation of entire technological systems that have interdependent design elements and components, right from the energy source, energy distribution, marketing and bill collection, to the electrical fitting, the luminaire, the type of bulb, and the space being enhanced, and all made functional by light. Object design is simple; systems design is complex. If one part of the system is missing, the entire interconnected framework may collapse. There is no sense inventing the radio without a broadcasting station, and one radio receiver will not support a station. So, thousands of radios must be mass-produced to sustain a broadcasting system. Unless advertisements are designed for broadcast from these stations, there will be no revenue to support mass media. The culture of objects has given way to the culture of systems.

Early in the 20th century the marriage between art and industry occurred through the German Werkbund movement, evolving into the Bauhaus and maturing into what is often referred to as industrial design. One of my gurus, Walter Gropius, brought this movement to America when he took over at the Harvard Graduate School of Design. The 'Bauhaus Approach' formed the basis of teaching at the National Institute of Design in India and permeates all basic design courses, be they in fashion, interiors or architecture.

However, industrialization has also uprooted and moved millions of people from their traditional occupations and habitats, bringing them into alien urban environments. The shift of jobs from rural production to industry has triggered a mass migration for which there was no design. It was chaos resulting in squalor. The results are unhealthy and inhuman. Design has not failed; it has been ignored.

We must create the scenarios where design can play a crucial role in uplifting the human condition. Design is the organizer that harmonizes thoughtless machines and raw materials to produce functional and beautiful artifacts. Design when wisely employed enhances the quality of people's lives. Our collective interest as designers is how we can create scenarios where design can profoundly impact the quality of life of average people.

Around the dawn of the 20th century business leaders in Chicago and San Francisco understood that there were no adequate city plans that would create order in urban life. In Chicago the railcar maker, Pullman, built a model town for his factory and his workers. The city's industrialists and traders floated a competition for the city's new plan. Within a decade the city came onto the world map as a good place to do business. Good design branded the city as a 'must see' destination in the world. By the end of the nineteenth century, the 'Chicago School' of architecture was synonymous with modernity and progress.

Design was an engine that drove an epic narrative. Design began to tell a story about the new life and the good life. Design created the futuristic image that inspired and catalyzed nations. Design created icons of 'what can be,' and then created the defining cultural artifacts of modern civilization. Design was integral to the process of urbanization and industrialization. An experiment in one city became the prototype for a dozen more, and then it became standard practice. This is what I mean by **epic** design, as opposed to **effete** or even lyrical design. Small ideas and little designs tempered taste makers, and then became the big story of life.

Too often designers focus on 'the pretty,' the 'clever', the 'cute' and the 'luxurious'. They start getting drawn into a partnership to promote conspicuous consumption and consumerism. They worry over what will sell and what is fashionable. By the time they do it, the fashion has turned stale and they are part of an outdated style. The glitzy small ideas, the fashions of a season and the gift

wrappings all hide the underlying truth that needs to be revealed. Each designer seeks attention; they become obscene and obnoxious just to gain notoriety, like a loud and demanding baby yelling and screaming, instead of an anonymous worker creating for the betterment of society. The result of such behavior is effetism and this little effete narrative, this tiny irrelevant story, begins to eat at the roots of the larger narrative. Design must get out of tinsel town, leave romanticism to Bollywood and shun the virtual reality of Hollywood. We need to recapture the Modernist mission, and focus on bringing ‘the good life’ to the masses. To this end we must employ appropriate technology.

Design today has become mundane and banal; frivolous and effete. It is playing on cheap emotions, like being the tallest, the largest, or the most stupid! Bright colors, reflective metals and using a surfeit of materials get crass attention. This is what I see in architecture, interior design, and in product design today. We must defy this.

I recently visited the Spanish town of Granada, where centuries of a city making tradition and effective urban design have tempered the inhabitants’ lifestyles for the better. The key to their success lies in the design fabric of separate templates for pedestrians and vehicles. People rarely walk across polluted and dangerous streets. They move down covered arcades, through human scale plazas, within pleasant gardens, past proportioned statues and harmonious fountains. One minute they are in luminous natural daylight; the next they disappear into dark shadows. Historic buildings are aligned along the visual axes of pathways. There are outdoor cafés and places for children to play and the elderly to sit. Shadows play through the glittering rustle of leaves of protective trees. Youngsters flirt and laugh everywhere. As the sun sets, calm light sets in the foliage creating a soft and romantic ambience. At each corner one is greeted by a pleasant, unexpected new experience.

Collectively it is our challenge to bring the benefits of good design to more and more people. To do this we must take on ever more complex challenges like the design of our cities, urban precincts, riverfronts, open spaces and affordable shelter. One of the simplest interventions in the urban scenario is the creation of appropriate public lighting for roads, footpaths, public gardens, statues and iconic structures.

City governments do not have the intellectual resources to make such plans, nor the vision to foresee dramatic changes. Urban planning legislation stifles any qualitative improvement of cities, forcing them into a step-by-step, knee-jerk method of identifying little projects arbitrarily lumped together and called a Development Plan. There is no design in all of this, just scheming and adjusting.

India’s designers, industrialists, business people and professionals need to engage with its urban scenario. But this should not be a cabaret where we hold useless meetings and ‘do-good’ seminars, just to watch each other dance and sing the praises of what we neither have nor can achieve. We need to study statutory barriers as well as plan options and work on a multi-level platform between policy, programs, projects and people. Our cities and metropolitan regions remain among the few mega-habitats in the world without even the gesture of designed urban environments. There is no opportunity for designers to play a role. What are we waiting for? Let us create that opportunity!

We must apply design logic, design processes, design techniques and design methods to the creation of artifacts that impinge on more and more people. We must employ design logic to correcting the environmental disaster facing us. We must employ design methods to create access to shelter for the poor. As designers, we can even create urban lighting scenarios for streets, footpaths and public spaces. We can do this!

Before our eyes we have seen the Mumbai-Pune Expressway emerge. We have seen the Hussein Sagar Lake transform from a polluted cesspool into a beautiful urban precinct of public domains. We have seen a dirty *nalla* in New Delhi transform into the Dilli haat. The landscape designer Ravi Bhan transformed a neglected drainage catchment in Ayodhya into a beautiful riverfront experience. A private developer, Harsh Neotia, in Kolkata turned a virtual garbage heap into a charming center for the arts called Swabhumi. In Pune's Koregaon Park a dirty *nalla* was transformed into the wonderful Osho Park. The examples of what we have achieved through design in India, and through private-public-designer partnerships, is endless.

In Pune, I remember the wonderful fountains which came up all over the city before the National Games in the early 1990s. TAIN Square has created public space for its neighborhood where everyone else is building right up to the road setback line. We are trying to create a youth plaza in the College of Engineering, linking the severed halves of a historic campus, over the national highway, cutting through and connecting them to the riverfront.

Why do we feel amazed when we stroll down the boulevards of Paris, stretch out on the green lawns of its gardens, and sip coffee at its sidewalk cafés? We feel amazed because we are a deprived lot. We are starved of the most basic human joys of life in a civilized city. We are hungry just to sit with a friend and sip tea in a cozy outdoor café.

We must rethink design; we must reconsider the role of design; we must redesign design!

Good design brings a better life to everyone. Good design is good business. If we want to do good things, we can do anything. Design is a process followed to reach our dreams. What are we waiting for? Let us design a better future!

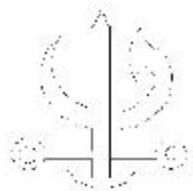
(Keynote address at the India Design Summit, February 2009)

Language and Pattern:

The Mahindra United World College of India

The Mahindra United World College of India is nestled in the Sahyadri Mountains between Pune and Mumbai in western India. It employs an architectural language that was set in motion by the Alliance Française in Ahmedabad (1974) and in the Bhanuben Parekh House in Bhavnagar (1972). The SOS Children's villages at Bawana, outside New Delhi (1976) and in Kolkata (1978) continued this line of thinking, which permeates the Centre for Development Studies and Activities. Thus, it is a language that grew and evolved over several decades.

This unifying language is composed of several design principles, of which the most important is the expression of materials in their natural form. This and the use of human-scale systems of proportion are legacies from my childhood infatuation with Frank Lloyd Wright. Another influence is my penchant for integrating the interior with exterior spaces using sliding glass panels, verandahs, low sitting walls, *ottas* and terraces. Classrooms and courtyards flow into each other, employing the vast borrowed landscape as a template, generating an immediate landscape, fusing the composition into the site. Over the years, my work has been characterized by an elaborate vocabulary of motifs such as *ottas*, ramps, *kund*-like steps, window boxes, round columns and waterspouts. These, along with murals cast in the exposed concrete, bring anthropomorphic dimensions and expressions into the work.



While the language and its slowly evolving vocabulary persist, I have rapidly changed my attitudes toward patterns and employed them in ever more radical manifestations. Starting with modular grids, moving to parallel lines, I later worked in angles and curves anchored in a strong language and design principles.

Once, asked about my conservative views on architectural language in contrast to my radical use of patterns, I replied, "Architecture is a curious craft. A structure may follow all the laws of design and yet be worthless. Another may break all of the principles and be profound. A building may be bad without doing anything bad. I cannot say that I am right or I am wrong, but it could be that one needs to sin against art to reach perfection." Somehow that statement has haunted me.

It is the deployment of a formal language in terms of strict principles that makes it possible for a large team to work in unison. An analysis of the Mahindra College reveals that totally new forms and patterns have emerged out of a gradually evolving language - the Art, Administrative and Student Centers being the most avant-garde. While they are part of the fabric, they are very particularistic statements. They speak the same language, yet they form their own unique poetry.

The Mahindra College campus is a counterblast to the fashion in contemporary campus planning, wherein each building sits by itself on its own isolated little plot screaming for attention. This trend is at once silly and boring. On the other hand I worry over the inane uniformity of generalized mega-structures. I feel one must seek a balance between entrapment in an endless monotony, and the Disney World of anal retentive screaming babies. Let me elaborate. When I move through a large metropolis, be it in India, Australia or America, I am numbed by the dull urban sprawl. Buildings are not really designed. They are conceived as 'statutory drawings', presented to a local body to gain permission and extract the maximum FSI. Can such an exercise ever be architecture? Never! Then, in this sea of ugliness, inanity and boredom, someone sticks their tongue out at you - 'gives you the finger' so to speak. They pull a stunt and do something really obnoxious in the name of being different. There is no order, poetry, harmony in the work, just an obnoxious insult. The habit of running nude in a stadium, called 'streaking', is a good equivalent. Showing one's ass is considered by these people as clever, unique and even spectacular. The only thing one can say is that this kind of seven-second wonder, or event, is remembered for some time, but it does not go down in history. Such acts cannot be called a great step forward for mankind. On the contrary they are insults to considered and articulate thought, analysis and structuring. What we can get then is a mundane urban landscape, occasionally punctuated

with an insult. There are appropriate times to take off one's shoe and hurl it at a symbol of inhumanity; but one does not scare a city forever by building monuments to stupidity. This is why 'language' is so important.

I am not a postmodernist. I look back with nostalgia to a modernist tradition filled with Wright, Le Corbusier, Alto, Kahn, Sert and other expressionists. Modernism was hijacked by bureaucrats and commercial developers to save and make money, and this boorishness was then appropriated by the postmodernists as their antithesis. The postmodernists have misrepresented modernism in order to appear 'new' to a gullible public, when in fact they are all guilty of an ongoing degradation of architecture. I insist that the roots of modern architecture lie in social purposes, in spatial movement, in nature, in urbanism, in technology and in visual and mental stimulation. Let me quickly look at several small design studies at the Mahindra College which have their own poetry, are aggressive pieces of art, but don't yell and scream for attention.

The *Mahadwara*

The *Mahadwara*, or 'grand entry', has its own unique shape and mass, much like the entrances to Egyptian hypostyles along the Nile. A directionality is set up within a complex composition, adding a 'sense of place' to the campus. It sets one in motion, establishes a landmark, fixes a cardinal point and lays out an axis. One enters along the auspicious north-south axis. The central Academic Quadrangle is like the inner ring of Jambudweepa in a *mandala*, and the various structures are like the rings of sacred islands spread out in the oceans of salt water, *ghee*, milk and honey. The portal thus takes on meaning from its greater role in a meaning system. By itself, it would be just a folly.



The Spanish Steps Amphitheater

The Amphitheater is a link on the north-south axis between the Academic Quadrangle and the Multipurpose Hall. This interlocking space is divided by a wall, placing a ramp behind it with cutouts to peep through. Young people like to look at each other, and to be looked at. They are at an age where physical beauty and beautiful ideas vie with one another for attention. It is like the Spanish Steps in Rome which are always crowded with young people sitting about. A college campus must address this urge to ‘be seen’ and ‘to see’, as it is an important aspect of personality development. It makes the experience of architecture very personal and very real. Perhaps only the Greeks understood this. Steps are more than a way of going up or down. They are an event. The Amphitheater is a popular venue for informal speeches, debates and drama presentations. Such elements are ‘land-locked beaches’ where people can gather to sun themselves in the winter.

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Site Plan

- LEGEND:*
- 1. Mountain Road from Valley*
 - 2. Entrance (Mahadwara)*
 - 3. Science Center*
 - 4. Administration Building*
 - 5. Academic Quadrangle*
 - 6. Library*
 - 7. Art's Center*
 - 8. Multipurpose Hall*
 - 9. Catering Center*
 - 10. Pathway*
 - 11. Play Field*
 - 12. Forest Land*
 - 13. Swimming Pool*



The Multipurpose Hall

This space is used for yoga, dance, music, drama, lectures, examinations and convocations. The high ceiling with 160 small coffers set within ten larger ones generates a hierarchy. There are ten skylights in the system. The result has a honeycomb effect, articulating the larger area into human scale modules. The four towers housing mechanical equipment, the sloped roofs over the verandahs and other elements reflect the dramatic landscape, and tie this large structure with the theme of the campus. All of these elements are deployed effectively to break down the mass, on a lower elevation, keeping the roof-line below the level of the adjoining building profile.

The Student Center

The design is based on a folded retaining wall, which holds the hill slope. The wall is composed of six vertical light shafts reaching up into the sky. Each shaft holds a room, or large niche, for activities. The harsh basalt stone wall is intersected by an entrance porch. Inside, a folded sheet of glass separates these rooms from a generous verandah, which frames a dramatic view of the Sahyadri

mountains. Large skylights are cupped in the stoneshafts, such that they follow the arch of the sun moving from dawn to dusk.

Aerial view of the Campus located in the cool western Sahyadri Hills

The Anjali Anand Art Center

While I belong to the Rationalist School of design, I want to extract tremendous variety out of very logical paradigms. The three studios of the Art Center fly out from a central courtyard like the huge wings of a mysterious, prehistoric bird, building up to a crescendo the visual gymnastics of the campus, yet maintaining a very logical organizational fabric. The studios are used for multi-media work – painting, print-making, and sketching. There is an office, a store room, a small pottery court and a kiln tucked into the composition.

The studios are made up of tandem walls culminating in grand glass windows, letting in tempered north light. These studios are joined to a low verandah, which wraps around a planted courtyard, stepping down to the mountain edge. Though the larger window masses are fabricated from modular pieces, each one appears to be a single glass sheet.

The result is a very powerful, apparently free form; composition which epitomizes my thoughts at the time, more than any other structure on the campus. It is functionally tight, while being visually free-flowing.

My work at the Mahindra College carries on a ‘language of build’ evolved over several decades. But the patterns and experiences are radically new. These spaces and forms provide a community with an underlying public bond – a ‘collective conscious.’ The recognition of a collective conscious is one of the intangibles that breathe life into true architecture. It turns spaces into ‘places’.

When I build in stone I place myself at the mercy of my craftsmen. Though uneducated in the arts, they have a sixth sense about the meaning of their work, good or bad. They are not fools to mindlessly place stones one upon the other. They also seek meaning. When they leave the site in the evening, they always pause and turn around to contemplate what they have done. If they smile in satisfaction, it sends a bolt of joy through an architect’s being. Though they are simple people, they know the difference between ugliness and beauty; between trivia and the profound!



NOTE : The Mahindra United World College of India has carved a niche for itself by winning the American Institute of Architects' 2000 Award for Excellence in architecture, sponsored by *Business Week* and *Architectural Record* journals. The award is given both to an architect and a client who jointly understand how effective design solutions support progress.

(An earlier version of this article was published in A+D, issue dated 12th December 2004)
Aerial view of the Academic Quadrangle from the Southern Side

De-schooling Architecture

(an interview by Brinda Gill in *Inside - Outside*)

“I wanted a different structure...one that would be like a monument without being large and arrogant,” said Dr. Gunvant Oswal, founder of The Center for Life Sciences, Health and Medicine (CLSHM), that treats brain and neuro-developmental disorders, especially in children, based on a holistic system of medicine. From 1968 to 2000, Dr. Oswal operated out of a 800 sq ft clinic in Pune’s overcrowded Bhawani Peth; however as word of the efficacy of his alternative system of medicine spread, he felt he needed a larger space to carry out research and treat children with special needs. For two years Dr. Oswal looked for an appropriate site, a place where children and their parents would feel at ease. Finally he found a quiet site on a hill in Kondhwa with a sweeping view of the city and adjoining forest land.

Then there was the search for an architect who would design a premises to suit his practice and patients. Seeing my design of The Mahindra United World College on the outskirts of Pune, Dr. Oswal approached me, carrying with him the tome *Frank Lloyd Wright, a Visual Encyclopedia*, with Post-it Notes marking pages featuring different architectural elements that he wished to have in his Center. His commitment to his cause of treating special children, not turning away any child for the lack of finances, and the fact that I was also deeply interested in Wright’s architecture, set the pace for the project. Dr. Oswal invested his life savings into the Center, supported whole-heartedly by his wife, as well as his daughter Pooja and son-in-law Shrirang, both doctors, who also practice at the Center.



I realized this was a project for very special children – eager to live and to learn; lively, loving and observant, but deprived of the normal joys of childhood, growing up and coming of age. My first intention was to reach out to them, rather than to draw them into a dull, rectangular, monumental institution which says: “You don’t belong here! You don’t belong in this world!” So I wanted to make a very different kind of building, but not in a patronizing way that mocks mental disabilities. I had to allow myself – the child in me – to emerge, to let go and to speak out. I had to de-educate myself from Cartesian ways of thinking: the X and the Y axis; the squares, rectangles and boxes, which for ‘normal’ children are called SCHOOLING. I realized that I too had been taught in squares and boxes; taught to think in parallel lines! It was very easy to stick to that tried and tested path, but the result would be a box!

By thus ‘letting go, in De-schooling myself of the Architecture I had learnt, I traveled a trajectory which intersected that of my user group. This is how I very consciously took on the Behemoth of Cartesian Thinking and tried to break it down the way an ancient army would demolish a fortress wall: battering the closed door of thought; pulling down the walls of false knowledge; destroying the culture of thinking which would put me into a box! In my struggle to de-school myself, I could come up to the beautiful, uncluttered level of existence of these special children where they can see things putatively, naturally and in the essence. I realized that seeing things generically, getting a glimpse of

the essence of things, is seeing beauty!

One day, after spending hours discussing the project over several meetings, looking at my proposed tile roof-courtyard scheme, Dr. Oswal asked, “But how does the wind travel through this structure?” The question itself led to the design solution. In the end it was the westerly winds which ordered the structure into a series of pathways for wind to travel in, which we would also walk in. Air and people would move in the same channels which, like the wind, would meander about. The high walls on the south would provide shade from the southern sun. There would be pocket gardens and secluded places. There would be plantation here and there, and each space would integrate with some outdoor space. The angular wind walls would form a honeycomb of indoor and outdoor spaces and places, generating a lot of energy.

A hint to the architectural approach of the complex is offered at the main entrance, where flower beds seem to define the compound and the main boundary wall is set back from the road. In Europe institutions are filled with people in the evenings. I wished to offer a similar expression. So the main wall is set within the premises and there are low, broad steps for people and passers-by to sit. Beyond the steps, transparent gates offer entry into the complex, graced with pristine white walls creating a sense of peace and space. White walls are something I have always loved since my youth in the Aegean Sea where the azure waters, shaded white walls, a touch of blue woodwork and shadows everywhere had grabbed my attention. I felt in this project – a small one – stone might be dark and oppressive. So I used white walls. Dr. Oswal shared this concern and the outcome followed naturally.

The ground floor serves all the needs of patients – reception room, waiting areas, doctors’ clinics, dispensary, green spaces to relax, a pantry, an area for patients and their parents to dine, a lotus pond, statues of the Buddha as well as of a mother and child in open spaces. These spaces are interconnected as the building is organized along two meandering west-east movement lanes that offer shade and catch the cool westerly breezes and direct them through the structure. The first floor has a bedroom, a guest bedroom, terrace and lobby; while the basement, with direct access from the roadside entry, as well as a ramp, is a venue for seminars. Along with the pockets of flowers and foliage within and around the structure (including a variety of indigenous and exotic plant varieties), the all-natural flooring of Jaisalmer, Dholpur, Red Agra and Kotah stones makes for a tranquil ambience. The gentle mist of water droplets sprayed on plants cools the temperature, offers a soothing sight and its soft murmur calms the nerves. The bonding with nature is conveyed in the slightly sloped water spouts that return rain water to the earth.

A walk through the building reveals spaces merging with one another. The idea was that each outdoor space would relate to at least two or three indoor spaces! Each indoor space, in turn, would relate on its sides with sequential outdoor spaces. Thus, there evolved a number of sequences, links, chains of experiences which would always be differently iterated, depending on the way you moved in the labyrinth. The skylights, the light wells, the light courts are vertical and horizontal means to achieve this. The columns are triangular, placed to turn spaces into arcades and integrate the spaces as one. They are fitted with small, colored ceramic tiles to enliven them and make them playful.

Apart from designing a center that would be child-friendly, there is also a commitment to being eco-friendly. Nowood is used in the design, solar panels warm the water; and most importantly, the

spaces designed and materials used minimize dependence on electricity. Tall glass windows bring in natural light, breeze and outside views. There is hardly any glass on the outside. Users are not inconvenienced when the electricity goes off, because the rooms are all full of natural breezes and light. What do we need fossil fuel energy for? Maybe the gadgets that clutter our modern life need power. But the architecture here is 'energy free.' This can always be accomplished if one leaves openings on several sides; uses light shafts and wells; mingles nature in courts and walkways. These things come naturally in India where the climate is salubrious.

(Interview by Brinda Gill in Inside-Outside, July 2007)

Themes and Motifs in Architecture: The Dilemma of Style

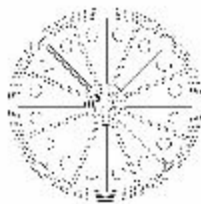
One of the characteristics of being human, a characteristic not found in other species, is the ability to use symbols and signs to manipulate concepts within one's mind. Here I do not mean using symbols for the mere communication of ideas. We go beyond the intellectual life of fish and birds and formulate ideas, concepts and constructs.

Our ability to conceive things is critical to human development. Symbols are used in human thought to stand for things which are not present. Words are symbols we constantly use. Imagination is the human function of making images in the mind.

Human beings can imagine situations which are different from those before their eyes. A child can remember things not present, but only later in its development can it manipulate them, even adding components it has invented but never seen. We explore a fantasy world and experiment with the rational world in our minds.

As architects we are interested in the rational exploitation of future experiences. We want to visualize in our minds different images and alternative situations in terms of built form, which respond to the same given conditions (site, regulations, program, geo-climatic context, budget, etc.). Though the constraints are a limiting factor, the variety of images is great.

Our *language of build* is full of symbols which allow us to create fabrics of build in great variety.



Caught in a world of vast choice, how does a designer go about deciding which mental image to pursue through an investment of effort in design? Unfortunately, like a child, most designers can deal intellectually only with things they can remember having actually seen, or something in front of them. They have not developed the ability to manipulate absent symbols. Creating new symbols – perhaps a third stage of imagination – is beyond their capability. Only education can overcome this gap.

The above lacuna underlies the need for style. Styles present the designer with a ready-made 'kit' of images to choose from, in which different assemblages appear 'new' or 'different' according to how imaginatively they are used. At best the designer puts together in his imagination bits and pieces of absent things that he has already seen assembled according to simple rules, usually seen in magazines.

Post-modernism is the currently fashionable style for the simple-minded. It is a system of symbols (Greek pediments, Classical columns, Palladian rose windows, 'period' windows, etc.) which can be thrown together to make 'interesting' façades. Even images from Disney World have been recruited into the lexicon of ready-made, post-modern iconography. Whatever weakness this style may have is overcome by the application of expensive materials (granite, Italian marble, mirrored glass, tinted metals). A kind of make-up, like lipstick, is applied as if buildings, like an unattractive person, can be 'treated' for defects according to the occasion, or the time of day.

The legitimacy of symbols is an area of debate. I believe that our architectural language must emerge from the **themes** of construction. Quite simply these themes are **Support**, **Span** and **Enclosure**.

To explain this let us consider the theme of **Support**. We are limited to bearing walls, on the one hand, and columns on the other. There are geodesic and hyperbolic alternatives, but their application is limited by cost, labor and constraints of technique.

We basically have to choose between a frame structure and a bearing wall. But within these limited options there are numerous choices of material, geometry and configuration. At the CDSA campus I chose a simple system of parallel stone bearing walls. But their orientation, rigor of spacing, and play against one another build a higher order of positive-negative rhythm.

Likewise **Span** is a simple system of beams running across these walls with tiles above. **Enclosure** is in the form of sliding glass panels. It is in the simultaneous choice of themes and their interrelationship that imagination is required. Motifs are stuck on later. At CDSA the motifs support the themes by locating vistas (windows), modulating wall planes (window boxes) and directing movement in

space (*ottas*, stairs, low walls). Direction and orientation are confirmed (only confirmed, mind you) by statues, decorative pots and various antiques. But all of the motifs applied are incidental to the overall effect of the building cluster. We could have successfully used a totally different set of motifs while maintaining the essential themes.

Architecture – true architecture – emanates from a language of themes, not motifs, decoration or applied styles. Post-modernism is constructed on a language of motifs. It does not qualify as architecture. It is exterior decoration wherein motifs are applied to wall surfaces just as interiors are ‘finished’. Architects are not in the business of decoration. God knows, however, that there is a great need for many buildings (inside and out) to be hidden under decoration. But this is a kind of cosmetics rather than a search for raw beauty. Intellectually, the manipulation of motifs is child’s play. It would be better to design as bees and birds do: they use a single-minded fabric of build (wax honeycomb or woven nest) and stick to their theme. Yes, bees and birds who are not ‘rational’ thinkers, instinctively build architecture, while the thinking mind makes a mess out of motifs – decoration!

We are not the doyens of a fashion industry. We are not the slaves of an ignorant quick-rich clientele that knows nothing of architecture. We are the guardians of an intellectual tradition in which principles of proportion, structural systems, human scale, appropriate use of materials, and choice of meaningful motifs are the essence of art. It is from the ability to make ‘components of build’ into symbols and configure them through interrelationships that architecture emerges: architecture of some lasting value; architecture which represents man’s higher aspirations.

*Architecture – true architecture – emanates from a language of themes,
not motifs, decoration or applied styles.*

Style is the sickness of the feeble mind, the opiate of the tasteless. Be it post-modernist, ‘Punjabi Baroque’ or Ethnic – style is merely an excuse for something which has not been conceived.

Back to Basics

We live today in the new economy based on the ‘bottom line.’ The bottom line means profit. No matter what unique selling point city builders advertise, be it green buildings or hi-tech environments, their bottom line is harvesting the maximum profit, even at the cost of the public good. Paying lip service to Corporate Social Responsibility (CSR) is part of the new public relations strategy, while the reality is cutting costs and increasing Floor Space Index at the cost of society. This ‘new economy’ has spawned a new architecture (quite different from the ‘New Architecture’ I have myself advocated elsewhere, in my letter on *Architecture as a Social Tool*).

Like all living creatures architects are driven by the quest for survival and the urge to dominate. There are two paths they can take:

- Promote their own value-based professional agenda, creating the ‘best fit’ between it and the agenda of the new economy; or
- Degrade themselves into a vocation, where their skills are offered to the captains of industry to consolidate the bottom line.

Willy-nilly architects are taking the second course, perhaps without even realizing what they are doing. Young professionals watch their peers in the IT and management sectors jump to high salaries soon after graduation. They see their own classmates joining MNCs and bringing in large pay packets. What they do not realize is that they are comparing themselves – professionals – with skilled workers practicing vocations. They are comparing professions with service industries.



We must get back to basics and ask ourselves fundamental questions. Who is a professional? What distinguishes a professional from workers in vocations? What is vocational education and what is professional education? At the same time, let us not fool ourselves. Vocations are needed and we must respect them. But we have chosen a more difficult and a more arduous path in life. As professionals we ‘profess values’ to which we are bound. This means that we have a professional **credo** (Latin for *I believe*); that there are fundamental values and principles which no professional can breach. We have an unwritten code of practice which we have to stick by. As professionals we have locked ourselves into this belief system, and we have to navigate our work within it.

The most important characteristic of a professional is his or her intellectual honesty. All professionals, be they architects, lawyers, doctors or accountants face a continuous and painful internal ‘dialogue with self,’ challenging themselves to be truthful to their core principles. The world’s most respected accounting firm went bankrupt and closed its doors within days after it was revealed that it put the bottom line of its clients before its professional commitment to society. As corporate auditors they cooked up annual reports to wrongly project as hugely profitable an energy company, Enron, that was in huge losses. While the accountants were doing this the corporate managers quietly sold off their worthless shares at inflated prices. Their vocational bookkeepers, software operators and managers all kept quiet. No one blew the whistle until millions of workers lost their pensions when their funds were brought down to bankruptcy along with Enron as the true share values were exposed. All of the vocational managers, software engineers and bookkeepers quietly shifted to new jobs. The professionals, the auditors, ruined their careers and professional reputations. Why? They lost their professional credibility when they sold out their credo, their professional values, and their intellectual honesty to an employer to help protect the bottom line at the cost of the society to whom they must ultimately answer. They put the bottom line above the **social contract** that binds all professionals to serve society, above those who pay their fees.

As professionals we ‘profess values’ to which we are bound.

Like those in vocations, we professionals also have technical responsibilities, procedural responsibilities and a duty to continually increase our awareness and knowledge. Like those in vocations we have to answer to clients, employers and seniors. Like vocational

employees we have to deliver cost-effective solutions that meet performance standards. But we are not just producing deliverables and making something bad work better, or making something started with the wrong assumptions reach optimality within a flawed problem solving environment. We always have to go back to question the underlying assumptions and the starting points. If these starting points do not fit our credo, or if our clients really do not want professional advice, but merely want vocational servants, then we have to opt out. Quit!

We must be clear about ourselves. We are not a service industry. We are not delivering goods and services at the doorsteps of our clients. ‘Profit’ is a business bottom line, but we are no more ‘in business’ than is a heart or brain surgeon. Like surgeons we have to put the hard facts before our clients and tell them the correct path to follow in order to achieve the best outcome. What we tell clients may not be sweet words. The procedures we recommend and the technical mechanisms we propose may not be what they want to hear. Our deliverables are the physical manifestations of our professional values and advice.

Many young architects and other professionals in the construction industry are opting to work under non-professionals in MNCs, real estate firms, and investment companies as their personal bottom lines rule over their professional bottom lines. Often we see young professionals with two or three years of professional work opening small practices, in which they lack both the experience and the confident maturity to convince clients to change their concepts of what the bottom line is. When dealing with life-threatening medical challenges patients seek the most seasoned professional advice. For a common cold they go to a MBBS round the corner. They tell the young doctor what their illness is and ask for the prescription they think is right. They are happy with the young doctor because he does what he is told like a waiter in a restaurant or like a computer operator.

Young architects and designers must realize that they too are prey to business whims and preconceived solutions. As youngsters they lack the credibility to be taken seriously when balancing social costs and benefits before clients. They may lack the finesse to illustrate options where public benefit becomes a factor in bottom line calculations. Senior architects need to create career options for juniors within more established professional firms, making it economically gratifying for young professionals to spend a decade preparing for private practice, or even a life-long partnership within a branded design house.

Neither our educational system nor the design profession is addressing this issue. It is high time we got back to basics and saved our profession.

(Lecture delivered to alumni of Malnad College of Architecture, Bangalore on April 12, 2008)

Looking Back and Forward

In this interview Benninger answers questions about the role of the client in the evolution of an architecture for the future.

The new millennium has raised a number of questions and a lot of expectations. In a field like architecture, what should we expect?

Architecture does not change overnight. It drifts – behind techniques, behind economics, and behind social trends. But most of all it drifts behind the ‘vision’ of clients. We should be more concerned about the kind of clients emerging in the new millennium than about the kind of architecture. Architecture will follow. But good architects would be good soothsayers. They would be in tune with issues and the social and economic trends the issues generate.

You would not say this unless you considered it a crucial point. Can you elaborate?

Unlike many arts, architecture requires clients as a starting point! Painting, music, poetry, sculpture and many other artistic endeavors are driven by the artists’ efforts, and can even be based on their personal whims. They get inspired, they create something and someone later buys it. Not so with the mother art. For architecture to flourish, a society must have patrons, not clients.



How would you distinguish between a client and a patron? Isn’t the latter a manifestation of the former?

Clients merely want a vehicle to achieve their functional ends, at the least cost. They lack ‘vision’. They fail to understand the potential of architecture to lend them identity, or for a key building to be a kind of icon which communicates transcendental values for which the client stands. Clients who want to use their structure as a vehicle to lend grace and poetry to the larger society are called ‘patrons,’ not clients. They repose faith in an artist and allow him the freedom to explore all the lyrical potentials tied up in the nature of their building.

By ‘freedom’ do you mean to say they just let the architect do what he wants?

Nothing of the kind! In fact they get involved in the design process with the architect. They know what ‘quality time’ means. They review the architect’s building program, schedules, cost estimates, site layouts, concept designs, Bills of Quantities, proposals for contractors, change orders, payment certificates... the works! But different patrons have different styles. There are patrons who are busy. They know that they will destroy a project by passing it on to a manager. So they just say, “Look, it is your baby. Make it great. You are a professional. Don’t let me down!”

That sounds very noble, but can you give examples?

Yes, there are so many. Nehru patronized Le Corbusier at Chandigarh, and Vikram Sarabhai patronized Louis Kahn at Ahmedabad. There are many more! No doubt different patrons work around different constraints, but the intention in all cases is to make a gift to society. That is the way Harish Mahindra handled the Mahindra United World College of India, and the way B.S. Teeka handled the Samundra Institute of Maritime Studies. I have been fortunate to have patrons like Harish and Anand Mahindra, Rahul and Rajiv Bajaj, the Kirloskar family, the Board of Governors of the Indian Institute of Management at Kolkata, Tulsi Tanti of Suzlon, the Tatas, B.S. Teeka and the Council of Ministers of the Royal Government of Bhutan.

Do you mean to say that the kind of architecture we will have in the new millennium depends on what kind of clients, or patrons, we will have?

Exactly! You've got it precisely! Unless clients have a 'vision' about their future – not only future profits but the kind of environments that they produce for their employees, for students and for future generations – they cannot be called patrons.

What about the immediate future? Should we not look at the lead sectors in the economy and see what they are doing? The information technology (IT) sector, which is growing by leaps and bounds, must be great for architecture. Would they not make great patrons?

The IT sector is an excellent case, even though it portends doom. Over the past decade infotech parks and infotech cities have come up all over. But the environments are, on the whole, dismal. The IT sector has an embarrassing track record with architecture. Since they are considered leaders in the emerging economy, they are bound to influence the way people think. Though they have invested a lot in new campuses over the past decade, none of them is notable. Where architecture is concerned they have literally 'missed the boat.' They are not the only losers; society has missed a great chance too!

Do you mean the new infotech parks and cities are bad?

I cannot really say bad because the firms are getting what they want. The designs are simply uninspired. They give nothing back to the society which created them. They say nothing positive about the future, or even something gracious about the past. At best they are comic, looking like flying saucers; at worst they mimic Roman Classical styles.

Can you be more specific?

Yes. Most IT projects are like babies screaming and yelling to get attention. They are mirrors of the clients who yearn for fame and popular acclaim, like rock stars. Even a single campus is a clutter of amusements, odd shapes: Greek, Roman, Early Ugly, Late Modern and Postmodern all cluttered into one walled-in and 'gated' campus. Even if one architect is the creator there is no thread of order and no harmony of elements; no architectural language. The architecture is desperately trying to be spectacular while actually being mundane. We get pyramids, globes, eggs, cones, huge pergolas and even a Roman Piazza. These are all jumbled together as the multiple personas of one simple technology company. This is architectural schizophrenia abusing the public. It reflects a lack of considered identity of the companies and the individuals who lead them.

Do you feel this is a sign of future?

I do think it is a sign, but it is not of the future. There are also people with vision, with traditions, and there are still people who understand something about art and architecture. I have had excellent experiences, and I know I will continue to have them. Patrons keep evolving and emerging. One cannot say where these patrons will come from either. They should know when to question an architect, when to praise an architect, and when to show a long face. Some institutes are also patrons. Who knows, some day an IT group may even build a marvel in architecture, or better yet, a humble, gracious campus? I think that will happen very soon; it will happen sooner than later.

Are there any signs of this?

If I study the evolution of IT companies there is a definite process of maturity. There is a growing interest in society, philanthropy, place making, humanity and architecture. The Azim Premji Foundation is a wonderful example.

What can architects do?

We talk too much about educating ourselves and have done very little to educate the public and clients. Architectural journals are trying, and I feel they have made an impact. But we are a very self-indulgent profession. We spend too much time talking to each other. We must learn to talk to others too. And we need to be tough. We need to be able to tell clients where to get off. We must enter contracts, and we must stick to the profession's fees. I feel by being honest and direct with clients, we educate them. They may be annoyed with the first meeting, but they will remember you, and they will either come back to you later, or try to show you later that they are not the nincompoops you have shown them up to be. But they will be back to architecture somewhere, sometime, because everyone has an ego; everyone learns; and everyone is on a learning curve.

(Discussion in A+D issue, July / August 2009)

Modern, Postmodern and the Intervention of the Effete

The spirit of a piece of architecture is the spirit of continuity: each work carries forward from an antecedent; each contains all of the previous experiences of the world of architecture. A good piece of architecture also contains the hints and the seeds of future works. Perhaps this is why a good piece of architecture is rare.

I wish all architects were on a rational path to excellence. But the spirit of our times is firmly focused on a present that is so expansive and profuse that it shoves the past off our horizon and reduces time to the present moment only. Within this system a building is no longer a work of art, or what the French would call an *œuvre*. It is no longer a thing made to last, or to connect the past with the future. It is just one current event among many, a gesture with no tomorrow. In our hyper-society no one has time to think any more or to contemplate. Just make something that will be noticed for a fleeting second. Make something queer, odd, unusual, but spectacular! This is the illness of our times. This is the reality of an effete society, with effete architects producing an effete clutter of inane objects that belong to no one, contribute nothing, and add nothing to the future.

This statement could be made today, or it could have been made in 1893, when the Chicago School – a school of modern thinking – was knocked into oblivion. It was sidetracked by an effete wave of mercantile architects backed by commercial clients and drawing on the bad taste of French academia. The International Exhibition in Chicago in 1893 was an *École des Beaux-Arts* extravaganza made of plaster hiding the actual structure. It had Roman domes, Greek columns and Baroque molding and details. The new technologies of concrete, glass and steel were thrown to the dogs. Patient search was replaced by greed for quick money. The exhibition was a rude lesson to modern architects. They knew they had been defeated.



It was due to the ‘modern project’ – the emergence of modern architecture as an aesthetic and social movement – that the thinking community of architects regained an image of themselves two decades later. A mission arose, allowing architects with vision of their place in history to take control of their art, their lives and their destinies. What follows is an attempt to put this phenomenon within the perspective of time and of history, in perspective at least up to the time the movement was again hijacked by French academia under the guise of ‘Postmodern’ theory in the 1970s.

Meanings of ‘Modern’

‘Modern’ is a commonly used adjective employed to describe many things. What does it mean? In architecture we all know of the ‘modern movement’ and we have heard of ‘modern architecture’, even if we do not necessarily know what it means. Similarly, we have heard of ‘Postmodern’ and we don’t really know what that means either.

If we don’t know what these movements are all about it probably means we are designing in a vacuum. Hopefully some kind of rational logic guides us toward creating functional and livable

buildings. Hopefully we are learning through our contextual experience how to solve problems that we encounter in our day-to-day endeavors, generating performance criteria, creating options and evaluating them.

But most young architects are lured by magazines and journals and the media into designing for the press, or maybe for competitions. We see spectacular building stunts on television and in the newspapers, and we think, ‘Can we ever create something like that?’ We get attracted to words like ‘cybertecture’ and believe it to be the future, though in fact it is just more of the same.

In the following discussion I argue that we are all barking up the wrong tree. We don’t know what we are doing. Instead of using our brains and thinking things out logically we are in effect looking at the social news and the fashion page to decide on the clothes we will wear, as if life is some huge fashion ramp, and we will be judged by the outrageous costumes we wear. All of us want to be ‘modern’, as opposed to ‘traditional’; we want to be ‘liberal’ as opposed to ‘conservative’; and we do not want to be left behind by history. In my argument I am stating that being ‘modern’ is not just being different for the sake of being different, but that we have to be a part of a value system, have a vision, know our mission, and set an agenda around these. We are architects, not a political party chasing votes. Our visions are evolving and each of us has to set our own agenda and assert our values through our work. Thus, my argument is not a prescription, but a ‘sifting of ideas’ that helps each one of us to settle into our own comfort zone regarding who we are and what we want to be in this great profession. I feel it is important that we start with a discussion of what the word ‘modern’ means to us. For me the word, and even the concept of modern, has several different sources and meanings. Let me share my ideas.

American Modern

In America the word ‘modern’ means ‘the latest’, something new or contemporary. It has a tinge of the innovative, of discovery. Equally it may imply a style, fashion or packaging. It could be just a ‘new look’ or the ‘in thing.’ Each year the American automobile industry changes the styling of its cars, and these are rolled out with great fanfare, as if last year died and the New Year’s birth is a world event. On the other hand European and Japanese auto manufacturers go on making improvements little by little, but the body of each car, its style, year to year, looks the same. In fact they may involve more unseen improvements under the bonnet than what the ‘new body’ of the American car offers. In a media-driven consumer market, what you see is what you get. Fashion shows have models walking the ramp, showing off preposterous costumes just to grab attention. Beefcake boys in little bikini briefs and anorexic girls looking bored, sashay in billowing outfits on the ramp. This is a game of style that we are all supposed to play. We are supposed to applaud preposterous ugliness. Socialites with no taste, scheme and maneuver to occupy the front row seats at the Milan previews. If they don’t know how to select their underwear, how will they ever be patrons of art or architecture! The new economy, the new urbanism, the new architecture and nouveau-riche are all mercantile sham. They are money making schemes driven by shallow taste and deep pockets. There are well-paid critics who write academic articles that reconcile bad taste to the contemporary scheme of things. Journals and curators of museums are all playing the same game, looking for cute things, tricky things, queer things, spectacular things, or blatant stunts, not for meaning. The new architecture is like a collection of *follies* in a nineteenth-century English garden; sentimental jokes and nostalgia at best.

Architecture is a more serious craft. Once built, we cannot just throw our designs into the washing machine, or give them to a poor aunt. Our efforts will be around for some time. Perhaps the word ‘contemporary’ is a bit kinder as it may refer to the era in which we are living and building, its technology, and its social structure, modes of production and machine processes.

American ‘modernism’ deserves a closer look. Most Americans carry with them the luggage of a foreign culture. They want to keep the good things and throw out the trash. They want to free themselves from the bondage of the past traditions and redesign themselves and be ‘free.’ Perhaps it is this parting with tradition and the exploration of self that makes American modernism attractive.

European Modern

‘Modern’ in Europe defines an age, or an era. In the sciences and philosophy the works of Galileo and Descartes tempered the birth of the ‘modern age’. God, testaments and religion were replaced by empirical observation and scientific conclusions, leading to axioms. It was now mankind that declared the truths of the world. Having its roots in Greek philosophy, the modern European spiritual identity found itself immersed in questions to be answered. It interrogated the world, not in order to satisfy any particular practical need, but because the passion to know had seized mankind.

Humanity desires a world where good and evil can be clearly distinguished as we have an innate and irrepressible desire to judge before we understand. Religions and ideologies are founded on this desire. This ‘either-or’ encapsulates an inability to tolerate the essential relativity of things human, an inability to look squarely at the absence of the Supreme Judge. This makes the wisdom of uncertainty hard to accept. The modern European journey is a narrative from a closed, traditional society into one of relativity and uncertainty. As God slowly vacated the seat from which he had directed the universe, declared its order of values, distinguishing between good and evil and endowing each thing with meaning, man began to redefine God in his own image. Europe set forth into a world it could no longer recognize. In the absence of the Supreme Judge, the world suddenly appeared in its fearsome ambiguity; the single divine truth decomposed into myriad relative truths parceled out by men. Thus was born the Modern Era.

The thinking self, according to Descartes, is the basis of everything and thus one has to face the world alone – *Cogito ergo sum* (I think, therefore I am). This anoints a heroic attitude to man’s personality. Cervantes takes this further, making each individual face the world of uncertainty; to be obliged to face not a single absolute truth alone, but to deal with many contradictory truths. One’s only certainty in this conundrum is the wisdom of uncertainty. Don Quixote, and the entire European village, departed from their rural, traditional belief system to search for empirical truth and a redefinition of self. Man awoke and he dared to reinvent himself.

Modern as Progressive

Distilled from both American and European ‘modernism’ there emerged a concept of ‘progress’. This concept sees history on a time line and assumes continuous ‘progress’ and improvement. The idea of progress got a fillip from the rapid stream of nineteenth- and twentieth-century inventions that redefined civilization – everything from electricity, light bulbs, telephones, cinema, to radio, television, cell-phones, digital technology, the Internet. In all fields technology drove human behavior, lifestyles, diplomatic relations and power.

‘Progress’ sees diseases conquered one by one; it sees democracy gradually replacing

despotism; it sees institutions becoming more accountable and transparent; it sees housing and consumer goods becoming cheaper and more accessible to the masses; it sees the rights of minorities becoming more sacred; it sees law and order and justice being more fair; it sees a growing reach of education, and more empowerment, through enhanced knowledge, skills and sensitivities; it sees a reduction in political polarity and more inclusive governance; it sees more sustainable and ecologically sensitive development; it sees the fading of borders, the integration of economies and the spread of social security. This broad concept of 'progress' dominated our consideration of 'modern' from the end of World War One (1918) to the present time. It tempered the concept of socialism; planned and mixed economies, and social security systems even in the most capitalist of nations. The roots of modern architecture lie in technological advancement of the nineteenth century. James Watt's long span wrought iron structures, Paxton's Crystal Palace, Eiffel's great halls and tower, the Chicago School and then the Werkbund movement, the Bauhaus and the CIAM, all brought together these isolated and diverse ideas into a unified movement.

Three strong threads define 'modern architecture'. These were maturing in the nineteenth century and grew more holistic in the twentieth century. The three threads are technology, social change and more recently, the battle against effetism. Effetism is the tyranny of the pretenders. Against the cheap mercantile architecture being cut and pasted on our cities, this triad of forces drove architecture as an integrated part of 'progress' until the advent of the 'new economy', the rampant mercantilism of the nineteen eighties, and the degenerate, mercantile architecture that emerged from the new economy and growing imbalanced concentration of wealth. The modern movement withered with the death of its founders and the rise of a new class of Atlantic-centered architects who were the torchbearers of the new effete, mercantile architecture. They disguised their work under the new French theory of Postmodernism. Of course in the process of looking to the French Academy they also claimed to be Structuralists and Deconstructionists – all styles with no social content and an expression of technology as opposed to an employment of technology. The new architecture has no putative beauty. It has to be 'explained', and there has to be an esoteric theory; the vaguer the better.

Modern Architecture

Modern architecture belongs to our times. But how 'young' is it? Surely Paxton's Crystal Palace created in 1851 was young? And Watt and Bolton's spinning mills in the first decade of the nineteenth century were new and dynamic? What about the Eiffel Tower or the series of Galleries des Machines built for the Expositions Universelles held in Paris in 1855, 1867, 1878 and 1889? All of these structures are 'modern architecture', not because they are 'new', or because they are 'contemporary', but because they address the human condition and the social and economic era in which they were conceived. They are 'modern' because they express themselves honestly through technologies that did not exist prior to their realization. Perhaps technology is the key to their claim to being modern. All of these structures are a counterblast to the fake and false Plaster-of-Paris neo-Greek, neo-Egyptian, neo-Spanish Colonial and neo-Roman buildings that cluttered cities – buildings that looked like one thing on the outside but were something different on the inside. Even today, the mercantile architecture of our times makes up ninety-nine percent of our urban landscape. Such false, untrue and debased statements are an insult to our intelligence and taste. Take the example of a state-of-the-art, cutting edge IT firm building its training campus in Mysore to look like a Roman Forum, with monumental columns, a great open piazza and pediments over a portico. I can imagine the leaders of

our IT industry dressed in togas, wearing centurion helmets and carrying long spears, addressing new recruits. Maybe the creators of this urban fabric also look different on the outside from what they are within? Maybe they seem to be what they are not.

Thus, modern architecture lies on the fault-line between ‘seeming’ and ‘being’ – an indeterminate space between lies and truth. This chasm yields the three-tiered agenda which must characterize ‘modern architecture’:

- The fight against the lie of effotism, consumerism and mercantilism.
- The search for improvements in the human condition.
- The employment of technology for the human good and for beauty.

Postmodern

About forty years ago, around 1970, architecture began to stagnate. This dormancy was anointed with the label ‘Postmodern’. It seems all of the concerns of the modern movement were put aside in a long sleep which engulfed the minds of architects. A French movement in literary criticism and philosophy became the opiate infusing illusions into the great art. Architects started fooling themselves and their students that they were philosophers. They imagined that through association with Jacques Derrida, or by quoting Michel Foucault, they would morph into Parisian philosophers. Ivy League school teachers started winding themselves up in French philosophy, like little plastic toys you wind up to watch them dance about! This spell of self-delusion and stupor coincided with the arrival of the ‘I, me, myself’ generation. It coincided with fast bucks, unearned increments on Wall Street and paper money from equities. Mercantilism, on leave of absence since the Great Depression, was back in town. Wealth, not creativity, became the new sign of status. Like the impact of the École des Beaux-Arts in the late-nineteenth century that smothered the modern movement in America and Europe, this hallucinatory drug captured the spirit of architecture. Quietly we left behind the search of function. Commercial ornamentation again crept into our language. Community design, affordable housing, open spaces and the public domain were quietly put on the back burner, and gradually out of sight. Honesty of expression, a dialogue with materials in the search for their capabilities, nature and expression faded. The aesthetics of honesty was replaced by consumerism and marketing. In its place came cute ideas, clever little stunts, even spectacular large monuments, and on Main Street, superficial packaging, fashions of the season, styles and billboard architecture. A huge chasm gradually emerged within the city culture of the modern era, through a growing alienation between individuals and their urban settings. Abandoned in heartless urban ennui, in a daze of sleepy acceptance, the consuming public lost touch with community, neighborhood and even neighbors. Television, the internet and shopping replaced conviviality. The rich forgot about the poor, and the poor got credit cards.

Hundreds and thousands of buildings have been produced in the past four decades, but few add anything to the nature of ‘being’. They neither inspire nor catalyze human interaction, nor do they sponsor ‘coming together’ – things that had happened naturally in the urban fabrics of small towns like Granada and Seville. Pedestrian, cozy, welcoming urban enclaves had been the stuff of life for centuries. All this was thrown out of the window for phallic symbols and meaningless stunts. These new buildings discover no new segment of human existence, only confirming what has already been built or said, thereby fulfilling their purpose. They confirm the stupidity of the life that everyone is living. By discovering nothing, they fail to participate in the sequence of discoveries that constitute

the evolution of architecture. They place themselves outside the history of architecture, or maybe in what is meant by ‘Postmodern’ – that they come *after* the history of architecture.

Postmodern Architecture – Effetism

The sole raison d’être of a building is to explore that which only a true work of architecture can discover. A building which does not express some unknown segment of existence is immoral. Revealing knowledge is architecture’s only reality. The sequence of discovery, not the sum of what is built, is what constitutes the history of modern architecture. The truth of architecture is contextual, but not nationalistic. There are analogues between meaningful work in India, Europe and Latin America. It is only in such a cross-national context that the value of work can fully be revealed and understood.

The rise of the sciences has propelled man into the tunnels of specialized disciplines. The more he advances in knowledge, the less clearly can he see either the world or his own self, and he is plunged into what Milan Kundera calls the ‘forgetting of being’. Architecture followed suit with modern man living in Spanish Colonial houses and working in monumental Roman IT centers. Perhaps at night he would buzz over to the Corinthian Club for an intoxicating Cuba Libre. Everything is false and make-believe. All is ‘seeming’; nothing is ‘being’. Imagineering has become the science of pretending and even one’s life becomes a pretense. In the modern world commercial building sells dreams, fashions, pretending and imagining what is not. Modern housing estates are investment parks branding fantasies. Shopping centers are becoming amusement parks for escape. When you are depressed and your love has left you, go buy some clothes, a CD or some software.

If Leonardo da Vinci and Michelangelo, along with Cervantes and Descartes, were modern men, then the end of their legacy ought to signify more than a mere blip in the evolution of architectural hype and forms. It should mark the death of the modern era. In fact what is happening is a form of terrorism. It is attempted murder, but the movement still lives. We know that architecture is as mortal as the human race itself. We have schools of architecture where there has been no birth, much less a murder. As a model of the human spirit, grounded in the relativity and ambiguity of things human, architecture is incompatible with the mercantile universe. Architecture is not the vocation of cutting and pasting, of advertising, graphic design or of branding. This incompatibility is deeper than the one that separates a human rights campaigner from a torturer; or a secular man from a fundamentalist. It is an incompatibility with the very nature of artistic expression, as opposed to just a moral or political paradigm; because the world of the various truths of architecture and the world of commercialism are molded out of entirely different substances. The new world of marketing, of salesmanship, of the new economy based on consumer products, and of the new urbanism, is a totalitarian world. This ‘Postmodern’ world deals with issues and decisions around them in terms of black and white, good and bad, right and wrong, and The Truth. Branding has no place for ambiguous messages. The branding experience is not an exploration, an adventure or a journey. It is a statement pounded into one’s head again and again through cut-and-paste graphics and cute ideas. Architecture deals with nuances, relativity, personal perceptions, human experiences and ambiguous lyricism. The commercial and the mercantile world excludes relativity, doubt, questioning. It can never accommodate the spirit of architecture.

The Modern Architectural Agenda

Modern architecture does not mean a bunch of modern buildings. It is a state of mind, conceptualized

within a social, economic and historical framework. Modern architecture is a reality only because it emerged through an agenda of change and actions with a mission and a vision. The modern vision of architecture is to create a better world, an ideal world, or even a perfect world, equally for all citizens. That mission can be seen in the spirit of Leonardo da Vinci's Ideal City designs and the designs of many of his predecessors and followers. Humanism, the human being in the center of things, has been the flag that rallied thousands of young architects to the cause of modern architecture. Civil life, city life, urban life and urbanity have been the central focus of this cause. Civic spaces, boulevards, parks, gardens, riverfronts and concepts for entire cities have been on the palette of architecture for centuries; but at the heart of these utopian dreams is simply a journey toward the good life. This is a journey that everyone has a right to experience.

Often this work invokes nostalgia for a simple, green, clean rural life lost in the rush toward industrialization and urbanization. Even through the design of sophisticated country villas, architects have attempted to illustrate a possible future. Arcadia, a romantic image of a lost rustic world of perfection, a world at peace within itself, has been a binding artistic force linking learned people in hamlets, farms and cities. The city planning and urban design agenda are not those of great design statements and heroic monuments, but the plans that fit 'everyman' into a world of beauty, work, recreation, household life and reflection. Le Corbusier's *Ville Radieuse*, or Radiant City, was an abstract concept where masses of people could live and work. It was a place where each sought her or his own individual opportunities. Wright's Broadacre City put the same search into an American context and made a statement of an ideal way of living, which fit everyone into the template of ideal life. Neither of these ideal plans was meant to be built as a solution to problems. They were emblematic gestures encouraging thought about new options. In the midst of the last century, Josep Lluís Sert sponsored some of the first charters of good urban design and planning. He founded the first course in urban design at Harvard. In my own studio we have promoted The Principles of Intelligent Urbanism through our planning work in Sri Lanka, India and Bhutan.

While creating a harmonious living environment for all is central to our agenda, technology is of equal importance to the agenda. What Le Corbusier said of the house – 'a machine to live in' – applies equally to the city. But again, this is a symbolic statement, meaning that if shelter (or place) is to be accessible to all it will have to be designed and produced like cell phones, bicycles and airplanes, not like a high art sculpture, a great painting or a handmade ceramic bowl.

To push this agenda is to fight other agendas. Mercantile architecture has its own rationale, its own framework and its own agenda. Commercial architecture follows the rule of Floor Space Index, cheap materials, flashy façades and creating false dreams. There is also the academic agenda of writing and theory, a museum agenda of the high priests of art, and a media agenda of making and breaking artists. All of these agendas share a commercial goal. All of these agendas make alliances and strategies for dominance. Thus, we cannot be silent spectators to life and the continuous changes going on around us. We must be participants in it. Architects are meant to be leaders, not followers.

Imagineering and the Creation of Space

A number of urban theorists have raised a core question regarding the determinants of urban form, urban planning and design. The most notable question is the assumption that rational decision making by professionals would continue to be the method of designing urban spaces. New theorists propose that urban form could become just another commodity – a product to be consumed if not produced for profit. Or perhaps, as illustrated by Blake in *God's Own Junkyard* (1964), our urban environment may become just a by-product or, worse still, the flotsam of the production and consumption process.

The Power of Design

In *Delirious New York* (1978), Koolhaas substantiates the formative role of ‘business’ and ‘the market’ in shaping large projects. No one really doubts that capitalism is the catalyst in molding its own artifacts and in guiding the plans of ‘people’s governments’ as well. But capitalism goes beyond just profits: it is about ruling and about the ‘practice of power.’ Capitalism is more than just making an efficient factory, or a profitable office building; it surpasses inventions, copyrights, packaging, marketing, sales and profits. It is about images that express decision-makers’ roles, and their domains of power. The idea of the Chrysler Building, the Bank of China and the Rockefeller Center is as much about imagery as anything else. These buildings create iconic images of New York and Hong Kong and the forces which move those societies. Without these images, aggressive competitors would well have swallowed up these entities long ago. The same is true of nation states. In international politics and multinational business alike, there is a hazy line between survival and successful imagery.



I propose we extend the argument into the realm of domains of power and how governments, corporations and other large institutions use urban spaces and urban places to temper these domains.

Autonomy and the Size-Hierarchy Scale

There is another issue that needs to be addressed here – the determinism of urban designers. The issue of artistic autonomy has been brought into question. While the Great Man Theory, according to postmodernists may belong in the trash-heap of history, there lingers an issue of the role of articulate and considered decision making by professional teams and their integrity in a process. Corporate Imagineering – the deployment of virtual reality versus the creation of genuine expressions – has been mooted as an integrated solution. I would like to propose that the larger the artifact designed the less would be the autonomy or the singular role of any one ‘creator’. For that matter, even the autonomy of any major professional design team would reduce in proportion to the size and scale of the artifact it creates. Opposed to this is transforming ‘designed experiences’ into ‘branding experiences’, devoid of human scale, proportion and cultural content.

I feel Team 10 was exploring this dilemma way back in the 1960s, and that they were saying, ‘If no one is going to be responsible, if no one is going to be the designer, then it would only be through the creation of a value system, with related principles, that we can get quality out of large, urban

infrastructure projects’.

Much of Team 10’s work was in the form of experiments with smaller projects that would generate these principles. Aldo van Eyck’s parks, his orphanage, and the Free University of Berlin by Candilis-Josic-Woods come to mind as significant experiments in this direction. There was also a concern that ‘methods,’ the International Style and other cookbook schools of thought were devoid of the kind of value base and lyrical expressions that urban fabric requires.

At the lower end of this size-hierarchy scale, an individual can still design coffee cups, chairs and houses. The issues arise in the design of larger slices of urban fabric. While an artist can design one’s own chair, or make a sculpture, they cannot compose a town. This size-hierarchy scale seems to make eminent good sense, because a town design impinges on more people than a chair, and there are more technology options that will affect the lives and consumption patterns of thousands of households, enterprises and individuals in a town. On the other hand, the likes of General Motors should not become the ‘artists’ either, effectively lobbying governments on the kinds of subsidies to be placed on energy, various transport modes, roads and urban layouts.

What is disturbing is when thinking trends, corporate interests, and political naïveté begin to converge. The American creed of the *New Urbanism*, like the creed of CIAM, carries with it the danger of cookbook rules for urban design. Even *Smart Growth*, while reaching back to the panacea of formulae, labels non-believers as ‘libertarian.’ In case you do not know, in Americanese that is a bad word for extreme individualism, or neo-conservatism in the sense of advocating individual freedom over the common good. There is indeed a deeper issue here which urban designers and planners must address. Is autonomy what we are really looking for in the design process? Or, are we looking for design that responds to some kind of social and contextual contract, to principles and to ways of thinking, and not bureaucratic rules?

Anything Goes! Ugliness can be Pop Art!

Robert Venturi and Denise Scott Brown began to look at urban landscapes in much the same spirit that Andy Warhol looked at cans of tomato soup – as a form of pop art, or relevant cultural expression. A Coke can is, no doubt, an important part of popular iconography. But we cannot call it ‘popular art’. The 1960s protest symbol, the raised fist, is ‘popular art’. Unlike people’s art, we are getting flooded with corporate, common images, which are thrust on the popular imagination. Times Square is a gross example of this, but it is happening in less obvious ways in every setting. I have always had a deep, intuitive sense of doubt about Warhol, Brown and Venturi. In their desire to be catchy – to grab the public eye – they were dignifying ugliness, aligning with a way of thinking. Warhol mimicked and mirrored corporate generated images, not art that emanated from folk ways or common people’s lives. Brown and Venturi glorified the ‘leftovers, junk and trash’ of America’s consumer society. Interesting idea, but that’s about all there is to it.

Koolhaas’ barons of New York City wanted their mega-projects to serve as icons of their families’ names and prestige while simultaneously profiting. But this was more in the spirit of Renaissance patronage for self-promotion than for advertising products or making spaces into products or mundane branding experiences. There is legitimacy in this kind of vanity and hubris – a facile marriage between art and ego.

Walt Disney Incorporated was, and is, on a very different path. It has designed several ‘brand name,’ spatial products, each with its own market niche and commercial value, packaged and

marketed with great success. They each sell under names we all know, ranging from *Mickey Mouse* to the *Pirates of the Caribbean*. The Disney Company has opened a real estate division, Disney Development Company, which has moved ‘Imagineering’ off the film sets and into the streets – the New Urbanism marketplace. The Millennium Village of Celebration, Florida, near Disney World in Orlando, was their first product. While Levittowns were in the same genre – packaging the American Dream into an affordable commodity – the Millennium project rests more on imagery than on mere functional factors like good location and affordability. Studies like the Tastemakers and the Levittowners explored the use of ‘packaging’ and marketing to create consumer products out of urban fabric. Our concern here is thus a long-standing one.

Given that urban design and city-planning fall at the ‘high end’ of the hierarchy of autonomy in art, it is clear that few individuals will sit alone and compose large-scale urban scenarios. What are the alternatives to corporate domination and its commercial iconography?

- Benign neglect;
- Participatory design;
- Indigenous accretion;
- Professional planning, and value based design teams; and/or
- The individual visionaries.

All five are becoming ever more illusive propositions. Most likely a combination of these alternatives would be employed by large corporate or government entities.

How Spaces Use People

In fact it is not so much the process of space creation as the way the spaces are used, which should really matter. In this sense we should be more concerned with ‘conception’ than production. Or, conversely, how spaces use people should be a concern to us. Do we conceive this at the outset? Disney creates the spaces, the characters and the storyline. Disney begins with terms of reference, performance standards and a clear brief on the product elements and characteristics, with a clear focus on the targeted consumers. In fact the consumers and what the product must **do to them** is the core of the brief. There is something here to be learned from corporate animations. As designers we must know what our compositions are doing, how they move people, how they play with emotions and experiences. What is objectionable, though, is that the Disney design method rejects context completely. If a lake is needed machines are brought in and one is made. If a lake is in the way it is filled! In a similar way, people are conceptualized and made into the set characters. While the project makes the same claims of higher density, footpaths and common open spaces that most New Urbanism communities do, one questions the kinds of social interaction that may emerge. The high costs, isolation from work places and limited housing design types lead one to conclude that the community will be one for older, well-to-do Anglo Saxons. The Millennium project raises numerous social issues about heterogeneity, about occupational and job opportunities and about variety in communities. American cities have always been cosmopolitan; a mix of Yankees and immigrants. In the Disney illusion there are no ‘new comers’, only old investors. It is a product, not a community.

Some spaces are convivial and catalyze social interaction. They make interaction happen. Some spaces temper one's curiosity and direct one's interest. Other spaces respond to the need for variety and diversity. A spatial system can 'set up' sequences of events, expectations and experiences which challenge the user's spatial intellect. As an urban designer, one can create 'hang-out nooks,' steps to sit and sun oneself on, corners to hide in with a friend, and low walls to sit on and talk. A courtyard can be an empty, dull shell, or a lively outdoor café. There can be a sidewalk, and then there can be a sheltered arcade, with interesting little vendor stalls. Some spaces are of human scale, making one feel a part of the ambience. Others are monumental and tell us of our insignificance. Their scale offends and alienates. Or they are grey areas, devoid of any character or quality and abusive to the human spirit.

Many urban spaces are bland, colorless and have no textures. They convey a message of neglect. They speak of an authoritarian attitude of governance toward citizens. I am reminded of a photograph in *The Natural House*, labeled 'Find the Citizen.' It is an aerial image of the East Side of Manhattan through the billowing exhaust of a thermal power plant.

Is Quality Measurable?

What should disturb us, as urbanists, is the quality of life being generated, and the scales on which we are able to conceptualize 'quality.' Kevin Lynch taught us that cities have several aspects, or elements, which can be enriched to improve the quality of urban places. He noted landmarks, boundaries and districts, amongst others. Lynch proposed that good urban fabric is not homogeneous; it is varied and articulated. In *The Image of the City* he emphasized boundaries and landmarks, which give further articulation and meaning to urban places. An urban core can have its own unique edge, can have distinctive entries and can sponsor movement through a network of walkways and paths. Small parks, gardens and courtyards can further accentuate these experiences. Exploring an urban core can be an odyssey through places, challenging one's senses, demanding one move further and deeper into unknown domains and precincts. Laying out such a scenario is no less than conceptualizing the cinematography of a film. We are designing experiences. There are urban elements, urban components and urban relationships, amongst and between them, which generate urban systems. It is essential that we identify these parts, analyze them in terms of how they work on us, and assess how we feel and how we think they should be used.

There are also systems of architectural values that are used and abused (contextual relevance, honest expression of materials, human scale, building modules based on anthropometric dimensions and production sizes, graphic proportions, etc.). All these come to mind when lamenting the banality of the emerging urban forms. These forms are more about 'appearances', skin, packaging, than about any of the concerns and values I have noted above. While we should be moving into the four-dimensional world of experience, such forms move us back into the two dimensions of graphics.

Most important are the unplanned, serendipitous and pleasant human interactions which happen and are enriched by catalytic urban spaces: a chance meeting; eye-to-eye flirting; boy meets girl; and boy meets boy. Good urban fabric leaves the parks and the boulevards open for all to walk upon, hawk upon and play upon.

Images as Antidotes

America becomes a focus of thought because it has a narrow vocabulary of traditional patterns from

which to evolve new forms. There have been a plethora of books on American barns; on highway hoardings; on shopping centers; on massive industrial complexes – all intent on proving that there is, indeed, an American urban tradition that we can learn from. While such studies are popular American doctoral thesis topics, they exhibit little virtuosity in terms of defining an urban language. The repertoire to draw on is very limited. It raises the question: Is *Learning from Las Vegas* possible? While bland America provides, so to speak, a ‘clean slate’ to work on, the reality is a milieu of ‘sameness’ or, at best, the trivia of endlessly repeated Disneyland imagery. The New Urbanism is a remake of the Levittowns of Long Island. We have added sidewalks, Victorian gingerbread motifs, front porches and declared that a kind of miraculous ‘smart urbanism’ has resulted. Indeed the sameness, the trivia and the banality of the Levittowns is more hurtful, because they have become the norm. Disney knew well the boredom of his compatriots as well as their lack of exposure to varieties of experience. He provided an antidote of sorts, in the form of packaged milieus, each with its own contrived traditions and fantasized geographical settings, which were then effectively marketed as lifestyle themes. The problem here lies within a kind of reality warp; a large and influential society began to gain its intellectual and emotional stimulation from fantasy and escape. Substance began to fade away and wither into a new virtual reality, created and produced by corporations.

One recalls with nostalgia that real places did exist in America as recently as the early 1950s, with their own styles, local dress mores, accents and even food habits. There were places like Cross Creek in Florida that ate its own alligator soup, Key West where Hemingway could escape to write, New Orleans with its own music and style, Cannery Row with its unique culture of poverty, Greenwich Village with real thinkers and painters. Even Faulkner’s hometown, Oxford Mississippi, has been transformed into a caricature of the Deep South, a stylized hyper-image of itself. Any ambience that had genuine qualities, or a unique character, was ‘made over’ into a kind of hyper-reality of what the place once was in the public imagination, depleting its authenticity. These ‘made over’ packages were then marketable – products for sale. Tourism became a vehicle to distribute them to millions of consumers. These hyper-real settings provide relief to the real urban ambience of Coca Cola signs, McDonald’s Arches, and curtain-walled buildings. If religion was the opiate of the masses in the nineteenth century, Walt Disney and hyper-reality are the opiate of the masses today.

Tourism/Urbanism

In such a confused milieu, it seems appropriate that the most talked of architecture is in the form of new art museums. And the most valued art is found in those museums. Galleries where something ‘new’ can be seen, either sell high-end ‘art investments,’ or trivial ‘arts and crafts’ bric-a-brac. Again these places are largely destinations for tourists, who are the consumers of these products. There was a time when people ‘traveled’ without any planned schedules or destinations. They were seekers – adventurers. In fact the entire concept of ‘tourism’ has emerged from consumer societies over the past several decades. The key requirement of the new tourism is that ‘nothing should happen!’ There should be nothing unexpected, unplanned or serendipitous. The new tourism that is preconceived and packaged allows people to consume places. Tourists use expressions like ‘let’s do Spain next year. Having ‘done Spain’ they will have to ‘do’ someplace else the following year. Again, consumerism. Tours have been designed, packaged and produced so that the essential qualities of a traveler, an explorer, or god forbid, an adventurer, are methodically distilled from the product. All risks, all dilemmas, and all strange people have been removed. Tourists do not need ingenuity to

solve problems, to mediate with people, or to just plain make friends. In fact they want to consume people, instead of meeting them. They feel uncomfortable unless they are paying something to the ‘natives’ to do something for them.

Tourism has become an analogue for urbanism. Variety, diversity and experiences are to be removed. Nothing unplanned, nothing unforeseen, in short nothing new, should happen.

Meaning Systems

Having thrown up that paradigm, I would now like to focus on my work in the Himalayas. Here we are planning a new capital city that is an overlay on an existing setting. To describe it fully would take thousands of words. So, instead, I will explain to you what a prayer flag is. In a way it is an analogy of urban design.

In its simplest form, a prayer flag is a form of votive offering. A very long strip of cloth is tied along a very tall pole. The color of the cloth signifies a mood. The mood may signify an event, like a death in a community, or the initiation of a new house, or the start of a new season. It may just be an auspicious omen. If one looks closer at the cloth, there are characters hand painted or block printed onto it, which are in fact words that form mantras. As the wind blows over these flags, it is believed that the mantras are endlessly let off into the breeze, and that they float about over the city.

Walking through Thimphu valley along the Wangchhu’s clear streams, one is surrounded by verdant forests, which stretch up the steep mountain slopes from the river. There, at the top – or rather the edge, making a silhouette of the hills against the endless blue sky – one can make out a strange articulation. If one looks more closely at that edge, one finds it finely articulated by rows of large prayer flags, of varying heights and configurations, presiding over the city, diffusing their favorable mantras. So we have this image and there is also this hidden meaning. The city is being protected, enriched and empowered by this guardian wall of auspicious prayer flags.

There are other artifacts, with other meanings. There are *mani* walls, or prayer walls; there are prayer wheels; there are *chortens* with prayers inscribed within them; there are *lakhangs*, or temples, and there are monasteries full of monks. There are also gateways, which welcome visitors. There are decorative signs and symbols, which emanate good vibes. And there are prayer flags that preside over the Thimphu Valley and gather the geographic space into a ‘place.’ Spaces are empty; places are full of meaning.

All of these artifacts – all of the meanings they communicate – charge the atmosphere with an *aura*. The mutual understanding of this meaning system, and the sharing of its *aura*, generates a deep form of conviviality. The artifacts are mechanisms created to generate meanings. These meanings are the shared feelings and sentiments of the inhabitants – the essence of their community. Thus, ‘places’ are imbued with the elements of ‘shared meanings’ and ‘conviviality’.

Urban Verbs

Just as Kevin Lynch defined districts, boundaries, landmarks, etc. as the nouns of urban design, I would propose that the meanings discussed above are the ‘verbs’. They begin to move feelings and sentiments in directions, just as static, immobile ‘nouns’ in literature need verbs to ‘get things going’.

All of these signs, symbols and elements become a language, which ‘speaks’ a knowledge system.

In this context decoration becomes important because different motifs become symbols of various intangible attributes: like ‘good luck’. By applying what appears to be decoration onto these components, additional meanings and emphasis are provided. Are these, then, not the adjectives and adverbs of urban design? All of these signs, symbols and elements become a language, which ‘speaks’ a knowledge system. The ‘auspicious’ is elemental to the Bhutanese knowledge system; just as the ‘rational’ is elemental to Western systems of thought.

The Urban Uniform

New York City, the Cartesian grid, the ‘x’ and the ‘y’ axis, are all our tools for thinking. We Westerners are mental animals of paradigms; we tend to think of one thing versus another, of ‘x’ versus ‘y’. We like a world of good versus bad, of polar views. We feel very comfortable with questions which ask if there is a god or not, but the idea of multiple manifestations of something, or many aspects of an idea, is not a comfortable proposition. Part of this emerges from our written, as opposed to oral, tradition. This means we must write things down, and that begins to mold how we think. For example there are thousands of Hindu gods. It is not really practical to write about several thousand gods – one god, with a few saints thrown in for good measure – that is within the bounds of the written word. Oral traditions are more expansive, flexible and imaginative. Pānini’s *Ashtādhyāyī* comprises close to four thousand *sutras* (aphorisms) on language, which were written down four hundred years after they were created. They were passed down those years through rote learning and recitation from teacher to student. Consider a mandala. It is a two-dimensional diagram of the universe, which describes matters in terms of mythological beings and places and relationships between places. Most important, every significant thing is a manifestation of something else and has hundreds of forms, which can be *avatars* or accretions. And these are not mere forms of things, but interpretations of feelings, moods and attitudes.

Most important, every significant thing is a manifestation of something else and has hundreds of forms, which can be avatars or accretions.

The experience of ‘this life’ then is an adventure, that of a traveler, not of a tourist. Nothing is certain or truly understood; or if it is, it can be looked at in many different ways. Milan Kundera, in *The Art of the Novel*, opines that ‘uniforms’ possess the Western mind. He explores the possibility of a culture of ‘multiforms’. He laments the fading of individual choice, the loss of the inner freedom; the absence of uniqueness. I feel we must address the same issue in urban planning and design. In another essay, *Slowness*, Kundera vents his anguish on the ‘hyper-experiencing’ that characterizes contemporary life. Everything is momentary, fleeting, at high speed; one image comes quickly over another, like the nervous clicking from channel to channel, from website-to-website, while one is bored even of the clicking itself. What is most disturbing about the emerging, consumer generated ambience is that it is a medium for Cartesian, monosyllabic kind of thinking, devoid of variety, differences and manifestations, fundamentalist in the worst sense. It is subtly fascistic. Boredom is the least of its sins; linear thinking, intolerance and a kind of mental blindness are its deeper pathologies, which give cause for concern.

The Ethos of Urban Space

Image-makers are media makers, and we define and design the ‘ethos’ that controls essential feelings. ‘Ethos’, according to Gregory Bateson who coined the term, is the way a culture emotes about events and happenings. Bateson saw it as a tool to distinguish between cultures according to their defining elements. He knew that the way people felt about events and places was the way they were – their essential culture.

Different spaces evoke different behavior. In India, visitors to Hindu temples instinctively remove their footwear, regardless of their own religion. Entering a mosque will evoke hushed silence, while in a wedding *shamiana* there may be a lot of chitchatting. Places, then, give out signals, which request specific forms of behavior, let off an ambience. Imagineering, no! A thread of history woven into everyday behavior, yes!

A Design Approach: The Differentiated Web

The basic concept of the Thimphu Plan is to create a network, or movement system, which separates pedestrians from vehicles, and which promotes movement. By movement, I do not mean movement for fun or pleasure – I mean movement that engenders social interaction. The concept is not so much geographic as it is conceptual. If there are ‘server’ and ‘served’ spaces, as in Louis Kahn’s sense of things, then the web is a facilitator to various specialized modules of spaces that have to fit into it – houses, shops, religious and institutional structures. We decided at the outset to use the traditional building components of the Himalayas as a kind of Lego Set to play with. The Served Spaces, or Buildings, could be plugged into or ‘set-into’ this network. We see the network as a ‘differentiated web’. One line becomes a long corridor, or as Shadrach Woods would have said, a STEM. The stem runs parallel to the riverbed and is so planned that over decades it can adapt to newer and varied technology. Trunk infrastructure would also run along the corridor. The corridor will be differentiated by nodes and by hubs. The nodes and hubs are points in the system that are in fact public transport stops, places of modal split, as well as the centers of various types of pedestrian precincts.

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Why Europeans Sleep with their Dogs and Other Architectural Theories

Modern society has brought young urbanites economic independence from their parents; for the elderly it has brought relative security of health care and income. Working women, equal opportunities and fast-track professions have made the traditional family redundant in many societies and in a fair segment of India's urban youth. The glue that bonded society has lost its adhesiveness. Maybe a more sticky glue has taken its place?

Architecture is as much an engine of this change as it is a result of this phenomenal transformation in our cities. Indian architects are quick to fall in line with the new world order, eager to appear creative and different by copying the banal and the mundane. The cell-phone has replaced physical neighborhoods and the internet is the street-corner for gossip. The surfeit of media information has made news boring, and just to catch attention one has to yell ever louder. The design profession is likewise promoting sensationalism and 'the spectacular,' rather than good urban form and human values. True, this trend is found in only a fraction of Indian society, but most of the new built form in the metros has turned its face away from community building. Moreover, this model is willy-nilly the road we are pursuing in every aspect of daily life. It is the reality of the urban niche that is in the limelight, growing day by day and setting the trends.

City form has responded with a myriad of branded eateries that are replacing kitchens. Multiplexes, lounges, cyber cafes, bars and discos are replacing living rooms. Beauty parlors and spas are replacing our bathrooms. Practically all that is left of the traditional home is the bedroom. Every building wants to spread over its own full city block; each plot is walled-in and guarded; gyms and health clubs are replacing neighborhood play areas; buildings are becoming monumental and impersonal, with harlequin façades. Roads are widening, sidewalks and cycle paths are shrinking, and the scale of cities is morphing from human to the machine in motion. More is more and big is beautiful in the new city.



People whose parents survived in comfortable simplicity on modest incomes, feel a pinch in their 'lifestyle' earning anything under four times their dad's last pay check. They covet and protect every rupee, counting up who pays what share on each outing. Habitat is no longer a home; it is a 'pad.' Protecting one's wealth from parasites, opportunistic relatives and hangers-on is a matter of daily management. The accepted dependents of the house are cell-phones and credit cards that eat up every paisa unnoticed. As the city, its architecture and urban society all change into a bland chessboard of stand-alone people and façades deposited in glass-walled blocks with no courtyards or street life, so too does the individual psyche, the persona and the personality transform. People don't like people any more. They love themselves. The word 'communities' is becoming as archaic as typewriters. 'Neighbor' is a bad word. Everyone is worried that everyone else wants their money; and everyone else does want their money!

Style, façades (personal as well as architectural), packaging, attention-getting stunts, fashion and

obnoxiously selfish behavior are all part of the new life that is a product of the new economy, new society and the new urbanism. Bland and ugly buildings merely mirror the people who live inside of them. In the emerging 'I, me, mine' culture, the only true friend is one's loyal pet dog.

Europe, which is six decades ahead of us in the search for self, has invented the 'single person family' as a demographic profile. It is the self-fulfilling prophecy of the paranoid urbanite who fears that human beings are predators and scroungers. At the same time, every average person craves a life partner who is incredible: great looking, super intelligent, professional, high earning and possibly even loving! Thumbing through Page Three they think, Wow! Fantastic! First Class Act! Spectacular! How Clever! The average person wants to settle for none less than the spectacular, who they know they will never meet and the attraction will not be mutual even if they ever do. But the media and the tastemakers tell them not to settle for less. So they cruise the streets with pet dogs in tow, looking around for companionship.

In the single person family what one is talking about is more important than who one is talking to. If your topic is not about the spectacular, your victim will feign busy and hang up. Family, close friends, and even lovers, are passé. That people are talking about you, good or bad, is more important than having a civilized conversation over a night cap. The weather and politics are no longer topics of discussion; sensationalism is the topic of catchy dialogue: Paris Hilton, a terrorist attack, the Bird's Nest, or upside-down buildings. Architects used to talk about community, engendering interaction between people and neighborhoods. Now they speak of new visual tricks driven by computer graphics. Where people used to discuss ideas they now talk about other people, software and objects. There is no time for quiet times at home. Even sex can be purchased off-the-shelf, or experienced in thirty second trysts in aircraft toilets 35,000 feet above sea level, but not with a long-term partner at home. The redefined human being is labeled a metrosexual. Yet, at the end of the day, the single person family too needs companionship, without the hassles of people and community. It still wants something warm, with loving eyes, waiting at the door to greet them when they return home after a long day.

According to recent census data on dogs in modern societies, the canine population is on the rise. Its ascendance shadows the rise of single person families. There are as many dogs in Amsterdam as there are people. This is all very important to us creatures who imagine ourselves as architects, as we are willy-nilly creating cities that not only respond to, but simultaneously catalyze the new social structure, culture and demography. Public screaming, posing, posturing and yelling are somehow the natural corollary to life alone with a dog. Just look at Europe, its new architecture, and its love affair with beasts.

Modern architecture, the kind of modernism that Jose Lluís Sert practiced and wrote about, was focused on resolving the conundrums of urbanism and our human condition in the beehives we call cities. Modernists dealt with urbanism, the aesthetics of new materials, and a rejection of effete styles and fads. Heading the International Congresses of Modern Architecture (CIAM, or Congrès Internationaux d'Architecture Moderne), Sert incorporated the Team 10 revolt within the movement, and then founded the first Urban Design course at Harvard, which changed the way designers thought about built form and community. Le Corbusier was equally concerned with issues of humanity in transitional societies, and Wright championed craftsmanship and integration with context. Aldo van Eyck knew that place was the realm of the in-between, and he created 860 small play parks almost out of thin air. All these men abhorred the effete.

While the modernists searched for human scale, proportion and social reality, the theorists were flip-flopping with new ideas and new heroes. Instead of evolving from a platform of ideas, a kind of incestuous love affair emerged between designers, magazines and architectural critics. Postmodernist theories in architecture attempted to piggy-back on French philosophy and the literary criticism of the late 1960s. Semantic Analysis, Structuralism, and Deconstruction that had come and gone in the arts in the early twentieth century, decades before 'liberal humanism' was debunked by French theorists, re-emerged as clichés of the vacuous elite. Philosophical and literary Postmodernism really shares nothing with architectural Postmodernism. Postmodernism in architecture seems to be some kind of neo-capitalist Employment Guarantee Scheme for a clique of academic theorists, journalists and designers, rather than a guiding charter of design leading to a better future. The prime beneficiaries have been the writers, publishers, magazines, media, and a few grandstand architects who vomit out the spectacular at the cost of good community building principles and practices. Honest expression of structure and materials has been labeled as passé. The word 'community' has evaporated.

Just as the 'Chicago School' of architecture met sudden death with the Chicago International Exhibition in 1893, modern architecture wilted to the blow of a few humongous projects in the 1970s and '80s. Feeble though spectacular architecture caught the public imagination. Sigfried Giedion cites America's cultural inferiority complex as the reason for the triumph of effetism at the close of the nineteenth century. He noted that the 1893 Exhibition sponsors turned to French academicians for advice, giving them, along with a thin theoretical base, a dominating role in the Chicago Fair. And again the world of architecture looked to France for intellectual reasoning in the last quarter of the twentieth century. Again a sense of inferiority drove bombastic exhibitionism and narcissistic isolationism. The terrorism of the avant-garde invaded the half-awake mind. The fear of being wrong and of displaying stupidity has made us stupid. The only creatures we dare talk to without recrimination are dogs. In fact in totalitarian regimes, where people snitch upon each other's wrongs and wield 'politically correct' thinking as a threat, pets are the safest bet for a trusted friend. So aware were the Soviets of this human weakness, that when they invaded Prague in 1968, their first act of state terrorism was mass canicide: the killing of all the city's dogs within the first week of occupation.

Lacking any complex mission of values to sustain thought, buildings provoke us to ask questions about the intellectual context, that is an abstract analytical context, rather than the content and the actual setting. What is being sold is a theory, rather than a design emerging from the physical and social context. Some theory about the building has to be made up to justify its existence. Success lies in evoking doubt: 'What is the reason for all of this?' Yet, this is not a reasonable question in our Postmodernist times wherein it is the institution of the media, the galleries and the critics who have declared a mound of nothing to be 'Architecture'. Postmodernists would like us to ask questions about the justifying ideas that surround the architectural monuments that they deem iconic. The buildings do not possess their putative poetry or beauty. You have to think over it and figure it out, and probably read several books by Michel Foucault and Jacques Derrida to fit things in. Like Duchamp's famous *Urinal*, or Gehry's *Fish*, these monumental clever structures test our patience and intellectual skills. As Christopher Butler ponders, 'calling into question', or 'making the viewer guilty or disturbed' seems to be a common element among the spectacular buildings being created. There is a neo-Marxist tint to it all as it makes everything from personal relations to buildings into political constructions and challenges. On the other hand the stunts created are to be consumed rather

than used. A fire station can be turned into a profit center 'museum' reaping income from visitor tickets. A project that is five times over the budget can be justified on the grounds that it was paid for through the entry fees of ogling tourists in the first year. The Wal-Mart business model has become the justification for art. Mercantilism justifies the tyranny of the spectacular and the plain ugly.

In the new world of cyber information, facts and figures are 'suspect' of being the manipulative image-making of mercantile forces, rather than the advancement of knowledge. To paraphrase Frederic Jameson, much built environment seems to be a mutation into a postmodernist hyperspace which transcends the capacities of the human body to locate itself, to find its own position in the mappable world, and this milling confusion is a dilemma, a symbol and analogue of the incapacity of our minds to map the great global multinational and decentered communicational network in which we find ourselves caught as individual subjects.

But yes, we like to talk about the **spectacular** and about things 'new'; we write about style, about fashion and about fleeting things that are here today and gone tomorrow. Cute and clever designs rule over context, community and the reality of materials. The victims are the people who live in little repetitive boxes and are told spectacular sculpture is their compensation. They are told good architecture is a stunt of monumental construction, which massacres anything inside of it, all in the name of a vague concept of art, with a little French philosophy thrown in for spice.

It seems architectural critics swallowed their own propaganda about 'modernism' and began to believe that 'modern architecture' was all about 'isms', great men, sculptural buildings and icons. In fact it was exactly the other way around. The Postmodern era we are living in is in fact a form of ersatz pre-modernism, which the early modernists ranted against. This is the same effetism which killed the Chicago School at the close of the nineteenth century, opening the door once more for the make-believe of the 'spectacular.'

The city makeover of recent decades is just cold façades; inhuman objects; machine scale monumentalism; stunts and spectacular structures and materials. Like the fashion ramp, our cities are getting cluttered with mimics of the Bilbao Guggenheim, the Valencia City of Arts and Sciences, China's CCTV Tower, the Vitra Fire Station and Amsterdam's Nemo. These are the scraps thrown out to the public for visual consumption. Museums have become the opiates of the masses. The urban landscape dies when the malls close, the lights go off and the pizzazz dims into darkness.

Like the spectacular, anal retentive buildings all alone in their own little city blocks, the city dwellers all go home to their little boxes to feed canned food to their cats and dogs, with whom they cuddle up and go to sleep. This seems to be the India of our dreams.

(Lecture at INDIA HOUSE for CCBA Staff, Saturday, November , 2008)

The Beautiful Velvet Box

At the World Architecture Festival in Barcelona in November 2009 I was highly amused. I learned that architecture has transformed from a complex, creative process of integrating diverse functions, structural systems, myriad services and utilities, interrelated spaces, and exterior-interior visual connections, into a game of appliqué where one cuts and pastes decorations and ornament onto four sides of dull boxes. Just as the École des Beaux-Arts of the early twentieth century provided mental ammunition to the mercantile juggernaut, so a band of editors and academics dominated the Catalan seaside conference hall (an ugly box too). Few jurors had ever put pen to paper, much less watch earth fly to the order of their drawings. Box after rectangle, each with a more meaningless and mundane interior layout than the one before, was showcased to the applause of the jury members and to the excitement of the participating cut-and-paste quacks. I learned that the mundane is in; profundity is out. Architecture is decoration, or as Farshid Moussavi, the Keynote Speaker emphasized, ornament is the function, and she spoke eloquently of the ‘function of ornament’.

Much in the same manner that degeneracy killed the early modern movement with the World Exhibition in Chicago in 1893, overshadowing the likes of Louis Sullivan and H H Richardson, we are back in a world driven by mercantilism and greed and where architecture is reduced to the commercial graphic art of decoration. Yes, the much loved dullness is accentuated by shrill and spectacular stunts. One stunt is the tallest, another is the most expensive and another is just silly. Most are justified by critics on the grounds that the hundred million dollar cost was recuperated through the sale of entry tickets to foolish tourists who came to gawk at artworks they could never understand. So in Barcelona I understood a few things that Giedion came to know more than a half century back, it seems, but that live on today as the ruling order of inanity. I learned that **Effetism is the Enemy**, and that the architectural Taliban hides out behind editorial desks and academic offices in London and New York. My life in fact could be termed ‘pre-Barcelona’ and ‘post-Barcelona’.



Before my latest jaunt to eat tapas and taste lovely wines, I thought that cladding was some kind of a finish, more specifically a material, which is stuck onto the façades of buildings for some functional or aesthetic reason. Post-Barcelona I now know that ‘Cladding is Architecture’. Cladding is It. What happens behind that velvet covered box is beyond the scope of total architecture, and is relegated to realms of structural and services designers, whom the architect need never meet.

The role of architects is becoming limited to the design of the outer envelope of buildings. If you do anything more you are insulting art! This aesthetic revolution was caused by several transformational factors that are redefining the meaning of architecture.

- Cold shells whose interior functions are not known, or even thought of during the design stage, are replacing buildings with specific functions.
- Air conditioning is leading to hermetically sealed buildings with no fenestration.
- Artificial illumination is supposed to be better than daylight and people inside working 24×7

aren't supposed to know what time it is anyway.

- Many new building types have no relationship between what is happening inside a building and outside. The clients want blank walls to be decorated later. These new building types could be shopping malls, multiplexes, business parks, museums, exhibition halls, or even libraries.
- The design profession is becoming atomized. Where the architect once created 'total designs', the trend today is to fragment design into an array of consultants: the structural designer, landscape designer, firefighting, water supply, drainage, sewerage management, IT and communications, interior designer, lighting designer, sound and acoustics, audio designer, signage and graphics designer, branding experience designer, etc.
- Finally, clients don't want the public to see their buildings anymore. They see their visible mass as a branding opportunity. Buildings have become billboards.

The architect is left only to create the volumetric box and to decorate the 'skin'. His sights should be set on creating the most beautiful velvet box. Here the function of ornament comes to play, as there is nothing else.

The entire game now lies in the cleverness of the skin: Does it hide LED lights inside? Does it reflect from different layers of mirrors? Are there layers of glass all etched in different patterns? Is there a *jaali* with different colored paints behind, preferably lit with flickering lights at night? Most important, does the designer have a 'theory' of how the skin and the integrated ornamentation reflects the culture of the setting and is even an iconic representation of the era in which it is created?

Speculative investors and the resulting development want things to be 'neutral' and 'blank', leaving a myriad of potential leasers and buyers to use modular furniture to shape the interior spaces and to use the building facades for 'branding experiences'.

More and more, the only role of architects is to create volumes that achieve maximum FSI and maximum saleable area.

Highly articulate and determined spaces, structures and elevations are often considered dangerous or bad investments, as they could limit the types and numbers of buyers, users, leasers or renters; or even reduce the hoarding space for lease. Architecture in the modernist sense is bad business. Not only does it limit speculation but it employs 'transparency' between interior and exterior spaces in a manner that leaves less of visible building surface to double up as hoardings and signage.

Even new environmental considerations have made designers see the skin of a building more in terms of its insulation, reflection qualities and radiation factors. To many architects these new restrictions are in fact a kind of liberation. They need no longer bother with the relationship between what is happening inside their buildings and what is happening outside. After settling on the most profitable volume they only need to decorate and ornament the façade.

This trend coincides with the emergence of new façade technologies. Stainless steel clips now allow us to paste thin sheets of stone across broad strips of wall. Aluminum and ACP sheets offer a variety of easy-to-clip-on, easy-to-clean surfaces.

Thus, there is collusion between need and opportunity. Just at the time when the market is demanding a blank wall, the market is supplying a plethora of cladding options to decorate blank walls. Architects are becoming exterior designers, while interior designers are thrust with the more complex tasks of creating moods, creating ambience and integrating diverse kinds of lighting, air

conditioning and utilities.

Instead of a trap, this can become a new challenge. As architects we must rebel against defeatism and the velvet box.

In retrospect I feel the World Architecture Festival was more of a memorial meeting bidding farewell to poetry, to art, to meaning and to architecture itself!

(Article for Buildotech magazine published in June 2010)

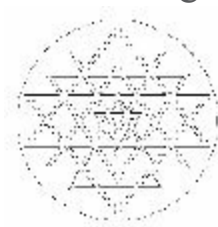
Le Corbusier: The Modern Project and the Challenge

My subject here embraces two centuries of technological, artistic, economic and social history. I refer to the modern age, of which Le Corbusier is a symbol and an icon. The connections between technology, urbanization, society and aesthetics are complex and lengthy, but our attention spans are short. Perhaps this is where Le Corbusier and his contemporaries excelled over us. They could compress into one page all these factors, trends and the artifacts resulting from them. My intention in this discussion with you is to bring us all onto that page and to pick up the journey where Le Corbusier left off.

To do this I must make sweeping assumptions about economic and technological developments that span the nineteenth and twentieth centuries, culminating in a movement that got formalized in the International Congresses of Modern Architecture (CIAM), of which Le Corbusier was an important member.

There were many other members of this movement including Hendrik Berlage, Walter Gropius, Jose Lluís Sert, Mies van der Rohe and Marcel Breuer to name just a few. Frank Lloyd Wright, Buckminster Fuller, Pier Luigi Nervi and many more were a part of the movement, though not formal members of groups. In India, Balkrishna Doshi, Achyut Kanvinde, Charles Correa and the devoted team in Chandigarh nurtured the movement in its early years.

One could even date the movement back to 1894, when Otto Wagner took over the Vienna School of Architecture and published 'Modern Architecture,' in which he asserted, 'Our starting point for artistic creation is to be found only in modern life.' He influenced a generation of modernists including Peter Behrens, Adolf Loos and José Olbrich. It was in Behrens' studio that Le Corbusier, Mies van der Rohe and many others found their moorings in the new aesthetic.



The underlying forces that drove the modern movement are more compelling today than ever before, yet we seem to have lost our bearings and drifted into an architecture that celebrates the effete and the spectacular, and relishes pure amusement park stunts. Even our view of Le Corbusier is one of a man who created monumental sculptures, rather than of a man with a social mission to drive design and technology toward the solutions of a rapidly urbanizing world. This is our self-inflicted ignorance, along with a fascination for things foreign, and things strange, that has diverted us from our professional mandate.

I want to drive home the point that Le Corbusier was not a 'stand-alone' artistic figure. He was part of a vast social movement composed of many revolutionary people and organizations. All of these people had a common goal of harnessing the forces of technology to the common good of common people, who were being uprooted from their rural existence and thrown into a new urban environment that could not fulfill their basic human needs. What was happening a century ago in Europe and America is happening in the emerging economies today.

This human crisis inspired Dickens to write of working class London in the nineteenth century and it is the condition of masses of Asians today. This situation inspired Karl Marx to contemplate the

organization of society around modes of production. All knew that a New Man and a New Society must be invented. Le Corbusier conceptualized this in terms of a New Culture, expressed through functional and efficient artifacts.

Perhaps the first instance of this concern being put in the forefront was the Werkbund movement in Germany. The movement understood that the role of the designer was to take products whose costly production made them exclusive possessions of the rich and to mass-produce them at low cost, bringing them to the doorsteps of the poor. Early modernists like the Belgian Henry Van de Velde, founder of the Weimar School of Art, stated, ‘A completely useful object, created by the principles of rational and logical construction, can only capture the essence of beauty.’

Le Corbusier was not a ‘stand-alone’ artistic figure. He was part of a vast social movement composed of many revolutionary people and organizations.

Thus, there was a schism in the art world between those like John Ruskin and Charles Rennie Mackintosh, who saw the machine as an enemy of art, and those who saw it as a friend with great potential. The new breed saw the machine world as offering great opportunities for mankind.

In fact it was in the Weimar School of Art where the links between practical workshops, design and consumers was first created, as the products of the students were sold through commercial outlets. Gropius took over the Weimar School of Art in 1919 and soon after founded the Bauhaus there. Under his leadership the marriage between industry, art, architecture, design, and town planning emerged.

Thus, the modern movement grew out of a sea change in technologies and in modern production, which had driven rapid urbanization, pushing masses of people into unplanned and inept city forms that could not offer even a modicum of civilized living and culture to the New Man.

Early modernist concerns with the concept of ‘the city’ can be seen in Tony Garnier’s 1901 drawings for La Cité Industrielle (published 1917-18); in Antonio Sant’Elia’s utopian design of La Città Nuova (1912-14); and in Victor Bourgeois’ garden settlement, La Cité Moderne (1922-25), around the same year that Le Corbusier began releasing his images of Le Plan Voisin for Paris.

The modernists discovered that the solution lay in technology itself and not in sentimental reflections of bygone times. They knew that designs pretending to be Greek Temples or Roman Forums were lies. They branded these lies as ‘Effetism.’

Le Corbusier’s edict, ‘A house is a machine for living in’ articulates this new understanding and can be validly applied to the city. He understood that there was a complete integration between art, architecture and city planning. He saw the need for a new aesthetic, a new way of building and new assumptions about urbanism. As a group the modernists broadly envisioned that change could not be segmental, or part by part. It had to be sweeping, and in that sense revolutionary. The vision was to create a new culture, wherein an uprooted humanity addressed ancient needs in totally new settings and contexts, with totally new solutions, tools and environments. The movement saw ‘design’ as the critical intervention, not as an end in itself. They were driven to design a new culture that addressed a plethora of new behaviors. They set out on a mission to do this through a series of new design interventions.

The modernists also realized that ‘machine art,’ and indeed affordable art, had to be simple, minimalist and rationally produced through logical design. In this realization lay the genesis of the

search for a new aesthetic, mentioned above, where architects like Mies van der Rohe proclaimed 'Less is More'. Several decades earlier in 1896 another new aesthetic had emerged from America, through the voice of Louis Sullivan, who asserted that 'Form Follows Function'. The steel and glass structures, the horizontal lines, and the appropriate use of motifs established the 'Chicago School' of modern architecture. Sullivan's student, Frank Lloyd Wright used Sullivan's slogan as his motto. The 1910 publication of Wright's work in Europe invigorated Berlage, who directed the young Le Corbusier in 1912 to study it.

Thus, the Modern movement collected a band of like-minded revolutionaries who knew their history and where they stood in the light of history. They saw the light and they grabbed the moment.

Yet, we have lost sight of history and been blinded by false and frivolous fashions. That is why I have come here to Chandigarh today.

- I have come here to ask you, young architects, **to Grab the Flame of Modernism and to Run.**
- I have come to ask you, sincere teachers of architecture, to inculcate into the teaching of architecture an understanding of history, knowledge of social change and the meaning of aesthetics.
- I call upon all of you to stop the monster of effetism; stop making iconic blunders in the name of design and get back on track with rational and logical design processes.

We have the huge challenge of a mushrooming urban society growing right in front of our eyes and we are sleeping. Out of this sea change I wish to isolate two facts:

First, modernism, or the modern architecture that Le Corbusier championed, is primarily concerned with humanist values in enabling an agrarian society to adapt to an urban environment; modern architecture has primarily addressed itself to alleviating human suffering in overcrowded conditions with no educational, recreational, hygiene or health facilities that ensure a minimally acceptable milieu for human living. It is focused on the creation of a New Man through a New Culture. Modern architecture has also championed appropriate technology and a new aesthetic. That is what we see, but that is not what lies under the skin. What lies under the skin is an agenda, a set of values and a movement. What lies under the skin is the intention to create a new, urban, industrialized society.

Second, I wish to say that Le Corbusier was part of a movement which saw the potential of taking away the drudgery of mass-production and the evils of urbanization and turning them into a tool for the good. He realized that a new culture and a new civilization must be created around a new technology, a new aesthetics and a new social reality.

The idea that industry can mass-produce everyday necessities at a hundredth of the cost of hand-tooled elite artifacts, and thereby bring a better life to more people, fired the Weimar School of Art, the Werkbund Movement, the Bauhaus, the Art Nouveau Movement, the Modern Architecture Research Society (MARS), the CIAM, the Metabolist Group and Team 10. It molded the thoughts of Charles Eames and created the foundation for the National Institute of Design in Ahmedabad.

Moreover, the idea that design is a rational process grabbed the modernists. They saw a link between the stating of a problem; the stating of performance criteria; the creation of sketch options; the evaluation of those options and making rational design decisions through drawing and modeling simulations as a correct process. The later modernists, through Team 10, Harvard's Graduate School of Design, and the Metabolist Group, promoted the need for better contextual understanding and a

more specific focus on the users of buildings and artifacts.

Le Corbusier's legacy to us lies in his heroic efforts in writing, in painting, in furniture design, in designing buildings and in planning cities. His gift lies in networking, in coordinating and in organizing large groups of people in a manner that touched us all. His intention was the search for a New Culture that would generate a New Man.

Le Corbusier was neither an architect nor a painter, nor a city planner. He had no formal training in any of these disciplines. He was a modern man who saw the need to design a new culture and a new society. Design was his tool and his medium of change. More than anyone, he threw out effeteism. He replaced romanticism with objective reality as his life's narrative. More than anyone else he saw the dangers of nostalgic and romantic aesthetics as the Trojan Horse of reactionaries. Effete design is the thin edge driven by crass, self-serving minds to infiltrate the search for a new society. For Le Corbusier effete art and architecture were like a cancerous growth destroying society. I urge you, young architects, to pick up the gauntlet that he has thrown to you. Reject effeteism, reject stupid iconic designs, start thinking and be rational. Have a design process and follow it.

We must tear ourselves away from the image of Le Corbusier as being a stand-alone, one-of-a-kind man. This Great Man Theory has deflected our attention from his true worth with a false role model. It has led young architects to think that becoming famous is the goal of architects.

The yearning to become famous has lured Indian architects to cheating and to copying monumental, ugly stunts created by megalomaniacs in the West. A mercantile architecture driven by builders and developers has replaced the search for urban solutions and the creation of a relevant urban aesthetic. This trend takes one back to the International Exhibition in Chicago in 1893, when it destroyed the seeds of the modern movement in America, known as the Chicago School of architecture. Recovery from this devastating blow took thirty years of reconstruction and a new European leadership to move forward again.

In the second decade of the new millennium we find ourselves back in the 1893 conundrum. Doing something 'different', something 'unusual', something 'spectacular' has replaced our search for an appropriate aesthetic, appropriate technology and dreams of a New Society. This is the tragedy of our times and of our profession. Le Corbusier epitomized the heroic characteristics of his age. He exemplified courage and daring. He used publicity artfully to push the modern agenda, not his own image.

We must see Le Corbusier as a simple man who was part of a movement involving a large circle of devoted workers. We must see ourselves as his fellow workers. We must join the movement; get back to the basics of our role and our life's work. We have forsaken his mission, his values and work. Let us pick up the threads where he left off. I call upon you, young architects of India, to make

your own beginning and revive the spirit of the Modern Movement. Give purpose to yourselves and meaning to your work. Walk in the true footsteps of Le Corbusier.

(Le Corbusier Memorial lecture at CCA, Chandigarh, Saturday, October 10, 2009; published by Architecture+Design in its December 2010 issue)

By placing Le Corbusier on a pedestal we are separating his quest from our own; by idealizing him as a maverick and an oddity, we are distancing ourselves from his reality; by making him the one-off case, we are lowering his and our own place in history; we are estranging his duties and ours.

The Principles of Intelligent Urbanism

The Principles of Intelligent Urbanism (PIU) are a set of axioms, laying down a value-based framework, within which participatory planning can proceed. After being reviewed and amended by stakeholders, the PIU act as a consensual charter based on which actual planning decisions can be constructively debated, evaluated and confirmed. The PIU emerged from several decades of my urban planning practice in the Indian subcontinent and Southeast Asia. They formed the foundation upon which the new capital plan for Bhutan was based. The ten Principles of Intelligent Urbanism are:

Principle One:

A Balance with Nature emphasizes the distinction between utilizing resources and exploiting them. It focuses on a threshold which, if crossed, would lead to deforestation, soil erosion, aquifer deterioration, silting, and flooding. These, operating together in urban systems, would destroy life support systems. The principle promotes environmental assessment of ecosystems to identify fragile zones, threatened natural systems and habitats that can be enhanced through conservation, density, land use and open space planning. This principle promotes the employment of ‘green practices’ in city making.

Principle Two:

A Balance with Tradition integrates plan interventions with existing cultural assets, respecting traditional patterns and precedents of style. It respects heritage precincts and historical assets that weave the past and the future of cities into a continuity of values.



Principle Three:

Appropriate Technology promotes materials, building techniques, infrastructural systems and construction management that are consistent with people’s capacities, geo-climatic conditions, local resources, and suitable capital investments. The PIU propose integrating interfaces between the physical spread of urban utilities and services, watershed catchments, urban administrative wards and electoral constituent boundaries.

Principle Four:

Conviviality encourages social interaction through public domains, in a hierarchy of places, devised for personal reflection, for engaging friendship and romance, for household, neighborhood, community and civic life. It promotes the protection, enhancement and creation of ‘open public spaces’ which are accessible to all.

Principle Five:

Efficiency promotes a balance between the consumption of urban resources like energy, time and finance, with planned achievements in comfort, safety, security, access, tenure and hygiene levels. It encourages optimum sharing of land, roads, facilities and infrastructural networks to reduce per household costs, increasing affordability and civic viability. The principle finds a link between access to basic, ‘user end’, infrastructure services and the per capita life cycle costs of these services.

Principle Six:

Human Scale encourages ground level, pedestrian-oriented urban arrangements based on anthropometric dimensions as opposed to machine-determined scales. It advocates mixed use Urban Villages, with most day-to-day needs accessible at walking distance, as opposed to mono-functional blocks and zones linked by motorways and surrounded by parking lots. The principle prioritizes people over machines and pedestrians over the automobile.

Principle Seven:

Opportunity Matrix enriches the city as a vehicle for personal, social, and economic development, through access to a range of organizations, services and facilities, providing a variety of opportunities for education, recreation, employment, business, mobility, shelter, health, safety and other basic needs. The principle views human settlements as the generators of learning experiences promoting enhanced knowledge, skills and sensitivities.

Principle Eight:

Regional Integration envisions the city as an organic part of a larger environmental, economic, social and cultural geographic system, which is essential for its future sustainability. The principle sees economies of scale in clustering activities, services and amenities into nodal hierarchies, with basic services being supported by more specialized centers, which are in turn supported by still more focused services and facilities at more central locations.

Principle Nine:

Balanced Movement promotes integrated transport systems composed of pedestrian paths, cycle lanes, express bus lanes, light rail corridors and other vehicular channels. The modal split nodes between these systems become the public domains around which are clustered high density, specialized urban hubs and pedestrian enclaves.

Principle Ten:

Institutional Integrity recognizes that good practices inherent in considered principles can only be realized through the emplacement of accountable, transparent, competent and participatory local governance. It recognizes that such governance is founded on appropriate data bases, on due entitlements, on civic responsibilities and duties. The PIU further a range of facilitative and promotive urban development management practices and tools to achieve intelligent urban practices, systems and forms.

(Lecture delivered at the World Society of Ekistics Symposium in Berlin, October 2001)

The Challenge and the Response

Those from the early batches of the School of Architecture, Ahmedabad, will recall the guiding force of education when the school was set up in 1962. Later, the School of Architecture morphed into the Centre for Environmental Planning and Technology (CEPT), under which the School of Planning was set up in 1971 – a process in which I played a key role. In 1998, while delivering the School of Planning Silver Jubilee Lecture, I revisited those heady days and the values that drove us. To a great extent the School of Planning echoed the spirit of 1960s. Responding to a question about planning education in the twenty-first century, I suggested that we ‘look into the past to find the future’, for the context of 1972 was still valid. I repeat that lecture here to remind us all about the goals of our education and the context of our professional challenges.

In 1971, when Doshi asked me to give up my teaching job at Harvard and return to India to start the School of Planning, my decision was instantaneous. That was in the early seventies and we (Balkrishna Doshi, Yoginder Alagh and I) were driven by a vision. In our different ways we all saw a new India emerging, and huge possibilities for not only India but for mankind. We were not alone. India was just one of the main laboratories.

The New Model

Not only would we address the stresses of poverty, inequality and the deterioration of the environment, but we saw a chance for a different model than China and a different model than what was then the Soviet Union. There were aspects of them we liked, but none of us could buy the entire system. We saw a model which was



- low energy;
- community based;
- founded in democratic, micro-level institutions;
- based on the full economic engagement of local populations using local resources linked to industries which supported an urban-rural continuum; and
- a mix of public sector actions and private sector activities, with the public sector providing social and economic infrastructure (particularly networks) and basic services, while the private sector mobilized production.

We wanted to avoid a consumer-based economy based on credit and financial investments as opposed to a sustainable and self-sufficient pattern of growth. We were wary of the communalism that had reared its ugly head in Ahmedabad in the 1969 riots, teaching politicians that fear speaks louder than ideas. Most of all we wanted to avoid the institutionalized ‘uniform’ that the West and China were pushing themselves into. We saw India as a model based on a plurality of lifestyles, cultures and values all thriving in a hierarchy of democratic communities. We hoped India would avoid

becoming a military-industrial complex, with one ideology pushing people into isolated, mechanical lives, dependent on high incomes and huge emergency expenditures.

These issues seem even more relevant today than they did forty years ago.

The Scenario

Starting a new institute meant being a key player in a change of tide. It meant throwing in one's lot with success or failure. More importantly, we were all full of hope. We were sure that with directed intelligence and leadership a logical set of events could be put in motion, which would transform India.

Let me brief you on the scenario as we saw it. First, the crises:

1. More than half the nation was under stress in dire poverty. Survival was the key issue. Basic, minimum needs were not being fulfilled. A vulnerable group was emerging within the poor, and it was feared that the poor class itself was growing as a proportion of the total population. Existing everyday services and facilities were getting further stressed as more and more people became dependent on them.
2. About eighty-five percent of the people lived in rural areas, depending on cultivation for their food and any cashable surplus. The economic infrastructure required to expand and intensify production was limited, putting a stress on the man-land ratio. The population was growing. Every day, each hectare of land had to produce more food for more people.
3. The environmental resource base of these people was rapidly deteriorating, stressing even the existing, limited support systems. Water, biomass, animal life, soil and human resources were all under stress. All were inefficiently managed.
4. All of these stresses were compounded by subsets of stresses: high infant mortality rates; low literacy rates; poor health; declining levels of functional education; deteriorating shelter conditions; increased debt; inadequate transport; shortage of potable water; and more.
5. Cities, which were the centers of production, were under stress. Serviced land, roads, potable water, sewerage, power, communications and other essential economic infrastructure was inadequate and overstressed. Overcrowded, unhygienic slums were growing and becoming the major mode of urban living. If cities were to become the engines of growth, they would also have to receive a proportionate share of investment in their infrastructure.
6. The professionals required to assess, plan for, and manage the resolution of these crises did not exist.
7. The necessary financial, facilitating, empowering and planning institutions did not exist.

Stating the Problems

Even though our core group at the School of Planning was overwhelmed by these immense stresses, we felt confident that for each stress, a set of well framed problems could be stated. A problem was seen by us as 'a question to be answered'. For example, we could take the above stresses and turn them into problem statements:

- How to reduce infant mortality rates from one hundred and sixty deaths to one hundred deaths per

thousand in five years?

- How to increase rice production yields from ten quintals to fifteen quintals per hectare in five years?
- How to increase irrigation from eleven percent to fifteen percent of cultivated land in five years?
... and so on.

I found it important to distinguish between ‘stresses’ and ‘problems’. People were saying ‘health is a problem’. I said, ‘health is good; the problem is how to reduce malaria’.

Problem Set		Objective Set	
A-1	How to increase rice productivity per hectare?	A-a	Irrigate 1000 hectares
A-2	How to increase wheat productivity per hectare?	A-b	Supply X tons of fertilizers
A-3	How to increase cooking oil productivity per hectare?	A-c	Supply X bags of hybrid seeds
A-4	How to increase access to green leafy vegetables?	A-d	Provide E pesticide
A-5	How to invest in seeds and fertilizers before harvest?	A-e	Supply Y Rupees credit
A-6	How to increase functional ability to grow more food?	A-f	Teach methods in 250 one-hour sessions

Integral Chains

We were youngsters of the modern age. We were sure that if a problem was stated clearly, objectives could be set using available, appropriate technologies to find solutions. In our back pocket we had **integral chains**. They looked something like this:

Each objective was matched with a set of desired outputs and required inputs. After laying out these integral chains carefully, we would find intersections between stresses, problems and solutions. We would find that the same inputs and outputs were commonly needed for answering three, seven or even ten questions. We also noted that some outcomes were pre-conditions for many inputs, and that outcomes with the most dependent inputs were the priority, or ‘lead activities’. Thus, **integrated** planning arose. Some key actions would resolve or **temper** two to three stresses at once. A kind of Functional Linkage Matrix emerged in which key objectives, with inputs and outputs, became clear. For example, a road could link markets (increasing capital accumulation through the sale of surpluses); could link a health care referral system; could provide access to education facilities; could link milk production to milk storage, processing and markets; could provide access to a range of economic services and so on.

We were drawing on Central Place Theory from geography; Input-Output concepts from economics; Community Development methods from social work; Village Planning and Low-cost Housing from architecture; Production Concepts from agriculture; Statutory Reforms from law and similar other ideas from across various disciplines.

These actions were always linked to spatial areas. The first integrated spatial plans took form (District, Block and Regional plans integrating a number of previously isolated sectors). A kind of trilogy emerged involving (a) micro-level planning (b) multi-level planning dovetailing local plans into a hierarchy of meso- and macro-level plans; and (c) decentralized, participatory decision making which would link planning, administration and resource mobilization at different levels.

Ideology

I say all of this to underscore the fact that we lived in an Age of Reason with a strong ideological foundation. In one way or another we believed the State could be reformed to play a facilitative role. It would provide the **incentives**, the essential **economic infrastructure** and the **constructive regulations**. In fact these three areas constituted our own vision of public policy as the focal concern of development planners.

By ideology, I do not mean we were Marxists or capitalists, though we were alternatively accused of being both. But we were driven by the beliefs that:

1. To transform, India must focus limited financial resources on key economic infrastructure, evolving productive systems, a 'social net' and human resource mobilization.
2. Left to themselves, individual entrepreneurs would prioritize their own profit, ignoring the fiscal transfers required to develop economic and social infrastructure or to enhance human resources.
3. The state, guided by a democratic government and managed by a capable administration could effect a transformation based on macro-, meso- and participatory micro-level plans.
4. All citizens had a right to fulfill their basic needs, to have equal opportunity, to realize their own potentials, and to develop their unique opportunities.
5. The prime vehicle to achieve these ends was multi-level and interlocking plans, with the people participating at various community levels through self-governed and self-managed development institutions.

Professionals

So we were not only burdened with problems, we were also confident that we knew some of the secrets of success. But most of all we knew that this was not the job of one or two people. We needed a new professional army to tackle this crisis on a war footing. That is where the idea of the School of Planning came in (and later the Centre for Development Studies and Activities).

Let me be clear what we meant by a 'professional'. We meant a skilled, knowledgeable and sensitive person who **professed values**. I can list some of these values:

1. Intellectual honesty; applying logic (as opposed to personal gain, biases, irrational beliefs and prejudices) to thinking and decision making processes.
2. A special commitment to the vulnerable within the poor; i.e., a commitment to poverty eradication.
3. A commitment to environment: strengthening the earth's carrying capacity.
4. Creating equality in the way institutions handle a variety of people.

5. Promoting democratic governance, with transparency, accountability and effectiveness in administration.
6. Being competent: having skills, knowledge and sensitivity to enact the above.
7. Putting society before self; understanding that the 'social contract' with destiny must overshadow the 'business contract' for personal gain.
8. Always being a student: (a) learning by doing in the contexts for which we plan; (b) working in teams which nurture a variety of views and approaches.

To all of us, these professional values transcended ideologies. They were valid in capitalism, communism and socialism alike. If I look at the past, at the present and toward the future I still believe these values define our profession.

These values were espoused by the faculty and debated by the students. In fact it was the second batch of our students who initiated the National Organization of Students of Planning (NOSPLAN) with the first formal meeting in Ahmedabad in 1974. The seed idea of NOSPLAN was to frame a new value system for planners.

Ground Rules

The most interesting thing about our team, our school and our effort was our recognition of our limitations. We were basically people from architecture and economics, sociology and social work. We had been broadly based in the social sciences, or were ready to learn. We knew development was neither an exercise in economics nor one in town planning. In fact we were quick to lay down some ground rules, lest we became too romantic about our potential. These were:

- (a) Development is not an economic growth or town planning process. It is primarily a political and social process.
- (b) The political and social changes required for development are likely to be chaotic in nature.
- (c) Development would not be a process that necessarily bred social contentment.
- (d) Our efforts would not necessarily succeed.
- (e) The price of material development could be the loss of freedom and democracy.
- (f) The unexpected and counterintuitive aspects of development are rooted in the development process itself.
- (g) Development, change and growth are an inescapable reality which must be faced and not be sentimentalized.
- (h) Without effective mass education none of this would be possible.

Looking at the above assumptions we quickly gave up a search for paradise on earth; we threw out **comprehensive** planning, or perfect ends. We realized that any workable plan would have to be **disjointed** and **incremental**. At best a kind of **main structure** of key economic and social infrastructure could be laid out, around which incremental actions could be taken as and when the funds and the will arose.

Most of all, 'development processes' were fundamental, and the objects or products of

development were secondary. I started saying, ‘Development is a verb, not a noun.’

What clearly emerged was that development planning was not only multidisciplinary, it was cross-disciplinary: an architect must understand economics; an economist must know social change; a social worker must comprehend political processes.

For this reason, all our work – conceptualizing the future path, the working of our own group, the setting of a curriculum – comprised team efforts. There were always economists, architects, sociologists, political scientists, and many others in all the meetings and in our final structuring of the student body. This reflected our view that planning is cross-disciplinary. By the time we had built our little temple we realized that we had unknowingly created an outcaste – the temple of the untouchables.

- All the other schools were of ‘Town Planning’.
- All the departments in universities were for singular disciplines; you could not move from a BA in one to an MA in another.
- Planning courses were restricted to civil engineers, geographers and architects; planning was conceived as a physical planning exercise.
- Jobs that our students would apply for upon graduation would straitjacket them. In fact, the new professional we had defined generated new jobs. But we did not know this back in 1971.

What Happened?

We have always seen planners as people with a broad vision. They are **conceptualizers**. They have a strong grounding in social, economic, political and spatial issues. They know how to prepare plans. In addition there has been a trend for planners to specialize so that they become experts. This is good as they can provide expert inputs, for example, in transport plans, or shelter strategies, or urban or rural financial investment plans.

Yet, it becomes dangerous when planners fall into the slot of technocrats implementing broad concepts set by ‘higher ups’. They stop applying their minds and become ‘yes men’. They learn to please and connive rather than to profess values. Our planners have often merely carried out decisions like mechanics fixing automobiles designed by others. The planner must provide **vision**.

Where Do We Go From Here?

Major academic decisions are often made for the strangest of ‘economic’ reasons. The length of the postgraduate program in planning at CEPT was reduced from four to three semesters, to provide higher monthly stipends in the same budget! It was not done for academic reasons. **(Note: This statement was true when I originally made it in 1998. I am happy to learn that since then, under changed funding circumstances, the course has been restored to the original four semesters.)**

The training of a competent planner actually demands a full year’s grounding in situation analysis for identifying strategies, for providing plan options with social and economic costs and benefits. Planners must have a clear understanding of the users, their stresses, possible problem statements and the issues. The second year would allow students to focus on specific micro-level plans and project formulation.

A graduating student should be able to prepare a crisp Project Document including an outline of

Stresses, Issues, Problems, Constraints and Potential Analysis, Goals, Objectives, Inputs and Outputs, Budgets and Cash Flows. These specific methods would focus on a number of specialization areas, but would be common in their applicability.

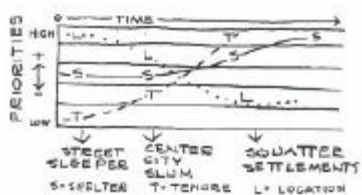
Yes, we must be able to provide experts focused inputs. But these must fit into greater visions.

(School of Planning Silver Jubilee Lecture, CEPT, Ahmedabad, February 1998)

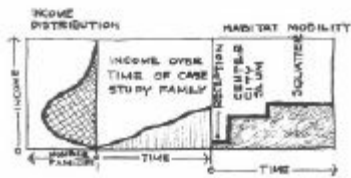
Models of Habitat Mobility in Transitional Economies

Way back in the 1960s John Turner developed a model of typical migrants to cities, based on Lima, Peru as a case study. The model used the variables of time (and an assumed socio-economic improvement over time), security of shelter (tenure), level of development of shelter, social status and the interrelations between these variables. In his model, Turner compared the priorities that the migrants assigned to location, level of development of the shelter and tenure and showed that the priorities which determine dwelling environment change as the situation of the urban dweller changes. This was done by tracing the priorities as they shift over time (as in Figure One). As the migrant's needs and means changed so did his priorities, which led him to change his dwelling environment. Thus, the migrant at first stayed with relatives as he could not afford rent, or slept on the street for ease of access to casual jobs nearby. Later, he rented space in a city-center slum to provide economical shelter for his young family while retaining work accessibility. Finally he squatted, in an organized take-over of land where enhanced tenure meant security and secure investment in home upgradation, improving his social status. With a slowly increasing income and with relative job security these need patterns could evolve.

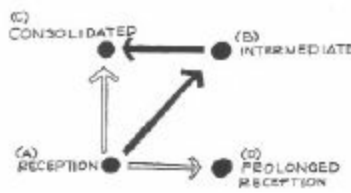
Turner's model, based on the situation in Lima, shows a direct response to the relevant human needs and household means. But the socio-economic movement Turner traced from a reception condition to the city center slum and finally to a fairly well planned squatter settlement in which consolidation takes place, is only one of several alternative patterns of habitat-mobility. In fact the 'Lima Model' is just one option.



This model does not represent what happens in South and Southeast Asia (and as conditions change in Latin America it also may not represent what will happen there). What is important about Turner's Lima model is that it teaches us to think of 'dwelling environments' as needs that change as a migrant grows from a bachelor to a young householder, to a family man, to a father who wants to make a good marriage for his children. In this scenario (see Figure One) his need for



status shifts from a minor to a major concern, and his 'level of development of shelter' bestows this status. Similarly, as a young man he is least concerned about the security of his shelter tenure. He can sleep on the street. But as he grows older to become a senior citizen the need for shelter security increases to become a necessity. Figure Two illustrates the total number of households in different income groups and their habitat mobility as their income increases.



Going beyond the Turner model, Figure Three designates four possible situations in which the poor urban dweller may live. These situations, familiar in the South and Southeast Asian context, are Reception, Intermediate, Consolidated, and Prolonged Reception. It should be noted that these are ‘situations’ in which the urban poor live. They are not environments. While an environment may be representative of a certain socio-economic situation, no single environment will adequately explain the general situation. Thus, the environments mentioned below indicate possible physical by-products of each of the socio-economic situations under discussion.

Reception is the situation experienced by the new migrant to the urban context. In this situation the newcomer – usually unskilled for urban jobs – needs maximum access to casual jobs and possibly proximity to a main market where he can obtain leftover food at the end of each market day. He has no money or time to spend on transportation and his need for shelter is minimal because he usually arrives without a family and carries what he owns on his back. Any independent shelter would in fact constitute a serious drain on his limited budget. The environment most amenable to this condition is the street, where in warmer climates the migrant may spread out a blanket, rags or newspapers to bed down for the night. Alternatively, he may camp in a railway station, where he can pass off as a waiting passenger. The more fortunate have family members who will allow them to camp in their homes while trying to find regular work. Whatever the environment, it is usually temporary and the situation changes fairly rapidly (within the first six months) to a more ‘urban’ existence.

Figure One: Lima Model

Figure Two: Optimal Model

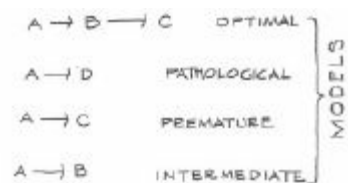
Figure Three: Situations

Prolonged Reception is the situation in which a migrant’s condition remains the same over a long period of time. It becomes a lifestyle rather than a temporary solution. It might be called a situation in which the Reception condition has become chronic. Generally it indicates a weak connection with the urban context, possibly a result of failure to integrate with the urban economy and social fabric. Usually the persons involved are engaged in casual manual labor, or in very marginal retail businesses. Although street sleeping is still the norm, it may have acquired a more institutionalized form. For example, in Kolkata shop-owners give certain people the right to sleep on the thresholds of their businesses against an agreement that the night-dweller will keep an eye on the place. In Ahmedabad and other cities in Asia, small lean-to sheds may be built, large enough only to store a small amount of cooking and sleeping items. At night a habitat environment is created by rearranging this equipment. Often whole families may be found in the Prolonged Reception situation, unlike in the Reception situation where only one member of a family has come to the city.

In the **Intermediate** situation, the migrant has become a part of the city. Very likely he either has a job with a regular income or he has established contacts with outlets for casual labor which provide him with enough assured income that he can rent shelter, even though the rent must be very low. Location is still of primary importance because he cannot afford transportation to his job, and his position with regard to job security has not advanced to the point that he can afford to be far from the places offering opportunities of casual labor. His family lives with him and they too may participate in small-scale economic activities. The environment most representative of this situation is the city-

center slum.

The **Consolidated** situation has been well described by John Turner and William Mangin. Both Lima style squatter settlements and the established hutments of India represent this situation, though they are quite different in their physical structure and organization. It is a condition in which the migrant has assessed his position in the urban environment and his possibility of making a permanent place there. His job and income situation have become relatively stable and his income is adequate for him to afford some transportation. The family size is normally large and some children are old enough to put pressure on the parents to acquire certain status symbols. By this time, the erstwhile migrant can visualize his security needs within the urban context and can explore ways to assure it.



The actual physical structuring of these situations depends on the timing of the migrant's movement through them. The environments will be different according to the timing and to the path taken by different segments of the migrant population through the possible situations offered by a given society. Therefore it is important to analyze the alternative models of movement between various situations and discover the types of environment which result from case study information. Figure Four is a graphical reordering of the situations depicted in Figure Three, showing the importance of sequence and timing.

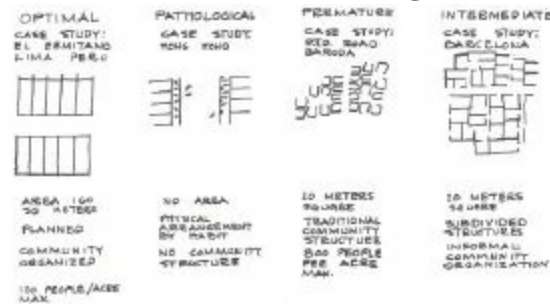
The Turner model is a movement from Reception to Intermediate to Consolidation. It could be called an optimal model because all systems respond to the needs and means of the situations, and the environment which results has great potential in terms of the physical needs of a growing, transitional city. As the situation changes, or is perceived to be changing, there are adequate resources available to make a change in environment. In this model (and in the Lima situation presented by Turner) certain conditions are assumed to exist, such as available open land, transportation at reasonably low fares, and a low level of police protection for private and municipal land. Moreover, it assumes that conditions in the city-center slum are satisfying enough to prevent people from undertaking a premature unplanned or haphazard move to the consolidated areas. Thus, Turner's optimal model assumes a great deal of planning in an institutional environment where land is up for grabs.

Figure Four: Models of Mobility

In Asia, and particularly in the settlement patterns I have studied in India, a model which could be indicated as **Premature Consolidation** is more relevant. In this model the migrant moves directly from Reception to Premature Consolidation. The resulting environment is quite different from the optimal model. The most noticeable difference in the resultant environments is that the premature model has no capacity for growth and change. Accessibility and utilities may only be improved by destroying some of the dwelling units, and plots are small and usually undefined. On the other hand the environment which results from the optimal consolidation model of an organized community shows a good structure in terms of the layout of streets and passages, with flexibility for the future. Drainage and utilities can easily be added within the network, and the size and configuration of the lots allow for growth over time and continuous upgrading. Low land values seem essential to this model, and forty years down the line since Turner published his findings, such low-value land no longer exists in most Indian cities that can offer livelihood opportunities to poor in-migrants.

Causes of Premature Consolidation include the highly unsatisfactory conditions in the city-center

slum; a desire to live with relatives and traditional social groups; good location of areas coming under consolidation; and lack of funds for rent. In Figure Five, environments are indicated that match



models of habitat mobility.

Figure Five: Situations and Environments

It is where the city-center slum is already overcrowded and rents are relatively high that Premature Consolidation takes place. In India, and other areas of the world where family, caste or tribal affiliations determine dwelling location for the poor, the Premature Consolidation model is very prevalent. In these cases a migrant will move as quickly as possible to the location of his caste group. If his father or his uncle is already in the city, he will certainly want to live next to them without considering an alternative better environment. This results in the construction of new hutments in areas where the land area is already overbuilt. In many cases a community of one caste is bordered by other caste communities, forcing it to increase its density and become overcrowded. In such areas community organization tends to be along traditional caste lines and oriented more to the ceremonies of life than to solving the pressing problems of an urban existence. Ties with the municipal and national political structure are paternalistic; quite unlike the Lima squatters who are highly politicized and who name their well-planned invasions and the resultant new communities after politicians whose favor they seek.

A third model includes people who remain in the Prolonged Reception situation. Although most common in the primate cities of Asia, it is certainly a worldwide phenomenon found in cities where a saturated labor market, causing underemployment and unemployment, permits many migrants only the most minimal personal income and where (the peripheries of growing cities in India and China are good examples) the available land area for building is already covered with structures, seemingly closing the option of moving on to the Intermediate phase. In addition to instances in which progress has become difficult there are constituent members in this model who can be called 'drop-outs'. These are persons who may at a previous time have achieved higher status and environmental conditions but, due to some misfortune, were forced to drop into this pathological model. While

misfortune may have thrust a person, or a household, into this model, the situation tends to prohibit them from moving on to a better lifestyle.

The Intermediate Model represents a movement from Reception to Intermediate that has become stabilized. For many inhabitants this can be a desired condition. The Intermediate situation may provide a level of shelter and access to employment not readily available in other environments. In a few cases, such as small families without children, it is in the most desirable location. Others who have had a long urban history may prefer the city center slum because it offers close family and friendship connections, which have economic meaning. They may consider that the security offered by home ownership in a consolidated settlement would not adequately replace the security offered in the existing close personal relationships. Referring again to India, where security is perceived to be in the village or 'native place,' and life in the city is perceived as a temporary condition (even though it rarely is), a movement to consolidate has much less meaning than in the optimal model, where urban inhabitants are building security for themselves within the urban context. The Indian migrant in the intermediate condition will send any surplus earnings to his family in the village, where his status and his security are founded, rather than move on to establish a new permanent home in the city. He will often return to the village to marry and believes he will go to the village in old age, and even his children born in the city know the village as their native place. Conditions which encourage stabilization of the Intermediate model are a large supply of slum dwellings in the city-center; a strong police force inhibiting consolidation; a lack of good sites for such consolidation; traditional sources of security other than the dwelling place; a lack of a tradition of planning; and a status structure unrelated to the physical development of shelter.

The four models (including the one postulated by Turner) presented here represent the predominant patterns of migrants' habitat mobility in many urbanizing centers of transitional economies. It seems possible to correlate environments with such models much more easily than to correlate them with situations, and further study of the determinants of these models may be helpful both in determining housing policies in such areas and in a better understanding of pathological environments and of what constitutes a 'slum'.

(Slightly modified and updated from the version originally published in EKISTICS, February 1970)

Visions and Devices of Urban Form

Urban form and patterns have been the subject of urbanism since the ancient *shastras* provided guidelines for laying out a city. Images of what a city should be have ranged from the practical to the romantic. The vision of Arcadia is a thread running through history, back to the Greek gymnasia in periurban gardens, away from the polluted thinking and degraded form of inner cities. The idea of ‘the garden’ as an escape from worldly chaos can be seen in imperial Roman estates, Mughal Gardens, and in royal Chinese urban precincts. The concept tempers the vast gardens which hinted at an urban design schema in the palatial royal estates at Versailles, Karlsruhe and elsewhere. The urban fringe has been the subject of both ridicule and nostalgia in painting, literature and urban theory.

Washington, DC links royal gardens to urban planning. The eighty-eight cantonments and hill stations created by 1875 in India were low density, open town plans, as was Lutyens’ Capitol plan at New Delhi.

In the nineteenth century the Garden Cities Movement attempted to draw gardens into the city and to modulate a neat transition to the rural countryside. Urban planning theory was actually a theory of escape – escape from the city and the realities of contemporary urban life. Frank Lloyd Wright tried to even-grain the transition of urban fabric to almost nil, reflecting an industrialized people’s fascination with the countryside as a place for escaping from reality. Broad acre City, as Wright fancied his dream, was an automobile-dependent, spread-out fabric of orchard plots, peopled by households very much like the multi-millionaires who were his patrons. Ayn Rand hallmarked this individualistic philosophy in her now classic demand, “Never say we, always say I”.



Le Corbusier’s Radiant City lifts the city on stilts above an undisturbed rural landscape that flows under the detached urban form. Thus the European abstraction of nature is antagonistic. It sees humans and nature in opposition, to be separated. The American construct integrates man and nature. At a more mundane level, Green Belts create a counter-intuitive no-man’s-land, where a jumble of illegal and unrecognized development occurs. Ironically these jungles of neglect are referred to as the ‘lungs’ of the city, trying to imagine an anthropopolis with its lungs hanging somewhere off on the side! Thus, in theory, law and practice, the fringe of the city is an ambivalent realm of the poetic ideal, and the reckless concrete jungle.

The modern movement, like the Modern Era, saw a marriage between a program of social objectives, technological applications and art. Early twentieth century urbanism, from Ebenezer Howard to the International Congresses of Modern Architecture (CIAM), was a statement about the social structure and the social organization of the city, equally as it was about efficiency and city form as promoted through the Athens Charter (written 1932-33, mainly by Le Corbusier, and published 1943-44). There were considerations for social justice and for creating social opportunities. The planning principles of the CIAM were largely misused in postwar Europe to create numbing, bland milieus of boring landscapes of gridiron streets and International Style look-alike structures. In the rush to rebuild cities, poetry and lyricism were neglected. As cities expanded, axioms were applied

as formulae. Theorists suddenly called upon to create new cities stumbled over the easy solution of repeating boring boxes laid out over chessboard plans. Team 10, the children of the Modernists and the parents of the Postmodernists, searched for abiding themes that would make cities work for people. The public domains and the movement stems, were among their central concerns. Those who followed them seem to have thrown out the baby with the bath water, favoring urban designs devoid of public domains, cut off from open-ended stems, shut in with 'gates' and entrance filters, lacking in any social theory of mobility, accommodation, or opportunity systems. Young urbanists were foolishly advised to go to American suburbia and to learn from its artifacts, as if it were the Florence of the twentieth century. Postmodernism ushered in the era of the big, personal, egocentric statement...as if everything were a Burj Khalifa, or the next tallest building in the world! "*Never say we, always say I*" is the motto!

Out of the void of Postmodernism, which was too individualistic and idiosyncratic to focus on something as complicated as 'the city', emerged the New Urbanists. The New Urbanism is neither 'new' nor in any sense 'urban'. Like the New Economy which follows Ayn Rand's motto, the New Urbanism is a mechanism for personal 'opportunity', created to appear as a vehicle for the public good. These isolated, inaccessible, resort-like elite communities have their roots in Seaside, Florida and in the Disney Development Company's Celebration adjacent to Disney World in Orlando. Both concoctions reflect the malaise of the late twentieth century. A kind of virtual reality replaces the actual. The Disney 'make over' mutating the New Urbanism projects into antique villages. The concept is to create stage sets (usually modeled on the mythical 'small American town') of make-believe streets and structures, much like Disneyland and Disney Worlds. The strategy is attracting buyers into the resale market. Instead of 'communities' these are created as investment opportunities. They are created as places to 'park money' and not to live in! These so-called urban communities have no jobs, no low-income housing, no workshops nor any other aspect of a diverse urban fabric. They are inhabited by white, Anglo-Saxon elites who, contrary to embracing urbanism, want to escape it. Moreover, these are gated communities with security guards at the controlled entrances. The past two decades have witnessed the demise of the 'public domain', with the rise of privately guarded, controlled and accessed spaces. Shopping centers, like in emerging economies, require a credit card as a 'visa' for entry. Economic profiling is taking over from the apartheid of racial profiling.

Urban planning, land regulation and development reflect the most civilized ambitions of any society. Therefore they also mirror the ugly realities of societies. Urban sprawl and 'strip development' plague India and the USA alike. The automobile takes command. Loans for cars are cheaper than loans for houses in most countries, and also easier to get. Roads are subsidized and petrol bills, insurance and car payments are all tax deductible! The government is paying a small minority to congest the roads and pollute the air meant for everyone. Public policy nurtures spread and sprawl. At least in India prohibitive taxes temper petrol consumption and automobile purchases. Yet, consumer loans and tax breaks make up for the difference. In any case the local petrol pump is a good place to dump black money. The automobile has had a greater impact on urban form than any other technical or policy intervention. Our policies on the automobile are our policies on the city pattern.

To a great extent urban form is a play-off between greed and regulation. In the New Economy where one is never to say 'we' and always say 'I', the plan is one to satiate desire and not to resolve

social stresses and find solutions to problems. Those who stand to benefit from a particular urban shape or form usually take on the 'burden' of responsibility to shape that form. Opportunities lie out on the fringe, where land is bought cheap, developed and sold expensive. The urban fringe is the Babylon of this new economic frontier.

A small university town, Gainesville Florida, with a population of 125,000, has an urban area slightly larger than the city of Paris! Why? The elected city and county commissioners are all land traders and developers, who personally benefit from the city services and amenities, which will spread over vast tracts of urbanizing land. They and their brothers across America are leaving their grandchildren a huge debt and an unmanageable network of roads, water supply, electrical and sewage lines.

All of the above has made city management relatively impossible, with land prices spreading horizontally out from the urban core, over an ever-urbanizing space. Artificially low FARs (Floor Area Ratios) help 'push' development out, by making habitable land more rare and costly. Land use planning stimulates a system of 'bids and prices', which directs users to compete for land in different zones. Low FARs geometrically expand the cost of laying storm and sewage drains, extending fire fighting, water supply, telephone and optic fibers. They make the maintenance of law and order near-impossible. Fleeing high prices, residential development leapfrogs from one pocket of inhabitation to the next, overstressing urban systems. Low FARs, vehicle subsidies and land use planning contribute to this sprawl and spread, where densities are too low to support efficient public transport, walkable compact communities, or even basic public health infrastructure. The Charter of the New Urbanism is a delightful document guiding planners and developers on how to plan and lay out 'good investments'. The people who will live there don't come into the picture until the deed is done. They come into the picture, along with financial institutions, as buyers and investors. These are high standard guidelines created to enhance investments.

Instead of master plans and development plans, we must start with a public Charter of Intelligent Urbanism. This would lay out a kind of 'covenant' of principles or themes around which the fixed components of the city structure can be fixed. Such a Charter is a kind of Robert's Rules of Order around which participatory debates on a consensual Structure Plan can take place. A Structure Plan includes the 'non-negotiable' parts and components of the city. These parts range from urban corridors along which future movement of people, goods, water, and waste will happen. It includes the establishment of high-density nodes and hubs that gather enough people to support public transport halts, amenities and services all at a walkable distance from compact neighborhoods. The Structure Plan includes a very basic set of regulations on uses that are compatible with one another, and on the connectors, safety and hygiene of the urban fabric nurtured.

The Structure Plan is basically the application of the Principles of Intelligent Urbanism as physical themes to a spatial arrangement determined by the geo-climatic context. Local Area Plans, such as Town Planning Schemes and private layouts, are kind of mini-master plans that plug into the Structure Plan through a participatory and transparent process of incremental and discrete planning. This 'process of planning' makes cities into vehicles for people to realize their dreams. Without such a process of policy, planning and participation, development is bound to be motivated and dominated by vested interests. It is bound to lack environmental protections, social facilitation, or the promotion of efficient infrastructure. The plans that we are producing and the cities we are creating are barriers to the people's development. These are not artifacts of public interest or in the interest of the public.

Only a handful of land traders will benefit.

There was a time when visionaries tried to ‘imagine’ the ideal city. There were times when cities were conceived as the habitats of societies and as vehicles for improving the human condition. As the New Urbanism is imported into the emerging economies it will become a device to divide society and a mechanism to transfer more wealth to those who already have it in abundance. What is needed is a more accountable, transparent and participatory form of city making. For this we require true principles of intelligent urbanism as a beginning of genuine debate. The Charter for the New Urbanism is an apocalypse.

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Five Myths which Plague Urban India

Myth One: Urban India and Rural Bharat are two conflicting worlds.

Many rural élites and urban theoreticians have seen cities as parasites on rural areas, sucking unearned profits off the backs of peasant labor. This was in fact the early colonial strategy to facilitate a wealth creation pyramid with the Presidency cities at the apex. Over two centuries, social, economic and political links integrated the forward and reverse exchanges, creating dependencies in everything – from consumer goods to urban labor force needs. The current exponential growth of the economy will accelerate dependence on an enhanced, and indeed a better-educated, labor supply.

As recently as a generation ago, most of us espoused a simple urban-versus-rural paradigm. This idea was tragically wrong. The first two Five Year Plans totally neglected cities and industrialization in favor of rural development. Other than a few mega-dams, fertilizer and steel projects and their related company townships, there was little investment in urban services and transport infrastructure. It was only in the early seventies that urban development corporations and the Housing and Urban Development Corporation (HUDCO) emerged (with HDFC entering a bit later), focused on housing finance for the middle classes. The National Housing Bank came even later. For two decades these institutions focused on meeting housing gaps, filling the missing numbers of projected housing needs instead of facilitating shelter processes and emphasizing the creation of basic infrastructure. There was never a concept of facilitating people and the private sector to solve their own problems.



In 1986 I was engaged by the Asian Development Bank to write a position paper for their Board arguing that the ADB should enter the urban sector. I had to serve them platitudes such as ‘cities are where civilization happens’. I had to tell them that cities are the economic engines which create products, jobs and a positive balance of trade. Throughout Asia (Singapore and Hong Kong excluded) there was a strong rural bias based on the myth that cities and villages are somehow at odds. To invest in urban infrastructure was wrongly thought to be taking from the villages. Or, conversely, there were those who thought installing one urban water tap, would cause ten rural families to migrate to use it! Like most paradigms which oversimplify things into binary opposites, the urban versus rural paradigm was taken, at a great cost to both urban and rural India, as a parable of truth. At the same time rural development policies were focused on economically advanced districts, leading to rural pauperization in most areas, and a ‘push’ of migrants to cities that were ill-prepared to receive them. This, too, was due to a costly myth.

In fact urban-rural regions are an integrated socio-economic mechanism that works holistically to produce raw materials, train pools of skilled workers and managers, and bring together clusters of economic inputs which create what we know as wealth. If they are integrated, each catalyzes and enriches the other. Without urban areas there are no markets for agricultural goods, and without thriving hinterlands urban products have limited markets. Economic regions have now become global in many respects – a realization that has reached us only in the last decade. For decades the urban versus rural myth starved India of adequate urban and economic infrastructure.

Myth Two: Illiterate rural masses flood cities, crowding and congesting them.

A thread of 'common wisdom' around the world is that rural-urban migration is the root cause of urban ills. This is wrong. Rural-urban migration is part of the process of industrialization, commercialization of crops, improved rural health and education, media and transport. The availability of cheap and willing labor has been one of the key factors in India's economic success. Policy makers took an ostrich-like attitude toward the role of cities in modernization and ignored the need for urban infrastructure. When the Municipal Commissioner of Mumbai declared war on illegal hutment dwellers and started demolishing their shanties and trucking them forcibly back to their villages, it soon emerged that about seventy-five percent of the civic body's own employees lived in these habitats! Had they been pushed out of the city, Mumbai would have come to a standstill. In fact all key sectors that contribute to the urban economic base would have faltered. Instead of tackling the problems of urban infrastructure, low income shelter and services, many of our urban experts pondered how to stop in-migration. In the early 1990s HUDCO set up a special Working Group to analyze ways to 'cap urban growth'!

Policy makers and planners had at hand a number of cost-effective strategic options which were only half-heartedly adopted. The most basic was the Slum Improvement Scheme, which brought basic hygiene and essential infrastructure to the 'end user.' The Busti Improvement Programme of Kolkata was an effective variation of this idea and it reached several million people. In the early 1970s we also used 'site and services' in Chennai to quickly produce about 15,000 serviced plots on which people could build their own shelters. Instead of letting local politicians colonize public lands (from which they harvested huge rents in the form of protection money and vote banks through sporadic promises) in an unorganized and chaotic manner, the government should have taken the lead and made access of the poor to shelter and urban services a key platform of their urban strategy. It was the *de facto* public policy to leave this task to party workers and solve the related problems thereafter in an *ad hoc* manner. The same combine of elected officials-colonizers discouraged the lower income groups from making fair payments for services and facilities, creating an impression that the rich subsidize the urban poor. Effective policy could have extended land tenure and essential services to these households, making them tax-paying, 'pay your own way' citizens. But it was in the interest of our system of governance to keep these households in a dependent and demanding survival state, rather than raising their status to self-supporting citizens. In fact these people were 'invisible' as illegal settlements were not even shown in most official maps, plans and documents.

Myth Three: Urban plans stifle economic development and growth.

It is a myth that urban planning is a barrier to economic development. In fact good planning is good business. The best planned cities have performed economically better than the unplanned ones.

However, the existence of urban plans in India is also largely a myth. What we label as 'plans' are not mechanisms to enable and facilitate development but mere statutory regulations. Pune city's last development plan was cleared for implementation in 1987 and it considered a population projection only up to 1997. It did not include the slum population – half the population of the city! This is criminal neglect. There were no demarcated plots and roads in the plan. It covered only the Pune Corporation limits, while the city actually spreads over nine local authorities and into the so called Metropolitan Regional Plan area. There are uncoordinated plans for the Cantonments, MIDC

areas, the Info-tech Park, two large municipal corporations, nearby ‘hill stations’, Business Parks and SEZs.

Capitalism thrives on a ‘defined, equitable and stable playing field’. Investors need to invest in land whose actual area, true ownership, land use, development control rules, and allowed FSIs are assured. Land sellers will hold on to their land if they think that every month will bring higher FSIs, relaxed standards and a better return on the sale of land. Urban planning in India has focused on colored Master Plans that designate single-function zones. Within the zones, Development Control Regulations determine the intensity of use through allowable heights, plot coverage and FSIs. Some road and trunk infrastructure projects, often called Development Plan Roads, may be shown. These two-dimensional, colored land-use plans, with reservations for limited roads, amenities and public areas, do not show plot demarcations. Within the statutory framework many changes can be made later through the discretionary powers of the Chief Minister’s opaque Sub-Committee on Urban Development. The system invites corruption.

These regulations and rules specify minimum plot sizes which are too large for the lower middle class or the urban poor to afford. Thus our miniscule efforts at urban planning marginalize about seventy percent of the population, making it impossible to acquire land for shelter. These so-called comprehensive master plans do not facilitate development. They do not allow development by small builders and individuals, and do not enhance the city’s character. In fact the eight- and eleven-storey apartment blocks where people are forced to live are based on a set of building laws created jointly by builders and pliable officials. Plans cannot be a time-bound static set of rules and regulations. I maintain that we do not have plans. These regulatory ‘non-plans’ give credence to the myth that plans hinder economic and social development.

We need Structure Plans that establish the essential networks and transport corridors, protect the environment and public assets, identify mixed use precincts and potential nodes where Urban Villages can be promoted. They demarcate areas where Local Area Plans can be prepared with the participation of the local land owners and other stakeholders. This is in fact very much in accordance with the provisions of the 74th Amendment of the Indian Constitution for the devolution of governance to the ward level. Local Area Plans should utilize land pooling where all the land is ‘banked’ momentarily and then redistributed, after deducting roads, open spaces, amenities and other common resources. Each owner then gets back their proportionate share of demarcated, documented land, not leaving any one a victim of ‘reservations’ or confused over boundaries and rightful ownership. If we utilize professional, facilitating, and enabling plans to guide urban development, urban areas would see orderly and enriching development.

Myth Four: Social legislation protects the marginal, the weak and the poor.

Under the argument that the State will protect the interests of those who are marginalized and powerless, a number of urban social statutory measures have been enacted. That they protect the weaker sections is a myth. In fact they exploit the poor and middle classes, pauperizing and marginalizing them.

Most notable of these legislative acts are the Urban Land Ceiling and Regulation Act of 1976 and the Urban Rent Control Acts. The retrogressive nature of the Town and Country Planning Acts has

already been noted above.

The Rent Control Act emerged in the 1940s when cement and steel were diverted to the war effort. At that time large cities had viable rental markets and the measure was meant to protect that market against the likely inflation in rents due to an expected hiatus in construction during the conflict. By extending these statutory measures beyond the war period, creating 'lifetime' statutory tenants and freezing rents for these groups, the government effectively killed the active rental market. Over the years a small minority of tenants were awarded highly subsidized units. For decades no new rental stock was created, as the returns on rental investment did not exist. By closing the door on the rental market the city center of Mumbai has critically deteriorated, resulting in collapses and fatalities. The meager statutory rents are inadequate to cover even the basic maintenance of these units. The civic authorities are unable to realize the needed property tax which these valuable downtown properties should be harvesting. The entire metro is the loser. The scenario is similar in other cities plagued with this 'social legislation'.

While the rental market was being squeezed at one end, the Urban Land Ceiling and Regulation Act of 1976 was throttling it at the other. This measure suddenly removed all large land parcels from the market. An arbitrary system of clearing lands added corruption and confusion. Larger plots were needed to accommodate parking, amenities and open spaces for new housing schemes. It took the relatively urbanized state of Maharashtra decades to tone down these measures, on which Gujarat and other states have effectively closed the chapter, or ameliorated their impact by modifying the rules on how new rental stock should be treated or how large plots should be treated under the acts. But we live on with this dark past.

The more recent scenario involves the release of the mill lands that spread over vast areas of the older part of Mumbai. In a generation when a continuous chain of four chief ministers have been urban developers, these lands were released into their own party members' hands. Worse, there was no plan to integrate all these lands into the city fabric and into one another. The result is that wonderful city center sites have focused on the extremely high income, high rise Mumbaikar and MNC tenant, excluding the average citizen from the new policy.

Another policy thrust is the Slum Rehabilitation Act where a developer builds multi-storied apartment blocks of 269 square feet each on forty percent of the land acquired and high income towers on the remaining sixty percent. People who lived on or near the ground, whose lives spread out and intermingled with common open spaces, now live high up in overcrowded cubbyholes. Better hygiene and more solid structures are a positive aspect of this development. But the strong-arm tactics adopted to gain community consent to participate in these schemes is a questionable expression of our democratic values. This scheme is builder-friendly and creates no new housing for the poor. Touted as a 'housing strategy', it is merely a strategy to house the very rich at the expense of a more democratic policy.

The result of these measures has been to make the supply side of the housing stock smaller and reduce the options and channels through which people can achieve housing. This increases the number of households forced to live in illegal slums, on the margins of society and dependent on criminals who control them. These measures must be immediately ended.

Myth Five: We have an army of urban development finance and development institutions to resolve urban problems.

Over the past four decades the semblance of a mature urban development institutional structure was put into place. The Housing and Urban Development Corporation and The National Housing Bank have operated for decades at the apex level. The Infrastructure Leasing and Financial Services Corporation and entities like the Maharashtra State Road Infrastructure Corporation have emerged as major players in India's urban infrastructure development. Regional urban development corporations, housing boards, infrastructure development bodies, agencies like the Delhi Metro Rail Corporation, the City and Industrial Development Corporation and many more across the nation appear to provide myriad instruments for the nation to solve its critical urban infrastructural problems.

The functions and scopes of these agencies often overlap and duplicate one another in conflicting manners. Many are dinosaurs from 'supply side' economic policy, attempting in vain to fill projected gaps. Others are part of the 'License Raj' issuing 'No Objection Certificates,' while harvesting bribes. For example, one needs to obtain seventeen NOCs in Mumbai to obtain an Occupancy Certificate. All officials are against a single-window approach as that would concentrate corruption in fewer hands! Their hierarchy and jurisdictions create confusion and unhealthy institutional competition. For years the apex bodies played the numbers game in generating housing units, neglecting urban infrastructure. Many stakeholders in the construction system have been ignored in the focus on large scale systems. For instance building material manufacturers and small housing contractors are marginalized in the focus on mega-projects. Large reservoirs and huge water supply pipelines feed potable water to the city. Yet within the city there are inadequate water taps in slums and high density tenements. The manner in which huge metropolitan transport projects have been forced down the throats of the local citizens who will pay for them is becoming legendary. These urban projects are well-tuned mechanisms for corruption, bribery and the enhancement of the black money economy. It is a myth that these agencies are addressing urban problems. They are creatures of a dysfunctional urban management system driven by ambitions of personal aggrandizement.

The sea-change in perception over the past decade is that we do need economic infrastructure and we need it fast. In this pursuit a few major projects like the Delhi Metro and the Mumbai-Pune Expressway have raised the nation's confidence level.

When a society is raised on biased attitudes and myths, it may introduce new programs and global slogans. But the underlying top-down, opaque and centralized decision making, fueled by a culture of corruption, continues to stoke the fires those myths ignited.

Poverty Alleviation in Cities

Throughout the world, preparations are afoot for the United Nations Conference on Human Settlements – HABITAT II – during 3-14 June, 1996 in Turkey. This ‘Rio-like’ meeting organized by UNCHS (Habitat), with its headquarters in Nairobi, will examine human settlements issues in the context of sustainable development; adopt a statement of principles and commitments; create a global plan of action; and review the implementation of past UN programs. It will also call for a focus on the specific needs of women in housing.

In India, too, several consultation meetings and conventions are being organized in preparation for the conference. Efforts are also on to bring about active involvement of Non-Governmental Organizations (NGOs) and Community Based Organizations (CBOs) in the preparatory processes for HABITAT II. According to me, all this is taking place when the world community itself lacks a clear statement of purpose. The UNO has a faint voice in the realm of poverty and people’s rights when it comes to basic services. I therefore feel that we in India should apply our minds to the stresses of people, rather than to physical things like houses and infrastructure, and must breathe a purposeful voice into HABITAT II, which would now pretend to be the archetypal forum, when in fact it could be christened a rudderless ship.

Since Independence, India has struggled with numerous development problems through its Five Year Plans, Community Development Programmes, Special Area Development Programmes and scarcity relief efforts. With more than eighty percent of its population dependent on agriculture over the greater part of the post-Independence period, it approached ‘urban problems’ with much hesitation. Moreover, it was felt that cities already had a disproportionate share of infrastructure and services, and that more urban investments would attract the rural poor to cities. Rural-urban migration was regarded as ‘bad’. Marxists, Gandhians and the feudal power elites alike shared this distaste for cities. Underlying this phobia was the realization that cities were the tax base of the dreamed-of socialist system. Cities were the ugly engine that pulled the beautiful train.



Regardless of how naïve such assumptions were, they did temper both government and international donor agency thinking alike. At the same time, a major proportion of urban public investment went into essential urban infrastructure, such as roads, power generation and transmission, sewerage, water reservoirs and supply networks. These investments tended to be on ‘trunk infrastructure’, not at the ‘user-end’. Housing boards created planned neighborhoods, whose costs placed their houses beyond the reach of even the lower middle class. Still, their efforts made a minuscule dent on shelter requirements.

Under World Bank funding the future populations of cities like Mumbai were projected and their water requirements were calculated. Catchments for rainfall, watersheds for dams and reservoirs, canals, holding ponds and pipes were designed. Large water purification systems were created. But the link between this huge public health infrastructure and the common urban household was missing. Stand-pipe taps in high density, often illegal, slums were missing. This lacuna was the source of

stress, the statement of the problem and the hint of solutions for all urban infrastructure issues.

The NGOs had a similar concern regarding the rural-urban issue, but used their limited efforts in urban areas to promote more participatory processes and enhanced user-end accessibility by the poor. The concern that a range of basic services do not reach the poor has emerged more recently.

Housing construction was popular with government and NGOs alike over several decades, because it was a visible output with reasonable financial accountability. Development could be 'seen'. Several fundamental barriers, however, plagued shelter inputs. In brief these were:

1. **Land Tenure:** The poorest households have illegally squatted on other people's land, creating a situation wherein qualification for loans, statutory recognition and essential services has proven very difficult, if not impossible, to provide. To some extent legislation in the mid-1990s alleviated the bottlenecks.
2. **Replicability:** Even when very minimal 'demonstration' units were affordable, hidden overheads, complex organizational inputs and special financial arrangements rendered these 'pilot projects' mere showpieces propped by hidden subsidies. These were never replicable models. At great cost a few households were subsidized. The subsidies were 'encashed' by the beneficiaries who sold the houses and pocketed the unearned increment.
3. **Ability-to-pay Constraints:** The cost of a housing unit must be estimated under the following heads: land for the shelter; share of public lands in the immediate neighborhood (roads, convenience shopping, open areas and footpaths) which support the house plot; civil works; trunk infrastructure overheads, professional design/management fees; and loan interest payments during construction itself. Not to be forgotten is 'profit'. If a builder is involved, or even a petty contractor, a minimum of fifteen percent must be assumed as a reasonable profit. These expenses add up to the principal package to be amortized over the loan repayment period. Principal and interest payments must be made monthly to pay back the housing loan. The poorest households simply cannot afford the resulting monthly installments, nor is strict monthly regularity in tune with their irregular cycle of earning.
4. **Target Groups:** In any slum a range of low income groups live together. Slums are not economically homogeneous. Programs targeted at the 'poorest of the poor' usually end up in the hands of the 'better-off' groups within the poor. Indeed affordability constraints tend to match the better-off amongst the poor with the benefits of shelter programs. A study carried out by Centre for Development Studies and Activities showed that it would cost a poor household about Rs. 600 (in 1986.) merely to do the eligibility documentation for a loan of Rs. 10,000. Thus, the actual beneficiaries tend to be the better-off households. Other than an escalation due to inflation in the amounts involved, conditions remain the same in 2010.
5. **Infiltration:** Because of the above, most shelter schemes are subsidized under almost every cost component. This makes it very attractive for a poor family to sell a subsidized shelter at near-market value, converting the subsidy into expendable cash. Thus, they escape the amortization rigor and make a windfall profit. If any of the cost components noted in (3) above are subsidized, even if profit is eliminated, that level of subsidy becomes a measure of the 'infiltration factor'. The goals and objectives of donors and local NGOs are subverted in the long run.
6. **Loan Recuperation:** The default rate increases dramatically as income levels decrease. This is

due to: (a) low ability to pay; (b) amortization systems demanding regular payments from households with irregular incomes; (c) lack of 'reach' of banking systems; (d) political instigation against making payments; and (e) inadequate, or unenforceable penalty systems. Other demands on poor households' limited incomes take priority over repayments to an unseen banking institution.

Community level banking systems, built on the Grameen Bank model, have had greater success.

'Housing', as such, never reached the poor in a significant way. If one looks at the numbers, it neither had an impact on the homeless nor did a significant process emerge that would bring shelter to the masses. It is often referred to as an 'employer of NGOs' as noted by R. I. Shah of Vikas Centre for Development, Ahmedabad. It is not the needs of the beneficiaries, but needs perceived by the donors that get realized. Perhaps it is the only 'skills' that a NGO composed of architects and engineers can provide. Worst of all, the truly poor can be displaced by housing investments. Either the shelter becomes too expensive, or default on payments makes an oustee of the householder. Even such a carefully tempered program as the Hyderabad Community Development Programme resulted in displacements; and it is always the vulnerable within the poor who are displaced.

In the mean time cities grew and in some states of India about forty percent of the population became 'urban' as early as the 1991 census. According to Amitabh Kundu, by 1994, for India as a whole, the proportion of the poor was larger within the urban population than within the rural population itself. By 1997 more jobs, in absolute numbers, were estimated to have been created in cities each year than in villages. Cropping patterns and off-farm activities in rural areas became dependent on urban markets and productive inputs, thus narrowing the conceptual difference between what was once called 'urban' and 'rural'. All settlements now clearly fell into an interlinked economic system where the neglect of some components would adversely affect the entire system. Being 'pro-rural' had become an increasingly quaint bias in an integrated economy dominated by a corporate-political constituency.

In her research during the 1970s Janice Perlman proposed that the dual economy had been a myth thirty-five years earlier. So had been much of the urban-rural divide. The rural élite invests in commercial crops and agro-industries with clear urban linkages and emerging corporate management styles. The employment absorption potential for poor workers in rural areas was approaching a threshold in the 1980s. These arguments aside, by the beginning of the 21st century, India had entered a phase of rapid urbanization. Yet in her cities one could still find the largest proportion of the poor, and indeed the deepest poverty, the country had until then known.

Let us consider for a moment who the emerging poor people are; what barriers they face in terms of access to employment and the income it generates; and, finally, what basic needs they find difficult to fulfill?

Who will be Tomorrow's Urban Poor?

By the early 1990s policy makers knew what today's scenario would be like. The urban poor now are primarily in their second and third decades of urban citizenship – the first, second and third generation urban immigrants and their children. The poor make up about half of the urban population. The truly vulnerable amongst the poor, that is those at risk of premature death, would make up about fifteen percent of the urban population using Sukhatme's nutritional analysis methods.

It is this last fragile group who concern us the most, as they will be the most vulnerable in every

sense of the word. They will lack even potable water. They will have very low skills, or no skills at all, and they will depend on their physical stamina for survival. Their awareness of the economic system will be limited to the modes of engagement of the labor contractors who employ them on a daily basis, usually in 'work gangs' or from informal labor markets. Their terms of engagement will offer no employment security, no unemployment benefits, no health benefits, no retirement funds and their incomes in real value terms will be among the world's lowest. On-the-job safety will be low and no accident insurance will protect them or their families. **India will compete in the global economy with an economic advantage underwritten by the social security deficit of the poor.**

These poorly compensated workers will be involved in unskilled construction operations, or as casual labor serving ancillary suppliers to larger, formal sector producers. In addition, a sizeable segment of the urban poor will be engaged in the tiny informal sector of the economy located within low-income settlements. There will be a link between monopsonist-exploited household production, slum-based workshops, 'illegal' small-scale industries and ancillary medium-scale operations supplying larger scale formal sector units. Examples of such operations are garment piecework, metal lathe work, small tanneries and related leatherwork. Traditional operations like *beedi* rolling will also be located in slums. More than fifty percent of slum houses will, in fact, shelter some type of productive activity, feeding the global economy.

These urban poor will enter the work force at ten years of age, either as helpers to their parents or as direct employees. Many will work in service industries as busboys, cleaners, messengers, helpers, stitchers, etc. As small children they will have accompanied their parents to worksites, worked in home industries, or taken major responsibilities in managing their own households. Most household members above the age of ten will work. But they will be marginally employed (working between ten and ninety days per year) or underemployed (working between ninety-one and one hundred and eighty days per year). During their working days they will earn slightly more than a subsistence income. According to UNICEF and the World Bank the infant and child mortality and morbidity rates in this group will be among the highest in the world, while their nutrition levels and health status will be among the lowest. For the few who find themselves in schools, problems of attendance, clothes, learning materials and competition will tend to push them back onto the streets. For those who manage to stay in school, theirs will be the worst learning environments and their teachers will be amongst the least prepared to impart education. Their education will be neither functional nor adequate. Dropout rates will reflect this.

In the same cities where the poor struggle for survival, a much smaller upper middle class community will enjoy participation in one of the most efficient educational systems in the world. Their parents will manage modern, formal sector industries supplying the global market. They will be engaged in vocations, professions, services and trading. They will be the operatives of a complex, 'post-industrial' type of service sector. Yet the struggle to 'stay middle class', and the competition to become middle class will create a major challenge to newcomers.

Additionally, an entrenched working class, employed in formal sector large scale industries, will enjoy incomes about three times those of the bottom two deciles of earners. They will account for about a third of the breadwinners in the bottom six deciles of urban workers and they will reside in slums alongside the poorest of poor households. They will be covered by pensions, health and accident insurance. They will enjoy annual paid leaves and bonuses. It will be their dish antennae, peeking out of tin covered shanties, which comfort the well-off who pass by. They will have attained

skills, education and an urbane ‘savvy’ about life in the city. They will belong to unions and political parties. It is this upper layer of the poor who will most abhor the squalor and poverty of their own slum neighbors. Desperation to survive will make the vulnerable eager to work for any wage. This weakness alone will pose an economic threat to those poor who have achieved a modicum of job security. The very poor will not be looked upon with compassion, but as a potential threat.

Barriers to the Job Markets and Income

Thus, when we speak of the ‘poor’ in cities, we are not speaking of slum dwellers. We are speaking of an underclass composed of the bottom fifteen percent. Given their skill profiles, their opaque image of the system, and their limited job opportunities, these truly impoverished people will be caught in a cycle of poverty.

They will not have the language skills even to carry out low-level clerical tasks or to provide basic services. They will be low on numeral sense, and even manual skills.

They will have no basic skills and moreover, according to the United Nations, International Labour Organization, they will lack the empirical logic necessary to make line decisions needed in rudimentary assembly tasks. Further, according to World Health Organization, they will be underweight, disease-prone and have high absentee rates, due to high morbidity.

Worst of all, their abilities to IMAGE will be very low. Visioning alternative roles, conditions, situations, contexts and scenarios will simply be beyond their capability. The GOOD LIFE will be out of their reach.

They will mainly walk to work. Some will cycle. Many will travel ticketless – illegally – on mass transit. They will live in families of five or six in single-room tenements of eight square meters each, lacking windows, water or sanitation. Some will have access to overcrowded water taps, shared (and unhygienic) sanitation blocks, and limited bathing areas. The land market will densify their environment, overstressing the inadequate basic services.

Liberalization and globalization will hasten this process, making it even more difficult to cope with the transition. Mass communications and consumerism will only make the divisions among the poor more obvious and painful. And with the new economy, new technology and new markets, ever higher entry requirements will confront prospective new workers. While all of those who live in slums will face inadequacies due to systemic dysfunction of income distribution and housing markets, the poorest families will bear the true burden of real poverty and hunger. The priority, then, appears to be livelihood sustainability.

How do we achieve this?

Basic Needs: Where will the gaps lie? The vulnerable within the urban poor, described above, will not have the purchasing power to acquire minimal levels of transport, primary education, basic health care, or recreation. The inadequacy of food intake, hygiene and preventive health cover will reduce their capacity to exert themselves. Lack of ‘ability to pay’ will exclude them from ownership of any form of private asset, including mechanized or electronic devices or a vehicle and land tenure. Clothing will be adequate, but will bear witness to their underclass status.

Their unfulfilled basic needs will lie in three areas:

- **Survival:** nutrition, health and hygiene;
- **Transformation abilities:** skills, awareness, and knowledge;
- **Support mechanisms:** shelter, transport, clothing and recreational.

Their basic demand will be for jobs, the income they fetch and the status they bestow.

Two strategies clearly emerge from the above.

Firstly, a kind of **social net** must be created to ensure survival. This is NOT development. It is essential crisis management. About fifteen percent of the population would need survival protection. The NGOs can only provide models and lessons; they cannot fulfill this immense gap. But by ‘doing it’ they gain the moral voice of advocates. They can speak with authority. A clear role for NGOs in urban areas is to ‘design’ a social net, experiment with its application, assess reasonable cost parameters and analyze the ramifications of ‘going to scale’. It must be seamlessly meshed with government apparatus to bring it to scale. Clearly, the government apparatus needs analysis and restructuring. In its present state it neither facilitates nor empowers low income households.

Secondly, a development strategy must emerge for **livelihood security**, inclusive of awareness building, group action and functional education. Awareness can be spread inexpensively through group meetings, learning materials and mass communications. Functional education and skill development cost more and have a less extensive reach. But the NGO sector has a wide network of training institutions, technical schools and job training centers, which could go double-shift, using the night shift for the vulnerable. Again, this is an advocacy function.

Most of all we must **guarantee livelihood sustainability** within cities. The right to employment within walking distance from the place of residence is a must. Work must be provided, whether in creating a public or a private asset. Land development is a clear area of mass employment. Unskilled casual labor on construction sites is another. Urban reforestation, refuse collection and basic maintenance are yet others.

What we must all work toward is a national human development program. First, such a program must guarantee access to SURVIVAL. A basket of essential needs must be provided. Sri Lanka’s social guarantees of the 1960s and 1970s are worth studying. A per-capita ration of rice, oil and *dal* was guaranteed to every citizen, distributed free through ration shops. India’s existing civil supply system can be built upon. Ration shops can be the outlets. Infant mortality rates in Sri Lanka were 36/1000 when in many Indian states they were as high as 160/1000 (in 1980). What policy measures made these two comparably poor countries so different when seen through Physical Quality of Life measures? Second, such a program must guarantee the RIGHT TO WORK.

Our existing work guarantee programs can be vitalized and enhanced. It would be relevant to study the Work Progress Administration introduced by the Roosevelt Administration in the depth of the US depression. Food for work, famine relief projects and the Employment Guarantee Schemes are other cases which require re-examination.

Herein lies our mission and our purpose. We must advocate the need for a national poverty alleviation program. Jobs for All is a goal within our reach. A social security net covering effectively the poorest of the poor is a must. Giving them the mental eyes to see the system they live in is our

duty. Most of all, in this age of ‘free enterprise’, we must reassert the sacred duty of the community and its use of public instruments. We must never privatize what is civil about cities.

HABITAT II is posed to be a ritualistic meet – over one hundred governments are the reluctant invitees. All that is needed for a grand success is a host of NGOs with their hat tricks and magic shows. The word ‘poverty’ has yet to find its way into the meet’s lexicon. Rather, their reports meekly refer to ‘those of limited incomes’. In its fumbling for ideas, HABITAT II will once more highlight old wine in a new bottle. Agenda 21 from the Rio Conference will be dusted off and an urban polish applied. The Global Shelter Strategy will be rechristened. Admitting that very little is known about what is going on, a rather academic plan for monitoring cities will be proposed. Be assured that all of this will be wrapped up in environment-friendly, gender-sensitive language. We cannot allow such an important event to slip past us as an empty spectacle, celebrating nothing but our duty to remember HABITAT I. Instead of worrying over what has in fact already been decided for 1996, we in India must have our own agenda, lest we waste our energies pandering to a system that lacks commitment and direction. In fact it is we who must provide the direction. Just as we cannot privatize what is civil about the cities, we cannot globalize what is humane and compassionate in our own communities. Yet we can work together to strengthen each other through a common cause and values. We don’t need another Earth Charter. We need a workable poverty alleviation strategy of our own. Let us hope HABITAT II would then follow us.

Channels of Access to Shelter

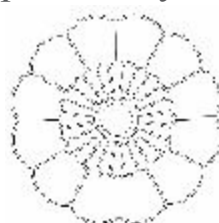
(Interview by Gargi Gupta, Business Standard)

There are so many buildings of so many kinds being built in India today. What do you think of them, architecturally?

Architecture mirrors the society for which it is created. It is no better or no worse. The new economy in India is based on its becoming a world destination for out-sourcing, where cost cutting is the client's objective. But they want to do this with a global touch. Ninety percent of the projects are 'cold shells' where low budget and fast-track schedules are the design brief. Quality and beauty and creating better places to live in are not what people are doing, because it is not the agenda in the first place. There is a kind of covenant between multinational firms, developers who house them and the mercantile architects to produce faceless urban fabric that belongs to no culture and respects no history.

You were probably one of the first architects and town planners to work with urban development authorities and state housing boards to develop new towns and large housing projects for low-income groups. What became of them? Do they continue to be used by low income households? Are you still working on such projects?

In 1972 I had the opportunity to design the first affordable housing scheme for the Economically Weaker Sections (EWS) for the Housing and Urban Development Corporation (HUDCO) in Jamnagar. It was a system of incremental courtyard houses, where every household had a WC and a tap, a room and a courtyard with a stair to the roof. The plots were about 28 square meters and the core house was just 18.5 square meters, with a small courtyard making up the balance. This project came to me through a series of accidents. While on a Fulbright scholarship to India in 1968, I focused on urban slums. This took me to a young labor leader, Sanat Mehta, in Vadodara. We envisioned a scheme to house the poor. Years later I returned to Ahmedabad to initiate the School of Planning and Mehta became Minister of Housing in Gujarat. About the same month the Housing and Urban Development Corporation (HUDCO) was founded. Mehta asked me immediately to take up a massive project in Jamnagar. In 1973 I introduced the concept of Site and Services (which I had developed as my thesis while a student at Harvard). The idea was that poor families could actually build their own shelters, but they lacked the land tenure to allow up-gradation. They also lacked rudimentary services like storm water drains, street lights, potable water supply, a sewerage connection and access lanes. My idea was that we provide what they cannot provide and they provide what they can. I was doing this with World Bank funding for the Urban Development Authority at Chennai. It proved very successful as we could provide about 15,000 serviced plots in what it would have cost to build 1800 unaffordable small houses. The idea was replicated by the World Bank globally.



This was a case of India exporting intellectual property to the world at no cost. The Bank just picked it up and ran. I worked with Mr. Dattatreya and Mr. Laxmanan (urban planners at the Madras

Metropolitan Urban Development Authority) and we did it only for the thrill of solving problems. Later we found a number of Washington based 'experts' claiming our invention as their own. We did not mind as long as the instruments we invented served a broad community of users. People today make a lot of noise about 'open software'. In our value system one created inventions to solve problems and for the thrill of seeing them implemented. In Washington our ideas became commodities on which people tried to build their careers.

In 1976 I had the good fortune to do the first project for the newly created Hyderabad Urban Development Authority in its first year of existence. I wanted to explore the concept that the secure urban poor (in this case, Class IV government servants) become the developers for the EWS for whom rental shelter made more sense. So we laid out a township of 2,000 houses, amenities, shopping and open areas where the owners had a small core house with essential services on one hundred square meter plots. These plots were five times the area we had used in Jamnagar. What happened was that the owners quickly built more rooms and rented them out to still lower income groups (often relatives or people from their own villages) at an affordable rent. The tenants used the public health infrastructure (increasing its efficiency several fold) and the landlords harvested rents for loan repayments and up-gradation. These petty landlords paid land taxes, thus playing their roles as tax payers contributing to urban resource mobilization. In the Hyderabad case the two thousand original plots served to create more than six thousand end user shelters for a very low income group – something a public authority could never have achieved. These were three different and unique channels which facilitated access of the poor to housing. This is what I call 'Social Architecture'.

I also carried out large scale planning operations for the Mumbai Metropolitan Regional Development Authority in the mid-1980s in Thane and in Kalyan where I emphasized both macro social and economic infrastructure and user-end access. Generally development plans only calculate the per capita needs for water, electricity and sewage disposal and provide these at the trunk infrastructure level. This leaves out seventy percent of the population who live in slums and subdivided chawls and old structures. We turned this around and went directly to the users and worked out participatory strategies. Common taps for potable water, bathing places and WCs managed by user groups were provided. At the other end of the scale we looked at the regional water resources, the regional storm drainage networks and transport systems. By addressing these from both ends a viable urban strategy was designed. In Thane, much of this was implemented in the early 1990s, which turned around the city's economy. This concept of slum up-gradation became a fourth channel, giving the poor access to shelter.

Why am I not still doing this? First, I am interested in designing concepts and inventing ideas and not in becoming a factory producing housing. Second, I realized that one could have more impact on access to shelter through framing public policy than through construction. Our urban planning legislation was modeled on the British Garden Cities Movement ideas, which included wide boulevards, huge parks, immense house plots, single-function land use zoning and low density as guiding principles. Our élite bureaucrats felt India should model itself upon the West. I knew that neither the government nor the urban poor had the resources to follow this model. The answers to our problems rested in new laws, new town planning standards, new philosophies and new principles. That is why I formulated the Principles of Intelligent Urbanism, which have become the basis of much of my recent planning work.

Can you give some salient details of the urban and regional development plans you came up with for other South and Southeast Asian countries like Sri Lanka, Bhutan, Malaysia and Indonesia? Have these been implemented?

I have been involved in the preparation and analysis of plans for existing and new towns in Malaysia (Terengganu); Indonesia, Sri Lanka, Nepal and Bhutan at different levels. In Malaysia I was pushing policy changes as the government was building new towns with over-designed, spread out roads and public health infrastructure. They planned such large plots for the poor and such high technical standards that these could never be handed over to the stakeholders for self management and local governance. They had priced out democracy. Only if a patron authority managed these towns could they survive – with large subsidies.

Again upper middle class garden cities designed to Western standards were dictating minimum plot sizes, setbacks, ground coverage and other urban development standards. This vision was wrong from the very top, where foreign models were drawn upon and dropped from the sky on emerging economies. In Indonesia I designed the shelter strategy for the rural poor as one of the cornerstones of their first National Rural Development Program. Here also the problem was wrongly stated as ‘how to provide a minimum, standard house to each rural household’, or about eighty-five percent of the population. Development was wrongly seen as the production of ‘things’, rather than of networks and processes.

The question was turned around by looking at health statistics in rural areas. The bulk of mortality and of morbidity arose from polluted drinking water, poor sewage management and low knowledge levels about disease and sanitation. The new approach was to see the ‘community as the family’ and ‘the village as the house’. Previously, community bathing ponds were used for drinking water and for bathing animals. Sewage leaked into these, making them cesspools of disease. The new house started with functional education, user-end water sources and sewage and sanitation management. The housing program was reinvented as a habitat program: the new house was a new village.

In Nepal, I designed an institutional system for local participation, micro-level planning and decentralized implementation. In Bhutan I have prepared the physical plans and economic strategies for three new industrial areas that will piggyback on the surplus hydel power and emerging labor force. We also prepared the new capital plan and the capitol complex – designs for the key buildings in the complex. These are now under construction. They will take years to complete.

Of greater interest are the Local Area Plans we prepared within our overall Structure Plans in four cities in Bhutan. These are for compact Urban Villages with all basic services and amenities. They are prepared with the involvement of the local land owners: they bank all of their land together, and after leaving aside thirty percent for common facilities, they get back well laid-out plots equal to seventy per cent of what they handed over. The trick? The new rectilinear plots with road access and basic services are worth five times as much per square meter as what their original odd-shaped agricultural lands were worth. All of this work is based on the Principles of Intelligent Urbanism. Our last plan in Bhutan was for a very small new town, called Denchi, in the far east of the country. This will act as an administrative center and as a small magnet to pull development to the east.

Indian cities are overcrowded, their infrastructure rickety. What do you think ails urban planning in India? And how can it be set right?

What ails urban planning is that we do not have urban plans. We just have two-dimensional plans marked with colors for land use restrictions and density restrictions. We link up building control regulations for different zones and call that a plan. These are merely controls on land development, not catalysts and enablers of urban development. A plan guides and facilitates development. Urban development is a partnership, not a tussle.

Not only that, for cities like Pune the last Development Plan was prepared way back in the mid-1980s. It is now three decades old. This is criminal procrastination. Fast-track cities like Hong Kong and Singapore, which are centers of economic growth, have used extensive urban planning to assure economic development. In Singapore all the land belongs to the government and is leased to the private sector for development within very clear urban design guidelines. A large percentage of the people live in public, government owned houses. In India we have had the worst of capitalism and the worst of socialism. We are confused and think that the two are mutually exclusive. Capitalism is sustained by good planning. What we are doing in SEZs and business parks is just the beginning of a more viable system. But the idea of appropriating land for these from farmers is a flawed concept. Thus, we do not have a viable system. There are rare exceptions to this, like what is happening in Ahmedabad, where a strong private-public partnership and Town Planning Schemes provide a rational pattern for urban growth, much like the land pooling I described in Bhutan.

Cities in India were totally neglected during the first fifty years of Independence. There was no understanding that rural regions and cities are one integrated system. Each feeds or starves the other. Cities are the engines of economic development. We starved these engines for fifty years. We are now paying for this neglect of basic economic principles through congestion, pollution and unsanitary living conditions. Good regional and urban planning is good business. No one understood this. We are awakening after a long sleep. In a way Indian cities are disproving economic theory. It was always thought that economic growth follows the emplacement of economic and social infrastructure. Cities like Pune and Bangalore are growing despite serious infrastructure gaps. But this entails a huge human cost.

A huge investment by the house of Tata was thwarted, shifting the production of the world's cheapest automobile to the better-off state of Gujarat. The events in West Bengal are not due to pig-headedness. We were all looking forward to the Tata project there acting as a fillip to turn around the economy. But the politicians failed Tata and their own people at the same time. Through inhuman land acquisition they ruined an important investment and employment generation opportunity in their state. We have to learn from that experience and put people first. As in Bhutan, so also in India, good planning has to be participatory, involving the land owners and the stakeholders.

Part of our problem today is the graduates of business schools who have a very narrow education. They are arrogant, with limited knowledge and poor skills for consultative decision making. Economic development, economic growth and social change all involve participatory and rational planning. We totally lack that today. We must understand that the WTO is not a development strategy. It is not a plan for social transformation, not even a plan for economic development. It is a vehicle for multinational companies to grow and expand. Our challenge is to seek how we can use this growth and this development for the average citizen; how urban planning can integrate a better quality of life with economic growth.

As a starting point, when we acquire land for large projects we have to do this through land pooling, land banking and rational redistribution. Next, in addition to Special Economic Zones we

need Special Habitat Zones that are part of, and complementary to, the Special Economic Zones. We cannot plan only for the machinery of development leaving out the essence of development – people.

You first set up institutions like the School of Planning at Ahmedabad and CDSA, and much later a firm of your own. Why did it take you so long to set up your studio and why did you set it up?

I never wanted to live off the proceeds of an architectural studio. Architecture is an art, a labor of love, not a business. All of my early works are for NGOs, voluntary agencies, or for myself and institutions I am deeply involved in. Harish Mahindra drew me into running a larger studio by asking me to design the United World College of India, and that won the American Institute of Architects Award in the year 2000 which fueled our client list. Architects need patrons to nurture their creativity. We need media coverage to meet patrons. Unless we are careful it can become a vicious cycle of greed and money making. I have never chased clients. I am arrogant with clients to the extent that they must know I am my own man, just like their heart surgeon who will not do their bidding. On the other hand I am the servant of my clients and I have to safeguard their interests above all else. But I never solicit work through free designs. I never lower my fees. I do not design for developers who will compromise on specifications and visit their sins on unknown end users.

It took me so long to set up my studio because I think it takes forty or fifty years to learn the craft of architecture. I think I am still a student. I think I am still learning. My teachers are my craftsmen, history and my team members in the studio. Every new building is a new invention. We can carry into that process the learnings from the designs before it and from our earlier projects. But each project is NEW and we have to be humble about it. Our clients are often represented by young managers who know nothing of the building process. They know nothing of art. They think one can create beauty in a month or a few weeks. This is an illness spreading like AIDS and we as professionals must educate the public about the disease. It is impacting on the very character of the urban fabric which nurtures us and which underpins the very creation of wealth from which we gain sustenance.

What has kept you in India?

I have always felt I am an Indian through and through. Maybe in my previous life I was born in a village and lived there in contentment. From the day I set foot on Indian soil in 1968 I never felt strange or in an exotic place. I love India's people, the landscape, the seasons, the dust, the heat, the dynamics, the food and even the chaos of the cities and the peace of the villages. When I feel out of place is when I step off a plane in America. Immediately I think, 'When can I get out of here?'

How far do you incorporate Indian architectural elements in your designs?

Architecture, good or bad, emerges from its context. One cannot really incorporate elements. Maybe

one can decorate a building and make it look indigenous, which is a sham. One has to let designs emerge from the landscape, craftspeople, materials and the client's needs. All of these variables are Indian and thus my architecture is Indian. Even if I experiment in glass and steel, it is my Indian curiosity about things new and things different. It is my Indian interest to bring in outside things and integrate everything into one huge pantheon of ideas and concepts.

(Interview to Business Standard, New Delhi, on 11th and 12th August 2007)

An Interview by Harsh Kabra of *The Hindu*

How do you look back on your many decades of association with India and its architecture?

Architecture is composed of many layers of reality and architects are trained to deal with such multi-layered complexities through templates and prototypes which can be mindlessly applied to typical problems. Architectural schooling can either open up multiple windows to self discovery (education) or pattern one to copy out the right prototype each time (training). After one's architectural schooling there are the onslaughts of fads, fashions and pop thinking that rein in one's creativity. The media conveys the messages of what is right and what is wrong. Young people try to find the thread of their future in the weave of the media. Then there are the frameworks of the 'gurus' who have projected their correct ways and manners of resolving complicated conundrums. Quite unintentionally, settling in India saved me from all of these forms of entrapment and enclosing paradigms. When I settled in India in 1971 there was no television, internet, affordable phone system or international journals. I was fortunate to 'miss' the entire love affair with 'Postmodernism'. I hardly knew it was happening until it was over. That was when we got an internet connection in 1998, twenty-six years after I left America.

That India has thousands, no millions, of gods only reflects India's intellectual bent toward multiplicities of interpretations, perceptions and conclusions. Seeing things in manifestations, rather than searching for the 'absolute' truth, is India's single most creative strength. Nothing is fixed or boxed in; everything is in flux and changing. An idea is seen in multiple ways, from diverse angles and in various mutations. There are avatars of even the greatest of concepts: never one singular 'correct' path. If I have any credo it is this continuous change, and the deceptiveness of the 'absolute' truth. I would rather search for the Good, than know the Truth. The Indian schema is more concerned about the core of a concept and its many physical and ideological interpretations, analogues and metaphors, than in axioms, principles, laws and rules.



India therefore has been my natural, organic home as an artist and architect. I would have been stifled by my own success in America. In America and Europe I would have always been seeking creativity and have been trying to find myself with blinkers on, leading me in one 'infallible' direction along with the mob, not letting me see all of the alternatives. The media, money and fame would have been pointing me to the pre-defined 'right direction'. India is my land of being. America is a land of seeming.

But isn't America a great country where you studied in great universities and taught at Harvard?

America is great because you can pretend to be what you are not; India is great because you can find yourself and be what you are. I learned a lot at Harvard and MIT because I had the singular good fortune to be taught by inspiring teachers. But when I started to teach at Harvard I realized I was falling into the trap of 'seeming'; I was hiding my true nature to seem like something I was not. The

taste makers were making me over. Madison Avenue left its calling card, and that I was getting seduced scared me. It was a fatal love that made me panic. I liked what was happening to me, but I knew it would kill my inner soul. At the first opportunity I fled to India.

People often ask you about your views on contemporary Indian or Western buildings that are coming up. What do you think of them?

I cannot address these queries, as two mutually irrelevant situations cannot raise a relevant question. Buildings are just the result of the events of building. They are a kind of cultural flotsam, or discharge, that rises up to the top where it can be seen. To be true architecture a structure must point to the future while reflecting the past. What is more interesting is the precursors to the events which gave shape to form, and the impact of the forms on future events. Then an analysis of the products of events becomes meaningful.

Just like cinematographers who plan out the sequences and experiences of people who view their films, I try to program the experiences of people who move through my spaces. This experiencing while moving through space is a kind of kinetic architecture. It is a preconceived scheme. Then the people who live in these cinematic sets become the role players. They give life and meaning to the spaces. Cold, artificial spaces then transform into 'places'. They come alive for their inhabitants as living organisms. That is what is so fantastic about any great boulevard, piazza, square, promenade or vista. It is the experience of the Taj Mahal, the multitudes of people experiencing it together, which bring life and eternity into the physical scenario. It is not just the amazing impact of the masonry. A human conviviality swells from the whole experience of being in the place. One feels proud to be a human being and to be a part of something greater than oneself.

The images of the Taj Mahal, or even of a more humble structure for that matter, are just analogues of the multifarious experiential systems operating within the context. Indian architecture, to me, is this fluctuating and ever changing context characterized by the happenings within, as opposed to the dull fixed images of packaged consumer items which form Western architecture today. Let us say that architecture is a shared memory of those who have experienced it. This memory uplifts the spirit, gives vision to the future and inculcates optimism. In a world where the essential struggle is between the optimists and the cynics, this role of architecture is very important.

To me architecture is not things, nor is it the process of making things. It is the experiences of the people who live in milieus, or enliven places, imbibe forms, perceive spaces and become lost in their interstices, accidental or intended, which impact on the emotions, sensitivities and

the memories of individuals.

In Europe recently I visited some of the new stunts which are parading as architecture. I saw a really great engineering feat, which was a bad building and a terrible museum. This bad building, made of a concrete frame structure, housed an ill-conceived and unworkable museum. Yet this entire disaster was covered by a daring steel structure that arched over it. The structure itself is a great feat. They should have just built the structure and forgotten about the terrible museum that was housed in it. It was a screaming child demanding attention. It was not a mature artifact of a great culture. Yet, here in India we laud such stunts without knowing their true significance. This is an example of the Indian romance with the West. We have great respect, but underlying that is suspicion. It is like the love affair between a patron and a woman of the night. There is a lot of passion and attraction all smothered in deception and distrust.

What are your views on contemporary Indian Cities?

Let me avoid the usual review of data on how crippled our Indian cities are. I'd rather point out that Indian cities represent the dynamism and energy which thrive on the periphery of the global system, which gets suffocated by the huge, hierarchical economy in which we live. In India we have regional literature, architecture, cinema and poetry. Have you ever heard about the film movement in Alabama? Well, there is nothing to hear. It is too close to the epicenter of world culture. Just as there is little that grows on the Deccan Plateau because it is in the rain shadow of the Western Ghats, so a good deal of the West is in the cultural shadow of The Great City. Pune is a vibrant city and few people in Paris, London or New York have ever heard of it. Yet they have all heard of Newark, New Jersey, which has no soul, no life – just empty shells and distant memories. Indian cities, like hundreds of cities in other countries in the Periphery, are full of chaos, fluctuations, uncertainties, contradictions and chance happenings. This is the raw material of creativity, the stuff of free thought.

We live in a world system where 'center-periphery' dynamics operates. The center is sucking and feeding off of the peripheral resources, and the periphery is buying what the center produces, including ideas, fads, tastes and habits. The dense, affluent central core gets packed into a tighter and tighter ball of pelf and energy. The rich center is a seductive trap. People who rush into it cannot get out; like moths attracted to a flame they go, never to return. The debts they take to get in inhibit their mobility to get out. Their ideas become a kind of debt too. Everyone is driven by 'right' thinking, 'correct' behavior, 'correct' taste, fashionable packaging and a few acceptable paradigms of what one's life can be. True art cannot emerge from such a maze. It is out on the periphery that life dwells. Art is just a reflection of life. In art one can make mistakes, explore options. Art must be an adventure.

But isn't India following the West?

It is true that India is grabbing at the 'latest' and attracted to what is vulgar, mundane and banal about American society. See the text messages which are written 'wit da' most banal of American English. But this is just a minor malaise inflicting a spoilt, privileged sliver of the middle class. Being products of the consumer-driven media, what counts for them is what's on TV and what sells. There is a growing sub-culture in India that is not looking for work, but for jobs. They are not interested in how much creativity they can garner, they are interested in how many 'Ks' they will earn. They feel driven to be 'in,' and what's 'in' is what sells. Karl Marx called religion 'the opiate of the masses'.

Today it is ‘buying’ that is the opiate of the masses. They need ‘Ks’ to buy. It is the privilege of these élite to buy into oblivion. They will all end up living in air conditioned little boxes, working in similar little boxes; and, driving between these little boxes in air conditioned little boxes on wheels. They will feel lucky to work in air conditioned little boxes; and when they get time off they can go out shopping with the entire family buying and buying and buying – in more air conditioned boxes. Their heads will be full of little ideas in little boxes, and they will find a drugged happiness living life in such boxes.

So it is true that the illnesses of the center spread out to the edges; but the periphery is penetrating into the center too. Even *Businessweek* had to acknowledge my design of the United World College of India as ‘One of the Ten Super Structures of the World’ – whatever that means. History is always a tale of creativity and strength at the edge overpowering the center. Take the case of Frank Lloyd Wright. From the middle of nowhere he moved further away from the ‘center’ into the isolation of Taliesin. Then his work, concept and ideas captivated the ‘center’. It is not always such finality, but rather a process of things less organized affecting the staid and dull central organization. Creativity at the center is akin to a bonsai tree – very organized and very interesting, but not at all creative; very interesting but surely not very beautiful. Of course there are exceptions, but there is also truth in what I say.

Recently in Madrid I was inflicted with a series of ‘happenings’ and ‘events’ called modern art. These crude inanities were not only uninteresting, but far from clever. One sees the same junk in New York, or in Paris and London. They mix them with masterpieces to anoint them, like the Pope’s blessing. These banal artifacts are so ill-conceived and thoughtless that curators think they must be very brilliant! The tastemakers at the center are confused. They lead, but they do not know where they are going. There is an incestuous relationship between Western artists, critics and the media who all chill out together. They cannot tell themselves from one another. This intellectual cronyism leads to an inane ‘Yes!’ Who would dare tell the king he had no clothes on?

Until a decade back in India we did not have mass media access to ‘the latest’. We had no templates to tell us what to create and what’s ‘in’. Skill, craftsmanship and hard work was what counted. What we ‘thought’ was order garnered from chaos; was filtered out of variety; was chosen from millions of manifestations. There were no blinkers of ‘fashion’ to tell us what to wear or what to pretend to be. Suddenly India is turning itself inside-out by trying to re-define itself. Unfortunately, this re-definition is based on consumption and the false sense of personal power it engenders. One’s self image is generated by the desire to earn and to spend, rather than to develop ourselves to our greatest creative potentials. One finds brilliant youngsters doing factory line functions in so-called IT centers; answering phones day and night in so-called BPOs; cutting and pasting solutions to non-existent problems, and churning out little nothings in the so-called ‘engineering units’ of MNCs. Quality and Values are being replaced by buying and consuming, moving in a vehicle and a kind of frenzy about nothing. We are moving from a low energy and low consumption society to a high energy consuming society. We are moving from high thinking and simple living to high living on mundane thoughts.

What are the strengths and salient features of your city plans for Bhutan and your architectural projects there?

Bhutan operates under a mindset which turns the Western paradigm upside-down. Instead of seeking Gross National Consumption, it is seeking Gross National Happiness. The essence of this is the balance in life, or what is called the Middle Path in Buddhism. The balance between humans and nature; the balance between the built fabric we lay down on this earth and the natural terrain it lies on; the integration of a passion for life and a passion for work. This search seeks conviviality within community; obligations and responsibilities as opposed to just seeking freedom. It is rooted in meditation, self-discovery and being, rather than media-driven frenzy. We discovered two ideas in our work which had a profound impact on the way we plan and design in Bhutan. One idea is called the Principles of Intelligent Urbanism. It seeks ten balances between city building/living and nature, tradition, technology, work, home-making, play, meditation, movement, governance, etc. These principles are a kind of charter which the urban community agrees to put up to any new ideas or projects being proposed. This has resulted in more than fifty percent of the urban land in the capital city being reserved for greenery, water bodies, natural habitats, fragile ecologies, and play areas. It has resulted in a concept of Urban Villages which fall in micro-watersheds between rivulets flowing to the main river. It has resulted in the creation of an urban corridor to which Urban Villages can be attached through inexpensive, low energy public transport. These concepts were then turned into concrete and mortar; into trees and water.

We are moving from high thinking and simple living to high living on mundane thoughts.

When it came to building new structures – the new National Capitol complex – we enriched the idea of critical regionalism. Looking critically at the traditional system of construction, which is based in an eternal logic, we analyzed the new functional and technical demands of work and living and explored the interpretation of the vernacular building language which would ‘fit well’ into the contextual setting. This does not fall into any current fad or fashion. It fits into a unique cultural setting and milieu. People often ask me if my new capital plan for Bhutan is not a reaction to the decrepitude of Indian cities. I see it the other way around: Indian planners have a lot to learn from our work in Bhutan. Learning is not about understanding what not to do; learning is about discovering new paths leading us to surety in what we are doing.

You were recently awarded the prestigious commission to extend the campus of the first Indian Institute of Management, which is a world class center of learning. The fifteen most celebrated architectural firms in India were short-listed for this project. What is your reaction to your selection?

Surely this was my most prestigious commission at the time. The Indian Institute of Management, Calcutta is a value-based intellectual center. The faculty there is very different from most management faculties. They have a vision of the role of enterprise in the formation of a new society and culture. They are very sensitive to trends and trajectories and where we are going. So it is an honor and a challenge to work with such erudite clients who are really looking for quality, and not just utilizing FSI, or not just being ‘cost effective’ in the banal sense of the concept. This project has been a joint effort to seek a new milieu for learning, for discovery and for creativity. The built fabric must create an ambience for interaction, for self-discovery and for the development of personal discipline too. It must satisfy the need for reflection and contemplation, while encouraging interaction on a number of

levels. We are working jointly with the Institute on this. It is not just the personal search of an architect to make some kind of statement. We are also working within an existing beautiful campus, characterized by the many water bodies and trees that mesmerize one's soul. We have to deal with a lot of old, uninspired buildings, but they have a great potential to be integrated into a new whole. We want this to be a place of inspiration and discovery. We want to further amplify a world class center of learning into the new business and cultural environment. We want to make an impact on that emerging environment in considered and articulate ways.

A campus, whether it is a monumental capitol complex, a small enclave, or that of an institute, must have its iconic qualities. It must give an immediate message about the values and importance of the place. There should be a sense of the triumph of the human soul. People who live and work there should feel transcendental about their mission in life. People who leave must carry eternal memories which help them overcome the mundane in life in order to reach for perfection.

(Interview to 'The Hindu' on 14th June 2006)

Acceptance Speech for the Great Master's Award

For an architect, receiving the Great Master's Award is a watershed in his career. It is a rite of passage few can imagine. First of all it is an honor bestowed by one's fellow senior architects, who are articulate critics as well as cautious admirers. Second, the award is unique. Over the past two decades very few architects have received this accolade and those who did, truly embrace the great masters of South Asia. They include Laurie Baker, Geoffrey Bawa, Achyut Kanvinde, Charles Correa, Balkrishna Doshi and Raj Rewal. Who could dare to enter such a pantheon of iconic, creative personalities? I feel humbled by the very thought.

All of these men were truly masters of our art and the visionaries of our time. They understood that Modern Architecture was not just an act of creating bizarre and exotic strange forms, but that 'Modern Architecture' is a social art bound within the craft of technology. They understood that it is also an 'ethical art' wherein there is a truth in its processes, and there must be honesty of expression to achieve transcendence. In many ways architecture is a search for the truth of a building within its setting and context. All of these former awardees fought against false ideas and bad architecture.

I reject Postmodernism as a frivolous and self-fulfilling ideology of personal aggrandizement. I see my personal agenda as a mere continuation of a great tradition set out by the masters who went before me in the annals of history. I was fortunate enough to have great teachers like Balkrishna Doshi, Jerzy Soltan, Jose Lluís Sert, Kevin Lynch and Fumihiko Maki, who laid out a strict path of struggle and self-realization. They set an agenda which I beseech all of you to make yours also. It is a mission worth our endeavors, our fellowship and our professional commitment. It includes an agenda with three thrusts:



First, the modern movement is focused on the **social issues of urbanization**, especially mass housing and the public institutions that create a civil society. A modern architect is an urbanist in this broad sense. His work must contribute to its context, be a part of its milieu and make life better for the neighborhood within which it participates. Buildings cannot turn their backs on their neighbors, be arrogant or be absurdly selfish.

Second, buildings must be **true to the technology and materials and craftspeople from which they emerge**. Materials must be expressed honestly and the technology must be appropriate to the context within which it is created. Modern architects, since the nineteenth century, have explored new materials and technologies, but nestled them within local conditions. They have also understood and employed ancient technologies.

Finally, modern architects are crusaders, spokespersons and even revolutionaries in their **fight against effetism and deceit**. In India today we are bombarded with false architecture 'cut' from bad buildings in the West and 'pasted' into Indian environments, ruthlessly and carelessly. Most of this crime is committed under the false ideology of Postmodernism and the cyberecture that followed; these are in fact creeds of greed and self-aggrandizement. It is the craft of anal retentive, screaming

and yelling babies out for attention. There is a wild grabbing for more FSI with no concern for the creation of civic spaces, human experiences and the making of a good life for the common man. I ask you to reject this. Speak out against this. All of our modern agendas should lead us toward more natural, more appropriate and more 'local' styles. The blind imitation of Western fads must come to an end. Modern Indian architecture must also be 'regional architecture' emerging from the climate, local materials, local traditions and crafts.

The rise of media and of science has propelled us into the straitjackets of specialized disciplines. The further we advance in knowledge, the less clearly we can see either the world around us, or understand our own selves. We have plunged into what Milan Kundera has called the 'forgetting of being'. True modernism began in an era when the 'passion to know' became the essence of spirituality. The essence of modern architecture is to explore that which only a piece of architecture can discover. A building which does not express some unknown segment of existence is meaningless. Revealing truths is architecture's only reality. Bringing the 'good life' to humanity is our mission. The sequence of discovery, not the sum total of what is built, is what constitutes the history of modern architecture; of all architecture. It is only in such a cross-cultural, historical context that the value of any work can be fully revealed and understood.

Friends, with these few words, I thank you for the great honor that you have bestowed upon me today.

(Sunday, December 21, 2008 at the Architect of the Year Award Ceremony, Kolkata)

In my life as an architect ninety percent of my work ended up in the

trashcan of my dreams. Some work survived in the form of models and drawings made to scale. What got built was a mere fraction of my life's efforts. When people praise that small evidence of my truth, I feel very nice. When they garland me, and call me a Great Master, I feel humbled, yet truly elated. All of those trashcan dreams get reborn and come to life again with new meaning.

What Made Raje Laugh

Friends, we have lost a great teacher and a great architect by the name of Anant Raje.

He is special to all of us who knew his work. He is still more special to all of his students who learned how to think from him, how to question and how to make decisions. He is very special for those of us who shared a beautiful intimate friendship with him.

He laid out an Epic Path in front of us and showed us the poetry of being a tiny part of it. Most of all he showed us how small we are, by painting on a huge canvas. He made the human condition real.

Man pretending to be God, while being a mere human, was Raje's endless joke. Man greedily seeking fame and money in the name of art was Raje's joke. Man reaching for greatness and instead grabbing "meaningless success" was Raje's joke!

The friendship was beautiful because he always made us laugh. Raje spoke to his friends through the medium of stories. These were epic stories about Le Corbusier, Louis Kahn, great projects, masterpieces of art and amazing people.

Raje called upon all of us to be great men, and we shared a dream I would like to call the Great Man Theory. It was a 'theory of the possible' and of the human condition. It was an idea that challenged us.

Many decades ago, in 1970, I was teaching at Harvard and Raje called me on a June evening from Philadelphia, inviting me to join him early the next morning. He wanted us to visit the Richards Medical Center, see Fuess Hall and share other architectural marvels of that city.



We walked for miles and talked for hours. We drank red wine in the evening, and we went to sleep charged with memories and dreams of great architects.

Waking on Sunday morning, Raje told me that he had a special gift for me! Kahn had agreed that he would spend Sunday afternoon alone with us in his studio.

You see, for Raje to share his great treasures; for Raje to share his knowledge; for Raje to place the window glass of life in front of us and to challenge us, was his personal mission.

Louis Kahn, for Raje, embodied all of the aspects of a great man that one needed to know. Raje used his iconic image of Kahn as an intellectual mirror through which he, and all of us around him, could see ourselves. Kahn made the complex simple, while we made the simple complex in the name of design. Kahn could quickly grasp the fundamentals of complex problems and interpret their complexities into simple forms and great spatial systems that all became iconic images.

Raje's Kahn became my Kahn. Like Mohammed about Allah, what Raje had to say about Kahn carried a profundity and a sense of the eternal Truth. Through these stories of truth Raje laid bare all our ambitions and our weaknesses. By rendering this epic image of Kahn, the perfect architect, Raje made each of us feel very small and fragile.

While I was listening to Kahn on that Sunday afternoon, four decades ago in his studio, he crumpled up an A4 sheet of paper, handed me a pen and said "Sketch it!" As I fumbled to pen down the incredible complexity, he grabbed the pen back and drew four lines, making the image of paper in

its true simplicity. Kahn smiled and we all laughed.

I realize that Raje played this ‘trick of truth’ on me continuously, making me sharper in my thoughts. In his generosity and insights, Raje the *chela*, went beyond Kahn the guru.

When Raje told a story about Kahn, Le Corbusier or Picasso he would usually end it with an incident where a well known architect misused architecture for personal glory, rather than as a spiritual path toward self realization.

Yet Raje also saw the beauty in this ‘wrong step’, because there is poetry in man’s weakness. There is a slice of each of us in every foolish act. Man seeks love, fame and fortune that weaves a lyrical story keeping the epic possibility just beyond his coveted reach. In a sense Raje knew the fundamental stupidity of mankind, understood man’s weaknesses, and by comparing arrogant architects, or fools, with his icon Kahn, he expressed humorously great errors, blunders and follies. Man approaches epic greatness, but trips over lyrical desire at the last moment, losing eternity for immediate gratification.

Raje used his iconic image of Kahn as an intellectual mirror through which he, and all of us around him, could see ourselves.

This ‘exposure’ was Raje’s own personal insight that he shared with intimate friends. Raje could analyze the essence of a problem; lay out all of the pieces before you; point to the solutions, and then humorously give an example of the wrong solution to the same fundamental problem, exposing why perfectly intelligent people would take the wrong step. In Raje’s unique wisdom about folly and weakness we could see our own predicament in life. When he made us laugh, we were all laughing at ourselves!

Raje knew that life was short and he knew that he would die. He knew that in this short life **truth** and the struggle for perfection was his path in an epic search. Raje also knew that all of the students, architects and friends he was talking to were potentially great architects. He told his stories not to mock individuals, but to call forth the profound in humanity. It was his love for humanity that drew him to teaching and story telling, and it was his love for humanity that made him laugh. I suppose, to me personally, Raje was the ultimate teacher. Raje always gave more than he took; Raje always shared passionately what he had.

What Raje never said, nor ever hinted at, was that he himself was an avatar of greatness. He never praised his own ideas or concepts. He just explained them, leaving each of us to absorb what we could. But Raje was indeed the essence of a great man on an epic path. It showed in his knowing smile, in his sketches, in his anger with bad details and in the wonderful compositions and epic designs he left for humanity. Yes, Raje was a Master Builder.

Raje never doubted that he had been to the Promised Land, or that he knew what heaven’s vaults, domes and arches looked like. He merely wanted to share his grand vision with anyone who he thought could understand it. There was always sureness and never doubt. When Raje told his stories, there was always Amita, an architect and his life companion next to him. Theirs was a partnership and a shared journey. Like Raje, she knew the humor of life. She was his secret sharer. This made him still stronger and still surer. And over a glass of wine he laughed and laughed. He laughed at all the funny people who could not understand how simple the Truth is! And Raje laughed and laughed, while he shared his Great Vision and inspired all of us fortunate enough to know and love him.

In Raje’s unique wisdom about folly and weakness we could see our

*own predicament in life. When he made us laugh, we were all laughing
at ourselves!*

And Raje is still laughing!

(Obituary note on Prof. Anant Raje)

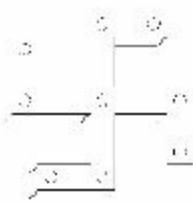
Doshi at Eighty

Doshi is both a man and an IDEA. I believe very special people are imprinted in our memories at birth. Even before the first time we meet them it is a kind of recollection from our memories! This is true only with a few unique people on this earth, and it was so when I met Doshi in October of 1968.

When one meets Doshi, even over a small matter, there is a glint in his eye that hints of the inevitable. It seems through mere glances and passing smiles that the larger concerns are demanded of us, which transcend the petty concerns of the moment. Rather than two people talking, Doshi is dealing with the collective concerns of humanity and thinking how this little problem is but a sliver – a sign – of the greater human condition.

There is a sense of vision, of the future and an excitement that we are not dealing with something small or mundane, but that we are unraveling the essence of the universe. The more one gets to know Doshi, the more his apparent contradictions seem to fall into an order and a unity. It is within these seeming contradictions that the essence of Doshi lies. What are these contradictions?

- Doshi is both simple and sophisticated! He tells his story in such a simple manner that his innocence obscures a great sophistication. Each building he describes and each question he answers is often analyzed through an analogy to a folk narrative, a riddle of life, or is explained through a passage from the great epics. His range and grasp of tales belies an underlying encyclopedic knowledge.



- Doshi is both a traditional Indian and a global man. He lives very simply within the great Indian tradition. Seeing his home one feels he could be in a relaxed village house lost in a rural place. Yet it is his great understanding of things which makes matters appear simple. He brings the reality of things down to their basics making them truly universal and global.
- Doshi is a wise sage yet he thinks like a child. Even at age eighty there is a child in his face; in the way he talks; and in the way he sketches. But behind that child-likeness, that playfulness is the ageless wisdom of a sage. Truth always presents itself in the simplicity of a child.
- Doshi seems as free as a bird, yet has the self-discipline to achieve. He is always relaxed, free and unfettered. He is not bound to any ideology, or to any ‘-ism’! He seems almost *bindaas* – like a free bird, or like a traveler without a destination; knowing only the joy of moving and exploring. Yet, the contradiction: he has labored to start institutions that live on discipline; create buildings that only hard work can bear; and create human relations which mature over decades of devotion. Doshi is free in his mind, yet a slave to his devotions!
- Lastly, Doshi is a Master of the Small, yet a Ponderer of the Infinite! If he draws a small bird, it will be in flight; it is all birds flying in one image; we too are watching it; we feel in flight; and we experience the transcendental beauty of flight, and the unimaginable! Doshi deals with the tiny seeds of things, yet in them lies the essence of all things.

The greatest fortune in life is to have good teachers. This fortune has indeed smiled on all of us who

know Doshi. He makes us aware of the good in ourselves and we feel very good about that realization. He excites some deep understanding of our essential possibility and who we could be. That is what is known as inspiring.

It is that good, our feeling GOOD, and our knowledge of ourselves that makes us want to celebrate Doshi's eightieth birthday.

The life of any person is a dubious experiment. Life can be fleeting, meaningless and insignificant. It seems so amazing that anything can exist or develop! Yet Doshi's life has been an epic journey:

- His boyhood in Pune in the old city;
- His student days at J.J. College of Architecture in Mumbai;
- A brief period in London with the good fortune to meet his guru;
- His years in Paris with Le Corbusier;
- His early days in Ahmedabad, moving about in the heat on a bicycle to supervise Le Corbusier's buildings;
- His marriage to Kamuben;
- Founding his studio Vastu Shilpa;
- Starting the School of Architecture at Ahmedabad;
- Work with Louis Kahn on the Indian Institute of Management;
- Wonderful friendships;
- Transforming a single School of Architecture into the Centre for Environmental Planning & Technology, and then a university;
- Making great buildings; winning prizes and awards;
- Being surrounded by a loving family and lifelong friendships;
- International recognition;
- Achieving contentment.

Doshi's life has been a psychic process that is only partly revealed and it still unfolds. Doshi is two beings inhabiting the same body. One being is the simple man, the friend, the husband, the father, and the architect. Yet there is another Doshi beyond the memories of encounters. There is the Doshi who is the *avatar* of imagination; there is the Doshi who is the manifestation of dreams; it is like two beings always walking together; inhabiting the same space; knowing us as a friend, but playing on our spirits like a phantom. On one level Doshi is an object, like a tree, a stone or a mountain or a human being; on another level he is ethereal, like a morning sunrise bursting over snow-clad mountains, awakening our inner spirit and making us question who we are. When we are standing next to Doshi we feel there are two beings next to us – one concerned with day-to-day life; the other drifting off transcending materiality. It is this second personality, this 'other persona,' which is a forming myth that carries within it the eternal spirit which lights up one's imagination; one's inspiration; one's desire to be.

Thus, on his eightieth birthday celebration we must consider Doshi's personal myth which will live forever. We must celebrate it without trying to understand it. We can only tell stories and recall incidents. Whether the stories are true has no bearing and is of no significance. The only important thing is whether we can grasp Doshi's story, and Doshi's Truth. The test of a man is in his myth; only his inner vision, which projects out to the vast universe and is etched into history, can have any meaning.

Every life is the story of the self-realization of the unconscious. Here Doshi's life is unique. Everything in the unconscious seeks an outward manifestation, and Doshi's personality also desires to evolve out of its unconscious condition to experience itself as a whole. Let us not employ the language of science or the words of measure to trace Doshi's growth, his contribution and his gift. Let us celebrate the myth which we all own; that is part of our being; which now passes as folklore and sets boundaries to all of our imaginations and possibilities. It is the myth of Doshi which allows us to set our own parameters; which has forced us to dream, which asks us to search and to seek again and again, that we can never forget.

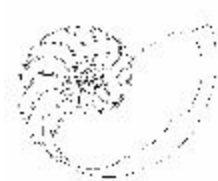
I came to India more than forty years ago in search of a guru; in search of truth, and in search of a believable myth. I was so fortunate to find all of these in one living being, who walks amongst us all here today: my guru, our guru, Balkrishna Doshi.

*(Talk on the occasion of Padma Shri Balkrishna Doshi's
80th Birthday – Saturday, the September 8th, 2007)*

The Beautiful Room is Empty

It is in the late 1960's. I search the side lanes amongst the haath-gaadi wallas of Bhadra in Ahmedabad for Doshi's office at Dhun House, a rickety old building. The air has an acidic stench making me walk faster. I climb up the narrow stairs with wires hanging and cobwebs dangling in the ceilings with paan stains decorating the corners. I find a door on the second floor. There is a young man standing there smiling, with baby-like smooth skin and amazing large dark eyes in white globes. It is Varkey in his twenties, with a receding hairline, warning that the boyish face hides within it an ancient man. Varkey invites me into the room. I enter the room where Kamuben welcomes me, saying Doshi is in Delhi. Our love of architecture and our youthful enthusiasm binds us into friendship immediately as Varkey shows me around Doshi's studio. The messy studio morphs into a beautiful space. It lives on today as a secret room playing about in the antique land of our memories; in the minds of those of us survivors who were blessed in its shadows and laughter.

Weeks later Varkey enters my room at "M Block" at the Gujarat University hostels. It is dark outside, but the room alights with the glow on his face. He had slipped on the mud outside and we both laugh. We are both students; he a trainee with Doshi; I on a Fulbright Scholarship with Doshi. The room is suddenly warm, glowing and full of ideas.



A month later I enter his room. He has invited me for tea. There is an electric coil heater glowing orange, with a pot of water on it boiling in the anticipation of a guest. That makes me smile. There are a few books on Kahn and Corbusier here and there. We look out of the window; the window of life; and we gaze into the infinity of possibilities. His room is beautiful. Life is beautiful!

It is 1971. I have returned to Ahmedabad. I search down a muddy lane for Varkey, amongst a chorus of monsoon's croaking frogs, for a small cottage in Azad Society where Varkey lives with his brother Thomas and the Kanade brothers. I look between the tall grasses down a long katchcha path ahead. I see the warm yellow light in a room. Through the darkness I see people sitting in a circle talking and laughing. They all work for Doshi now: Varkey, Shankar and Navnath. I need not guess the topic of their debate; it is ARCHITECTURE, and I enter the room of friendship, and join in the fellowship of laughter. It is the laughter of camaraderie around shared values and evolving ideas. There are issues demanding insight. We have all left our bodies on the door hooks with old clothes, and drifted off into our secret world of true believers. The room is beautiful. Life is beautiful.

It is 1973 and Varkey has been to Jaisalmer. He comes into my room, now an airy large studio

next to Gujarat College. In that space, filled with his positive energy, he shows me his sketches of a room in Jaisalmer. A few lines turn a flat white sheet of paper into a 3D view of an amazing place. In the middle of that static medieval room there is a static column! Varkey explains how as one walks around the room, the column stays in one place, filling the walls behind it with kinetic energy; the walls are moving, turning and twisting as one moves. The room is beautiful, and my studio room fills with positive energy. I am inspired.

In that beautiful room of Varkey's imagination I find a person; it is I! I am walking and moving, and my mind becomes a place! There is a model of my design of Alliance Française where Keshuvbhai Mistry is working, and I ask him to craft a round column and place it magically in the middle of the main hall of the model. As he does so, Varkey and I smile the bright smile of discovery and realization! A static room has been filled with the energy of moving kinetic space! The room is beautiful! Life is beautiful!

*In that beautiful room of Varkey's imagination I find a person; it is I!
I am walking and moving, and my mind becomes a place!*

It is 1983 and Varkey has a wife, Alice, and an apartment and a car! He is teaching at the University of Nairobi. Varkey is standing at the door with the interior light surrounding his shadowy silhouette in a saintly glow. I enter the living room, and Varkey switches on the fan. It is evening time and the sun is settling down through the ancient Jacaranda trees full of light purple flowers gathering illumination. Varkey smiles, the wise old man looking out of the young boyish face. We are laughing; he pours red French wine, and the room that was in faded light comes alive around a few candles on the table. Our mental energy leaves our bodies behind. Place making, low cost housing, the growth of slums in Nairobi, and the city's beauty, all absorb us in thought, and the room flows over with ideas that drift out of the windows into a night air full of singing birds. The room is vibrant and full of energy! The room is beautiful!

It is Ahmedabad in the 1990's and I enter a room full of students awaiting their jury. They are laughing nervously, awkwardly twitching and smiling! There are nine chairs in a row for nine wise men to sit and nitpick on the victims. There are chairs for Doshi, Raje, myself, Varkey, Kulbhushan Jain, Miki Desai, Chhaya, Muktiraj Chauhan and Kiran Pandya...all ageing relics from the school of architecture in the late sixties and early seventies; Creatures from an antique land where Kahn actually walked and talked. These were true disciples who had shared wine with and accidentally nudged the bodies of the modern apostles! They sat down and began their inquiry. As the jury proceeds, the rational triumphs over the iconic. Logic and order are the rules of the debate. The tyranny of the avant-garde is plundered like David slaying Goliath! Varkey fills the room with his love of students, spending more time elucidating the possibilities of the bad designs than the brilliance of the good ones! After all Varkey was a "people person", and architecture was just a vessel carrying the human spirit towards epiphany, enlightenment and self-discovery. The room is filled with laughter, love and beauty! Life is beautiful!

It is mid-September 2001, and the final jury of Dhananjay's thesis is quickly called to order. I,

Dhananjay's guide, Vivek Khadpekar the External Examiner, and Varkey the Internal Examiner all listen intently. The inquisition has begun. The topic spans the realms of architecture, planning, urban design and street people! Under the wrong hands it was a sure failure. Varkey grabs the moment, catches the core ideas and begins where the candidate must have started. His mind grabs the subject, the analysis and the findings like a super computer! The room awakes to his questions and to his insights. All are lost in debate! The room is alive with possibilities and shared conclusions. We are all lost in the world of ideas, considered thought and articulate reasoning. The room is beautiful! Life is beautiful!

It is October 2001 and the towers have fallen. I am sitting quietly in a room, on a hillock in Tuscany. I am looking out over the vast olive orchards and vineyards that are the ancient lands of my hosts, Rebecca Szabo and Vieri Salvadori. It is beautiful. The sun is setting, turning the landscape into multiple stage sets, turning my mind this way and that in amazement. I pause under the spell of the valley's mystery before turning on the computer. I get an e-mail that Varkey is no more, and I understand the meaning of emptiness!

Pity us.....

In the beautiful room.....

In its glow.....

So Briefly.

The beautiful room is empty!

Pity us...

In the beautiful room...

In its glow...

So Briefly.

The beautiful room is empty!

The Ordering Project

Orderly by Nature

Bringing things into order, finding patterns generic to things, and making templates into which things can be ‘ordered’ are the unique feats of the human brain, driving our strongest emotional compulsions. It helps us to understand and give meaning to complex phenomena and events. Human curiosity seeks the structure of things and their innate patterns.

Animals react to immediate stimuli and instinctively behave in ways that ‘make sense’ in assuring survival. Their curiosity is catalyzed by events happening right in front of them, while humans can create unseen scenarios. Humans are unique in their ability to ask questions about the past and future applying ‘fitness criteria’ and evaluations to generate new, alternative patterns. We are uniquely able to respond in a manner that brings complexities into our own desired scenarios, be it a battle plan, the making of a city or building a house.^[1] Imagination and imagery involve manipulating sets of orders that create new patterns and situations. The Greeks, in their search for balance and harmony, analyzed order in its explicit geometrical sense as well as its cosmic, political and social reality.^[2] Tongue in cheek, Plato advised young students of order to,

“Take in all the scattered particles of an idea, so that every one understands what is being considered, and then separate all of the little particles of the idea into ‘like parts,’ dividing them at their joints as nature directs, not breaking any limb into broken pieces as a bad butcher does.”^[3]



Collective Order

Ordering is not something we do individually. It is something humanity does collectively, with an unconscious desire to find the inherent fabric of things, to label them, and to control the relations between them. Social systems have emerged around the communal processes of managing common resources, creating public assets and forging new ‘stories’ for our lives to follow. Societies have evolved institutions that order collective decision-making, public policy, ‘the social contract’ and group action. These institutions transform norms into standards, codes and laws, with incentives for conforming to them and penalties for deviating from them.

The mores and norms of social behavior, right up to the post-industrial age in which we live, have all functioned within something I would like to call the ‘ordering project’. All knowledge systems are founded on some code of order, some system of labeling and placing each thing or idea, in a designated relationship with all other things. The concept of time was amongst the first epic ideas of ordering, enabling conceptualization of the cycle of life, imagining death as terminal, or just a transition, creating the afterworld and reincarnation. Sundials, clocks, seasons, festivals and calendars ordered the narratives of life into measures of time.

Early concerns with time and the earth’s dimensions and dynamics demanded an understanding of

‘measure’ leading to concepts of incremental lengths and heights, distances, scale, proportion and later harmony. Measuring turned sound into music, gossip into poetry, and graffiti into painting. Building became architecture. All of these measured systems sponsored ever more abstract meanings.

Animistic knowledge systems assigned meanings to mountains, rivers and lakes and wove these attributed spiritual qualities into folklore, investing these inanimate objects with divinity, emotions, magical powers and auras.

The Order of Meaning

As these animistic patterns developed they took on semiotic meanings through devices such as hieroglyphics and pictography, evolving into written scripts, or oral traditions in which aphorisms were memorized and passed down from teacher to student; from generation to generation. The great Tibetan civilization devised sophisticated mnemonics to store and explain complexities, establishing a highly evolved emblematic knowledge system, embedded on *thangkas*, within *mandalas* and through complex forms of iconography. These images were embodiments of ideas, not idealizations of events or of people.

Temples, or *lhakhangs*, were ordered as ‘body supports’ for meditation, while *stupas* were classified as ‘mental supports’. By the fifteenth century there were texts on iconometry and manuals offering iconometric lineaments, including definitions of iconographic parts and proportions.^[4] Standardized motifs, acting as descriptive invocations, heralded various deities and historical figures as precise definers.^[5] This graphic ordering system is easily shared within homogeneous cultural groups and is comprehended by each individual up to their level of knowledge.

The Vedic oral tradition sets out another ordering system of rote learning, employing aphorisms with embedded lessons or truths. The entire grammar of Classical Sanskrit was codified in the *Ashtadhyayi* by Panini (an approximate contemporary of Plato) in an unwritten set of several thousand aphorisms, incanted, memorized and passed on from generation to generation of Brahmin scholars.^[6] This esoteric and elite oral tradition offers yet another meaning system, and demands unique mental gymnastics. It assumes specialized scholars and priests who will hold, interpret and pass on an arcane knowledge system, making it accessible to the common man only through the medium of explicit stories or in the form of visible idols.

In the twentieth century traditional folkways and patterns of behavior soon became mechanical and monotonous. ‘Bird’, the counterculture hero in Kenzaburo Oe’s narrative, *A Personal Matter*, saw life as a trap. He found escape in travels that were never realized, visiting gay bars never picking up other men, and imagining a fantasy adventure in Africa.^[7]

Spatial Order

Architecture and art, operating within these great syntactic traditions, involve their own unwritten systems of ordering and manipulating spatial ideas in graphic languages. There are patterns, prototypes, thematic concepts and design processes giving structure to the development of order in design problems and solutions. Design emerges as yet another means of seeking order, employing its own performance standards as a criterion for evaluating appropriateness.^[8]

The Order of Science

The European empirical system of knowledge is centered on the ‘proof of truth’ through controlled experiments. A good experiment has a clear hypothesis to be tested and a replicable method of testing. The steps involved in arriving at an ‘empirical truth’ involve labeling the elements of ‘matter’, determining their dimensions and weights, measuring their interactions and defining their causality as a proof of their existence. Rather than depending on images or memories with embedded meaning systems, science is based on an ordering system of ‘empirical hypothesis’ that confirms truth through testing. Science poses to be objective and measurable, devoid of subjective meaning. These observed orders are ultimately transformed into mathematical formulae that simulate real phenomena. They can be digitally stored and uploaded into virtual reality in the form of sounds, images or even simulated nuclear explosions.^[9] Emblematic, oral and empirical ordering systems can be proposed as posited benchmarks around which further varieties of ordering systems could theoretically operate. Thus, the ‘ordering project’ has claimed humanity’s imagination since the dawn of time itself.

Diversity within Order

There are as many knowledge and meaning systems as there are tribes, and each of us has roots in some aboriginal system of meaning for which each sign and symbol have an etiology of evolution up to the present moment. Some *agarbatti* is circled over the image of Goddess Laxmi for good luck in a business deal, while *tikkas* are adorned as one passes through a day’s ordered phases. Urbanization and globalization are making us all cultural mongrels; marriages of choice, media influences and work cultures overlie our tribal influences. Hybrid cultures characterize the most educated and specialized households at the top of the earning pyramid, while homogeneous, traditional societies compose the mass at the bottom. Economic and social systems have divided and nucleated us over the past two centuries wherein every micro-region hosts a dual society interacting through ordering systems in much the same way as imperialism acted across the globe on indigenous societies. The sociometry of an urban hutment is analogous to that of a metropolitan region, or of a nation.

Ordering as Closure

With such diverse conceptual mechanisms and skills, almost all societies ironically came back to a common set of concerns in an interesting ‘closure of the human condition’, often expressed through rites of passage, or ordering *samskara* of specific types. Birth, naming, attaining puberty, marriage, and death are all part of a pre-ordered life cycle in all societies. There are analogous ordered ‘markers’ in the form of rituals and celebrations that persist through all cultures. Signifiers, be they birthday cakes, sacred threads or circumcisions, mark out time and life into some meaningful order of phases in a predictive narrative of events. Order posits one’s life into a prescriptive story, which is inescapable, with the transcendental euphoria of art being the only possible escape. Otherwise, life is a trap!^[10]

Evolved societies used their ordering systems to philosophize about the ‘meaning of life’ and ask the question: ‘Why are we here?’ All societies have pondered over the meaning of death and the nature of the soul. All texts, symbols, benchmarks, signs and numbers accordingly take on both implicit and explicit meanings, developing into ‘narratives’ in which we become unwilling actors, with pre-assigned roles. Everything therefore has a name, material reality and immaterial ‘significance’ carried along with it, allowing for intertextuality and nuances of meaning, adding richness to life and essence to art and architecture.

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Artist as Anarchist

Over the past century sculptors and painters used *collages*, *combines* and *assemblages*, often in *absurdist contrast*, to manipulate intertextual meanings and even to challenge the common wisdom considered an essential underpinning to governance, law, stable societies and order.^[11] Unlike the Romantic and Impressionistic artists of the nineteenth century, Modernist painters were looking 'behind the order' to expose realities, just as Marx called for objective reality to rule over popular, yet camouflaging, artistic romanticism.^[12] A wave of counter-culture art forms were catalyzed by the pioneering experiments in *collage* by Georges Braque, with Pablo Picasso, further exploiting these concepts as revolutionary new orders.^[13]

Collage, and later *assemblage*, essentially take fragments of things and stick them over other materials or images to discover new opaque and esoteric orders. In the process, *signifiers* get cluttered within the textual patterns. Small cuts of newsprint could actually be photos of inhuman acts by 'civilized' governments, exposing their inhumanity, and thus covertly questioning the ethics of the state. Robert Rauschenberg used mixed media of painting, *collage*, film negatives, pieces of this and that, pulling them together into complex *combines* mirroring contemporary life, while creating his own space in the trap of life.^[14]

The Eagles' epic song, *Hotel California*, has been accused of harboring subliminal messages in praise of Satan embedded within its text as a kind of counter-cultural *collage* in music. Jim Morrison challenged societal order through his lyrics and stage antics, attracting an arrest warrant in Miami for exposing his genitals, just weeks before dying of a drug overdose in Paris at age twenty-seven. His counter-culture myth, signified by his songs and antisocial behavior, emerged as a spiritual cult, making his grave in Paris a pilgrimage destination to this day.

Music as revolt against order attracts thousands of youth to mass concerts where cult followers, dressed in the 'uniforms of the nonconformist', dutifully stand with arms stretched toward heaven, fingers pointed upward, swaying rhythmically in mindless mob unison, reminiscent of the mesmerizing mass Nazi rallies of the 1930s. If I'd blink my eyes I could imagine Nazi Brown Shirts, arms raised skyward, chanting *Heil Hitler!* As a species we love order, even mindless order.

The Image of the City

This intertextuality emerges through urban evolution as each historic era has deposited a layer of artifacts that overlie those of previous times, creating a collage of meaning systems. Where diverse cultural groups construct their identities side by side along lanes and *gullies*, employing diverse signifiers within motifs of their various communities, the montage becomes more complex. All of these meanings are filtered over time up to the surface of contextual reality, expressed at any particular moment, like cultural flotsam rising to the surface of an ancient lake.^[15] Nuances of different urban policies, styles and personal pretensions coalesce into a contemporary urbanity. Kevin Lynch conceived a new pattern language to understand this complexity by defining relevant signifiers of order, much in contrast to earlier ideas of urban order based in architectonic vistas, boulevards,

grand gardens and palace complexes.^[16]

By the mid-twentieth century a privileged Western ordering system had overshadowed other knowledge systems and popular indigenous cultures, fomenting an urge for a reversal of these dominant values and cultural institutions. The ordering project was being questioned.^[17a/b] A counterblast of theory and propositions emerged, recognizing the diversity of ordering systems and the oppression of minority cultures. These new starts found their sources in French avant-garde philosophical and literary criticism reflecting discontent with modern postwar institutions that were elitist and stifled. Even the boring, grey new town grid plans were symbolic of this stagnation and the oppressive public mechanisms that applied numbing formulae to solve myriad problems.

In urbanism and architecture the allusions to sources are generally too putative in their blatant meanings to be eclipsed by the highly theoretical analysis of mosaics found in the analysis of literature and in philosophy. In painting and poetry one can argue, as Barthes does, that the meaning of an artistic work lies within the minds of the viewers and not in the textuality of the work itself.^[18] In the late 1960s and early 1970s postmodernism saw everything as potentially 'text', which is privileged and hierarchically organized, including the formalism of modernism. These elite 'texts' excluded the artifacts of the repressed and marginalized communities of the world.

Effete, formal works were to be attacked and subverted. Derrida and Foucault assumed all thinking and ordering to be based in verbal and written language, leaving out our non-linguistic ordering and designing. They saw signs and signifiers in everything, interpreting each gesture as a calculated act of manipulation and control. Opportunistic academics saw an easy road to fame by feebly appropriating the postmodernist label into architecture, as if their intellectual gymnastics were some sort of branding experience exercise. This shaky theoretical acrobatics seems suspect as an alternative ordering system, or even as a deconstructive analytical framework that could temper the direction of architecture, or the way one goes about ordering space through arranging materiality. But it was a successful power play in grabbing the center stage of architecture and denigrating it into an esoteric, academic debate. By making theory more central than the process and reasons for creating, the entire modernist agenda was derailed into a cheerleading club for postmodernist stunts and amusements. 'Talking architecture' replaced 'making architecture'. The art historians and critics dedicated a generation of postulating to applauding the magicians of tricks and stunts.

Christopher Butler finds a self-contradicting irony in postmodernist analysis:

"Everything [in postmodernism] from furniture to clothes, to buildings, had to be seen as part of a 'language' whose social structure could be investigated and then shown to be susceptible to some kind of disruption or reversal, away from the suspect hierarchical ordering it had received in a 'bourgeois' society. And if everything was part of a language, and if language just disseminates, and if the discourses of art, like the discourses of medicine, law, penology, and so on, actually transcend the individual, then even the notions of authorship, creativity, originality were suspect and could not be 'privileged'."^[19]

Formal art and architecture, like formal systems of governance and ruling, find this all a bit messy. Vernacular architecture, or 'architecture without architects' gains an enticing order organically from a paucity of materials and limited structural possibilities, but takes on the ambience of collage from factors of climate, contours, unusual plot shapes and indigenous functions.^[20] Robert

Venturi appreciated this as the ‘richness of meaning, rather than the clarity of meaning’, preferring the ‘both-and’ to the ‘either-or’. Venturi exclaimed, ‘Architecture involves many levels of meaning and combinations of focus [with] space and with its focus being readable in several ways at once.’^[21] He proposed more articulate, considered and well-tempered designs, rather than a grandstand theory regarding the fate of civilization.

Orders of Discovery and Orders of Control

A major factor in the pursuit of civilization has been a hunger to create order out of nature’s chaos. Early scientists were concerned about the shape of the earth, the nature of water and fire, and the reality of the universe. Socrates paid with his life for his questions and Galileo barely kept his head on his shoulders after his heretical proposal that the earth revolves about the sun, rather than the other way around, which had been the prevailing belief.

The *ordering project* could be seen as one huge scientific work nurtured by an empire of ‘rulers’ to cement their position of advantage on the world, much as the Manhattan Project set out to create a nuclear bomb on a specified timeline. Such a proposition may be as surrealistic as the genius in Fellini’s *8½*, struggling against all odds to confabulate his epic project.^[22]

The *ordering project* proceeded from one of ‘discovery’ to one of organization and control, shadowing the geometric growth of the New Imperialism that emerged in the late eighteenth and early nineteenth centuries.^[23a/b] This was the time when the Old Imperialism of European ocean empires, driven by trade and supported by guns and sails, was overshadowed by a quest for vast land empires.^[24] Science and technology proved decisive in the consequent conflicts and victories. This new era was driven by the industrial revolution and the spread of its agro-based industries creating a necessity to control vast land empires.^[25]

The Social Sciences

The seeds of anthropology, sociology and other social sciences saw their beginnings as a support system to European colonial administrators who needed to know the order and structure of the vanquished peoples over whom they would rule. Ruling over water and fish, coming into conflict only with fellow European competitors, was much more simplistic than ruling over diverse peoples, ancient cultures and complex civilizations. Ruling over things was quite different than ruling over meanings and manipulating them to one’s own advantage.

In the eighteenth century Carl Linnaeus classified all living things into their genus and species, introducing the concept of binomial nomenclature in taxonomy. In the nineteenth century order seeking became hyperactive and intense, with myriad inventions driving the industrial revolution, demanding raw materials to feed expanding industries. Both were generating new knowledge that had to ‘fit in’. Knowledge of a ‘summative’ nature was pouring in chaotically, while formative knowledge was required to make things develop and grow in the planned direction. Knowing and ordering everything became a compulsive, summative preoccupation driving the invention of ‘ordering devices’ such as the encyclopedia, dictionaries, the Dewey Decimal System and the Periodic Table of Elements, attempting to bring all knowledge onto the same page.

In the early nineteenth century mechanical musical instruments (known by a variety of names such as street organ, fairground organ and several others) stored music scores in the form of paper rolls or folded cards with programmed holes punched in them (precursors to software). These, when pulled through a portable organ or harpsichord, played prerecorded music, as if from the memory of a person. The Maratha army invented a code based on the flickering light reflected from mirror to mirror between the six hundred forts along the west coast of India, informing the rulers at Satara or Pune within minutes of the Portuguese naval movements in and out of Goa. The telegraph digitized language into the Morse Code, allowing ‘instruments’ to transfer coded data, decoding it at remote destinations. Artificial Intelligence could now use orders in new and amazing ways.

The concept of the ‘university’ was crucial to the ordering project, as these institutions would collect varieties of ‘order’ and research their patterns. Universities could discover new information, ideas, concepts and design systems, storing the catalogued ‘parts of order’ and teaching a geometrically growing class of technocrats to understand and apply the ordering systems.

University campuses were, and are, laid out in the pattern of colonial cantonments with race, gender, class, position (and other ordering factors) generating the circulation patterns, land use plans, the sizes of living and working spaces allotted to various players and restrictions on movement. Universities subdivided knowledge and ordered it into ‘faculties’ that hosted ‘schools’, which in turn hosted ‘departments’, each hosting research cells and ‘laboratories’.

Finding Orders of Control

The New Imperialism was nourished by discoveries of empirical order in the fields of medicine, ordnance, marine architecture, material sciences, new and more efficient industrial processes, steam power, mass pedagogy, engineering, and much more!^[26] The employment of rifled guns and malaria prophylactics were adequate innovations for a small imperial army to vanquish the vast army of the Maratha Confederacy, gifting virtual control of the entire subcontinent to alien rulers. Exploiting vast lands and ruling over huge populations proved more complex than being traders in garrisoned, port cities with walls around them. That was based on the Hellenistic and Roman ‘garrison city’ prototypes and was perfected by the Portuguese, the Dutch, the Spaniards and the English through their global archipelago of ports and forts.

The Templates of Globalization

Nineteenth century Imperial Britain created many of the prototypes, or ordering templates, that persist with us today in the form of uniforms, language, education, locomotion, communication networks, medical systems, commercial cropping systems, land revenue hierarchies, trade, finance and the manipulation of knowledge and meaning systems for political manipulation.^[27]

Constructs: Order through Meaning

Creating social orders, like the ‘heterosexual construct’, molded the emerging urban, middle class householder into a new and binding lifestyle supported and implemented by law, police, media, entertainment, medicine, education and religion.^[28] Deviant behaviors were declared both as ‘abnormal illnesses’ and criminal acts.^[29] The ‘heterosexual construct’ prescribed monogamous ‘Christian marriage’ between a man and a woman as the only legitimate emotional relationship, and determined the purpose of all relationships to be the procreation of more dutiful subjects for the

empire, placing women in a subordinate ‘domesticated’ position. Men were actors and women were at best beautiful objects. History was ‘his’ story, not hers. This Victorian ordering system created a horizontal layer of order within a vertical, hierarchical order of classes and positions inhabiting the order of the city.^[30]

The cantonments of India, emerging immediately upon the capture of new client states, were prime examples of the new ordering system, which has unfolded until today in the form of ‘garden cities’, tram suburbs, university campuses, dormitory communities, gated communities, satellite cities and the ‘new urbanism’. India’s traditional neighborhoods and caste-based *wadis* continue to be replaced by ubiquitous ‘flat schemes’ devoid of meanings and personal signifiers, alluding to the brave new world that the *Generation X* imagines as the new Utopia.^[31a/b]

This ‘plan’ and other designs are not artifacts of accident, but rather arrangements of contrivance. Division, dividing and subdivision order and manage social and physical space simultaneously. Even the rulers trap themselves within a hierarchical, compartmental system of restrictions; a microcosm that mirrors the larger world they dream of creating. Empires, both corporate and governmental, have treated the people engaged in cities – their way of life and culture – as a matter of insignificance. What is important to planners are the inanimate aspects of cities: land use zones, vehicular circulation networks and infrastructure webs. Organization and control have become the overriding goals of the new prototype that morphed from cantonment, to garden city, to university campus. The Raj city plan, like the homogeneous dress code, ‘put people in their places’. The plan was a system of exclusive cells, between which only a few could move freely. The ordering project morphed from a summative to a formative endeavor; from describing to controlling.

Uniforms and Pluraforms

The strategy was to replace ‘vernacular culture’ with institutional culture, right from the plan of the city to the architecture of the buildings. Uniforms replaced *pluraforms* that gifted people the diversity of their personal cultural richness and identity.^[32]

Formal architecture, the kind taught in schools of architecture, relies on ‘orders’ and formal patterns of construction to gain clarity. As signifiers these styles become analogues of political orders, or ruling systems. They are metaphors for social constructs for controlling large, potentially anarchic populations. Discovering and applying these orders is what empire building and ‘civilization making’ are all about.

Bernard Rudofsky proposed that architectural history, as written and taught in the Western World, has never been concerned with more than a few select cultures, emphasizing that this history covers only a tiny sliver of time and a minuscule geographic area. Besides architecture, he notes,

“History as we know it is equally biased on the social plane. It is little more than a who’s who of architects who commemorated power and wealth, an anthology of buildings of, by and for the privileged, the houses of true and false gods, of merchant princes and princes of the blood, with never a word of the houses of the lesser people.”^[33]

Defining factors in vernacular orders emerge from within diverse societal components, expressing themselves to society as a *collage*; in institutionalized cultures the uniforms are imposed from the outside inward, suppressing down from the top in to the parts. Vernacular dress and built

forms are ‘expressions’ while the institutional uniform is ‘suppression’. This system of articulated stratification lives on with us today. It tempers the way we think, how we deal with ‘others’, and our self-images.

The Campus as Ordering Prototype

This concept of space allocation and restriction pervades contemporary global, corporate thinking in how we organize people in space. Modern university, factory and information technology campuses are variations of the ‘cantonment model’ where the military area created a unique social space, defined by occupation, gender, race and rank. This was a precursor to ‘globalization’ in urban planning, where gated communities and malls have replaced lively, informal public domains. The loss of social space is a metaphor of the loss of community and indigenous identity.

Collage as Escape

In the end analysis we are all servants of ‘the empire’ that is now an integrated poly-nucleated global web of interrelated corporations and governments. We are a part of the ordering project and we reinforce the fabricated constructs that make life a trap.

Artists and architects wear the black uniforms of the ‘creative’, concealing the reality that they have no new ideas. The empire commissions their work and pays their fees; in the same way they buy governments to bend public policy. The empire buys what is drawn and written, builds and publishes it, and give awards if they like it. All of us – architects, poets, artists, doctors, accountants and sales girls – play our roles in the big scheme of things. If we don’t like it we can read a novel or embark on an imaginary adventure into the heart of Africa ^[34] or maybe create a *collage*! But in doing so we shall only be following another preordained pattern of escape. Life is a trap!

But for the true artist there is the secret of transcendence, that moment of inspiration where the profound lives unhampered by the day-to-day trivia of reality. Within that momentary sliver of self-realization lies ones eternal truth and being. Only true artists exist in a liberated state, free from human bondage.

Note: *The word ‘empire’ is used in this paper as a general reference to the formal, top down organizing world and not necessarily to the Raj, though it is specifically referred to.*

Citations:

1. Bronowski, Jacob (1978): **The Visionary Eye: Essays in the Arts, Literature, and Science**, MIT Press, Cambridge, Massachusetts.
2. Sekler, Eduard F., et. al. (1965): **Proportion, a Measure of Order**, Carpenter Center for the Visual Arts, Harvard University.
3. Waterfield, Robin [tr] (2002): **Plato: Phaedrus**, Oxford University Press (Oxford World’s Classics), Oxford.
4. Jackson, David and Janice (1984): **Tibetan Thangka Painting: Methods and Materials**, Snow Lion Publications, Ithaca, New York.
5. Chakraverty, Anjan (1998): **Sacred Buddhist Painting**, Roli Books, New Delhi.
6. “Sanskrit Literature”, **The Imperial Gazetteer of India**, Vol. 2, p 263.

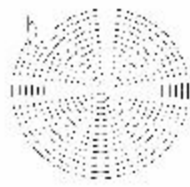
7. Oë, Kenzaburo (1969): **A Personal Matter**, Grove Weidenfeld, New York.
8. Gropius, Walter (1956): **The Scope of Total Architecture**, Harper, New York.
9. Thompson, D'Arcy Wentworth (1959): **On Growth and Form** [2 Vols.], Cambridge University Press, Cambridge, UK.
10. Kundera, Milan (1984): **The Unbearable Lightness of Being**, Faber and Faber, UK.
11. Elderfield, John, Peter Reed, Mary Chan [Eds.] (2002): **Modern Starts: People, Places, Things**, Museum of Modern Art, New York.
12. Marx, Karl (1867): **Das Kapital**, Gateway Edition (1996), Regnery Publishing, Washington DC.
13. Anderson, Donald M. (1961): **Elements of Design**, Holt, Rinehart and Winston, New York.
14. Kotz, Mary Lynn (2004): **Rauschenberg: Art and Life**, Harry N. Abrams, Inc., New York.
15. Benninger, Christopher (2010): **Gleaning Pune's Future from Pune's Past**, 13th June, Sunday Pune Mirror, Bennett Coleman Group, Pune.
16. Lynch, Kevin (1960): **Image of the City**, Harvard-MIT Joint Center for Urban Studies, MIT Press, Boston, Massachusetts.
17. a. Foucault, Michel (1970): **The Order of Things**, Pantheon Books, New York.
b. Saïd, Edward (1979): **Orientalism**, Vintage Books, New York.
18. Barthes, Roland (1975): **The Pleasure of the Text**, Hill and Wang, New York.
19. Butler, Christopher (2002): **Postmodernism: A Very Short Introduction**, Oxford University Press, Oxford.
20. Rudofsky, Bernard (1964): **Architecture without Architects**, University of New Mexico Press, Albuquerque.
21. Venturi, Robert (1966): **Complexity and Contradiction in Architecture**, Museum of Modern Art, New York.
22. Reid-Paris, James (1993): **Otto e Mezzo [8¹/₂]** in James Reid-Paris (Ed.), **Classic Foreign Films From 1960 to Today**, Carol Pub. Group, New Jersey.
23. a. Headrick, Daniel R. (2009): **Power Over Peoples: Technology, Environments and Western Imperialism, 1400 To The Present**, Princeton University Press.
b. Morrison, Elting E. (1966): **Men, Machines and Modern Times**, MIT Press, Cambridge.
24. Cipolla, Carlo M. (1982): **Guns Sails and Empires: Technological Innovation and the Early Phases of European Expansion 1400-1700**, Thomas Y. Crowell Publishers, New York.
25. Benninger, Christopher (2010): **'Poona Papers'**: The Margaret Mead Lecture, World Society of Ekistics, Athens.
26. Freire, Paulo (1968): **Pedagogy of the Oppressed**, Penguin Books, London.
27. Benninger, Christopher (2010): **A Tale of Two Cities**, 22nd August, Sunday Pune Mirror, Bennett Coleman Group, Pune.
28. Pagila, Camille (1990): **Sexual Personae**, Penguin Books, London.
29. Benninger, Christopher (2007): **The Science of the Absurd**, in *Biblio*, May-June, Vol. XII Nos. 5&6.
30. Metcalf, Thomas (2004): **Ideologies of the Raj**, Cambridge University Press India, New Delhi.

- 31. a. Sarukkai, Priya (2009): **Generation 14**, Penguin India, New Delhi.
b. Benninger, Christopher (2002): **Imagineering and Creation of Space**, Graz Biennale.
- 32. Kundera, Milan (1986): **The Art of the Novel**, Perennial (Harper Collins), New York.
- 33. Rudofsky, Bernard (1964): **Architecture without Architects**, University of New Mexico Press, Albuquerque.
- 34. Benninger, Christopher (1995): **Hope in the Midst of Despair**, in Biblio, May-June, Vol. I, Nos. 5&6.

The Importance of Curiosity

My passion for life has not really come from architecture. It has come from adventure and exploration, and I see architecture as just one vehicle through which I can explore the human condition. It is a way to study culture, history, society and our place in civilization.

I feel one can be an explorer sitting in their study, or in their living room. Exploring is searching within unknown contexts, finding things new, different, or maybe just appropriate. A train of ants moving up the wall could lead to speculations and evoke curiosity. Where do they come from and where are they going? What brings consistency into their work and processes? What are they carrying and who tells them to do so; maybe no one? Maybe I am like one of the ants climbing up some wall mindlessly doing the work of civilization, thinking it is very important, while maybe that is not so? As with those ants so with peoples. I am driven in my inquiries by a curiosity about their traditions and the environments that support them. Architectural design is a composite analysis of those things, leading to alternative options for sheltering and maybe inspiring people. The work of writers and journalists is very much like mine as an architect. They too are looking at life, analyzing the human condition and chronicling it. Maybe curious people wonder if we are just trying to amuse them through small gestures of uniqueness or something 'new'. If life were so mundane then there would be no purpose in our being or in our existence. As people who do things and 'make things' we should be asking many questions whose answers would temper our thoughts, ideas and designs. We are not factory workers, bankers, or accountants who just sit at a work place and repeat routine functions. We are supposed to be designing a transforming culture and tempering the structure and nature of civilization. Our tiny gestures hold vast implications for mankind.



When I take up a new design, it is that same sense of challenge, of curiosity and exploration which drives me. The people who patronize my studio are my real work. They are the 'stuff' of my passion. Their design briefs, their sites and the patterns of human behavior expressed in their lists of functional spatial requirements, and the relations between them become a kind of pattern which organizes complex materiality and building systems. It also raises questions about the implicit functions and purposes that are not explicitly stated in the brief, the building programs and the budgets. A good functional solution for an office or a factory becomes the template through which non-programmatic ends are realized. These are the unseen truths of architecture. They are the nonverbal, unspoken necessities of humanity; call it conviviality, community or the social patterns that our template facilitates and empowers. Call it a sense of place, a feeling of belonging or gifting identity, but these are all part of the measure of immeasurable architecture. No one is going to tell an architect about these hidden dimensions and secret attributes. An architect has to be driven by curiosity to observe the ants and wonder what drives them. An architect has to look behind the material requirements and create immaterial experiences.

A curious person who is creating spaces for many users has to imagine what it is like for the users to enter and to walk through his building. Then the designer has to become the man who cleans

the building and the secretary who smiles at the boss as he grimaces on his brisk morning walk to his chamber. He has to be the visitor who is asked to sit and wait and think what that visitor will think while he is sitting alone, concerned about the impending meeting. He has to be the youngsters working in the building and consider whether they can feel the seasons change and live the day with the sun as their timekeeper. Do they know it is raining and does it bring them a sense of joy? Do the spaces of the inanimate structure come alive as one moves through them, under a low ceiling that emotionally explodes into a large hall, and then directs one's attention to a landscaped garden where a patient, alert kingfisher waits to swoop on his prey in a silent pool of water? Did all of these experiential emotions just happen, or were they the orchestrated choreography of some thinking mind that was curious to know of all of these imaginary people and their behavior in the designed spaces? Did the designer talk to them in his mind and walk with them through the spaces, like an actor who plays all of the roles in a one-man performance? This, my young friends, is the essence of being an architect. It is grounded in curiosity and in imagination.

My passion for life has not really come from architecture. It has come from adventure and exploration, and I see architecture as just one vehicle through which I can explore the human condition.

I fear that the internet generation is being robbed of their curiosity, discovery and wonderment. Instead of having to search for things and the connections between them, young minds are overwhelmed by images, emotions, ideas and facts. As every young man knows, whatever is left of his curiosity is instantly gratified by the click of his forefinger and like magic the most forbidden fruits can be tasted, at least in his jaded imagination. The luxury of boredom does not challenge him to think and to seek. He was born into a hyper-world. Like a chef in a restaurant, he no longer yearns to taste food; only to produce a flood of new menus and exotic dishes and thrust them *ad nauseam* onto a client's plate. Drawing and drafting software have robbed our youth of the talent and the skills of sketching; the ability to instantly juxtapose one design idea over another on transparent sketch paper has gone. Architects have ceased to be designers and are now clever machine operators, unless they arrest themselves and start to think with their hands. The real problem is that drafting is not architecture and buildings are not architecture either. Architecture is the magical spatial aura and the experientialism contained in the material shell; it is not the shell.

In the United World College of India the Mahindras wanted to create a new learning culture in which people from various societies would seek common values and a common vision to which they could devote themselves later, in their adult lives, through various callings. On the face of it they wanted some buildings in which they could teach some courses.

Though they could not articulate the other dimensions, they were sophisticated patrons who could step beyond materiality and imagine the unspoken, the unseen and hope for transcendental experiences in spatial complexity. This shared curiosity about the nature of a campus, of a designed community and of a little society we would all create generated a vibrant dialogue and an adventure.

In the new capital of Bhutan and the buildings that make up the National Capitol Complex, the Royal Government seeks an abode for democracy organized within the Buddhist and Bhutanese tradition. It is a country with living traditions of Himalayan culture. It is unlike many of its neighbors where more powerful nations have taken over and created fossilized 'museum piece' cultures – a kind of a static amusement park for tourists; a physical shell with no soul. The bane of globalization is

that it tends to homogenize dynamic vernacular experiences into preordained amusement park experiences, like the New Urbanist townscapes in America referring back to a romantic small town past that existed only on Hollywood sets. In the valleys of Bhutan culture is alive, with its own style and vernacular. Vernaculars are always changing, morphing and unfolding as new technologies, information and functions diffuse within the society. What is important as democracy and its related new institutions emerge, along with the buildings that house them, is to conserve the essence of Bhutanese culture, while expressing it through contemporary buildings. This challenge was not articulated in the architect's brief, but this *problematique* was certainly in the mind of Lyonchen Jigme Thinley, the country's first elected Prime Minister. It drove my thinking too. But none of this was ever written down or expressed in budgets, bills of quantities or specifications. This most important aspect of the work lay in the realm of curiosity, speculation and a bit of magic. Much of the work we are doing employs wonderful traditional woodcarving, massive white masonry walls, overhanging roofs to protect the walls from rain, and the traditional iconography that instills meaning. We are speaking a clear Bhutanese architectural language that has grown out of a particular history, climate, economy and society. We have to speak that language while designing.

In the mainstream, 'cutting edge' architecture of the West these concerns are no longer central to architectural dialogue. Architects there are arguing about *isms* – modernism and postmodernism. The debate does not conceal the fact that most of the architects are into self-projection and bent upon becoming famous by making spectacular stunts, like a fire-eating trapeze artist in a circus, or like a thrill ride in Disneyland. Put together, all of the notable buildings of the past two decades are nothing but a giant amusement park littered across the global landscape, or perhaps a junkyard of worn out stunts and wrecks left over from megalomaniacal follies. The issues of vernacular, of experientiality and of just plain conviviality are lost to these creatures who are very concerned about their visibility, being watched, being talked about and just plain being famous. It is good to be well known, but let us not smother humanity with the garbage dump we are creating.

I must confess that at a vulgar level I also enjoy a bit of pornography and spectacularly contorted structures doing things that are not meant to be possible or affordable. I also derive a perverse pleasure from the garish displays of ill-earned wealth and ill-spent income flaunted by rude film stars, corrupt politicians and spoilt princes. But something inside me makes me catch myself and ask, 'What am I doing?' An inner voice tells me to step back and think. Architecture over the past few decades has lost that inner voice, and all the critics and art historians have become sycophants of the tastemakers. Even the universities and the museums have become stooges of the global amusement park.

This vulgar aspect of the human spirit, call it a need for 'thrill', triggers a rush of excitement that replaces true curiosity. The crowds that mob to the wreck of an auto crash are driven by a crass desire to see spilt blood, not by any curiosity of what could be done to prevent more such accidents, or even a concern for the fate of the victims. The thrill of the tallest tower, of the most sensuous and seductive curved form and the most audacious structural feat are all ghoulish responses from the animal in us, not the refined considerations of a well tempered mind and a focused society.

Architecture is not a structural feat, an amusement or a spectacular personal statement. It is a considered, thoughtfully articulated, well-tempered response to diverse desirables, not the least of which is uplifting the human spirit. To discover this truth one must be able to analyze and understand complex situations. The cut-and-paste and the rubber-stamp solutions that typify our urban landscape

and the stunts that draw crowds are a travesty of architecture.

What I am hinting at is what I described earlier as the element of exploration and discovery that architecture entails. It could be the intense curiosity of solving a specific problem right on the drawing board, or it could be the thrill of a serendipitous discovery while browsing through books. Intense curiosity can emerge from travel, writing, reading and just meeting people and exploring one another's minds. Meetings can be causal encounters. Intellectual companionships can emerge, grow and later fade away. But the element of mutual discovery is what it is all about. In architecture there is always that moment of discovery in each project, where one defines the essence of the work of art, experiencing transcendence and achieving a moment of epiphany.

My personal interests and hobbies have all revolved around this central theme and passion. Architecture is just my formal work, merely one form of this search that comes in different manifestations. So my personal life involves a number of compelling initiatives in my need to understand what I do not know and do not understand. I want to enter unknown territory, dissect it, take it apart and put it back together again. That 'territory' could be a new friend, a vast sand desert, or a novel. Surely each new design problem is a passionate form of exploration, driven by curiosity.

I find people my most intriguing pastime. Every person presents a new conundrum or puzzle: what are the values which motivate them to be passive or active? What makes them reactive to their context, or proactive to change it? What are their personal visions and their objectives, and what path have they chosen to get there? What gives them a 'lift'? What makes them laugh? What makes them cry? Why are they afraid to be themselves? I like hard working and focused people who know where they are going and are developing the skills and techniques to get there. I like people who have a grasp of who they are and what their living and working environment is actually like. I like 'in-your-face' kind of lively people who wear their emotions on their sleeves. Friendship becomes a kind of mutual exploration of values, patterns of thought and structure of behavior. Friendship is emotional attachment, but most of all it is intellectual companionship, driven by questioning the nature of human existence and by an honesty in proposing answers. From a base of honest expression one can share ideas and study how one idea relates to another. One can see such relationships between ideas as concepts; concepts about the human condition and society shared by intellectual companions. Friendship is my main hobby and I have friends from all walks of life: owners of great industries; drivers; socialites; cooks; professors; attendants; artists and inventors. What they earn, where they come from and how much they are worth are just interesting facts that decorate their search for meaning. Income and social status can be very boring attributes if they constitute a person's totality. The 'search' is where the fire of friendship burns and what we share. At India House that search is a quiet one; it is in the whispers over a drawing; in the annoyed glares exchanged over errors; and in the smiles confirming something beautiful. It is through work and sincere effort that companionship is shared. Everything else is irrelevant.

My interest in people leads me to a love of reading about them, because good literature is a study of human nature and of perseverance within the human condition. I like works like *The General in His Labyrinth*, *Love at the Time of Cholera* and *Memories of my Melancholy Whores* by Gabriel García Márquez; or Milan Kundera's *The Unbearable Lightness of Being*; or Kenzaburo Oe's narratives, *A Personal Matter* and *Nip the Bud; Shoot the Kids*. These are profound as they drag one into new worlds, new situations, sentiments and nuances. Through their characters we find ourselves dealing with their world. Through our curiosity we evolve a worldview.

Friendship becomes a kind of mutual exploration of values, patterns of thought and structure of behavior.

I also like the study of economics and behavioral sciences, because they speak of collective human aspirations and action; of how larger groups of people organize themselves and their behavior for the common good or even self-destruction. Literature and the behavioral sciences often deal with collective amnesia regarding the huge failures of the human race, and the horrid stereotypes we cast on groups in order to ignore and suppress them. If I pick up *The Economist* I first look in the back to read their page long obituaries. I learn about such fantastic lives, times, struggles, achievements and failures. I always finish the article thinking ‘I wish I had known that person.’

Reading about people leads one to writing about them. I yearn to be a writer and that would logically be another of my hobbies, time permitting. My short story *Akhada* appeared in *Femina*, which made me happy. *Akhada* is about a Nepali woman who mistakenly wanders into the all-male world of a traditional gymnastics ground, hidden within a deep overgrown forest. It is an encounter between the fragile personas of the immodest male ego and protected female pride. It explores societal boundaries and limits, which are things I do not like. I have written a novel over the past ten years just for the fun of it, not to publish it. It is called *Samsara* and deals with the evolution of a woman’s soul and her self-discovery, against great odds in a traditional patriarchal society.

Writing is an opportunity for me to explore the relations between the central, controlling mainstream of society, and its less defined creative edges. It allows me to analyze things strange, queer, transforming, and unusual and perhaps to discover the unknown. Art lies out there on the unfettered periphery of the mainstream. The core in metropolitan centers appears stagnant and polluted, the more so as one approaches the Eurocentric universe.

The literary journal *Biblio* gave me the opportunity to publish many of my ideas, including a review of Kenzaburo Oe’s literature when he won the Nobel Prize, and an analysis called *Queer Words* that reviewed the rich history of gay and lesbian literature in India. Vikram Seth, Bupen Khakkar, R. Raj Rao and many others have contributed to this very Indian tradition, allowing multiple images of normal, yet ‘different’, existences to emerge, unlike the cutting-edge West that has stereotyped gay life into a stale pop culture. Again everything becomes Disney World, an amusement, a stereotype and a trap. By exploring the ‘edge’ one gets back to the potentials of the center. India offers its people a huge variety of personal identities and paths in life. Our culture in India is not uniform, but what Milan Kundera calls a ‘pluraform’. What we wear and build is expression coming from many pluraforms deep inside pushing up and out, while the global culture of ‘the center’ is a uniform pushing in onto people, smothering their unique identity and sense of individuality.

Such playing about with ideas about people leads to critical analysis and proposals, and new models and paths for looking at the human condition. In *The Science of the Absurd*, a critical review I wrote of Ruth Vanita’s translation of *Chocolate* by Pandeya Bechan Sharma ‘Ugra’, I could employ her wonderful introductory essay to explore ‘common wisdom’, stereotypes, prejudices and biases to explain how empiricism and science have been perverted to the misplaced cause of suppressing and marginalizing minorities. Medicine, law, the family construct, advertising, religion, politics, education and other ‘institutions’ are more often used to ply untruths about minorities than truths. We started the twentieth century with erroneous ideas about women, blacks and Jews, and we ended it

with erroneous views of Muslims, gays and Africa. Through reading and writing one can climb up the tree trunk of life and walk out to the ends of life's branches and twigs, exploring worlds and realities hidden from the average mind. Young architects, go out there on the fringe and explore yourselves.

People, reading and writing and a little daring leads to my most favorite hobby: being a traveler. A tourist is not a traveler. A traveler has no plan, no bookings, no train reservations and no guides. A traveler moves on instinct and according to intuition in a general direction toward a vague destination. Fate and turns of events shape their paths. The movement and journey is their search. The people met and encountered, the incidents and the experiences are the goal. When I first visited Bhutan in the late 1970s I was the only Caucasian in Thimphu and maybe one of five or six in the Kingdom. I was able to traverse the entire mountainous country over dirt roads and mule tracts. I slept under vast skies filled with a blanket of glittering stars. I fell in love with the land and with the people. Its culture, for me, has been a deep well of learning and a source of wisdom and mental peace. The greatest honor in my life was being commissioned to design their new Capitol Complex, within the Capital City Plan I prepared. In this work I am dealing with the most sacred artifacts and icons of Bhutanese culture. I am overwhelmed by this. It is a continuation of a search that began in the Himalayas many decades ago. So, to be commissioned to design a new building is like embarking on a new adventure. It is setting off on a search for the real and the true. A good architect must be a good traveler, at least in his mind.

Another expedition, referred to in an earlier letter, was my travel overland from London to Mumbai in 1971. I had no itinerary or clear idea of how I would do it. I learned survival along the way and took the help, protection and shelter of pure strangers. This adventure was my education about the greatness of humanity and our common values and concerns. It was my rite of passage from a boy to a man. I flew from Boston to London and took a train down to Dover, then the Channel ferry across to France. From there it was mostly overland: trains, buses, walking at times, camels, vans and hitch-hiking. I just traveled from one city to the next and made friendships along the way. All I knew was that I was moving toward India and that I must 'move' and I must 'survive'. I came to love the Turkish, Kurdish, Iranian, Afghan and sub-continental peoples. They are at once earthy and erudite. They are steeped in great cultures fostering a love for humanity; a will to persist and to survive. These are the people from whom the West can learn a great deal, but there has to be humility and submission to learn anything from anyone.

To travel like this you have to put your fate totally in the hands of others – unknown others! You are not a tourist; you are traveling in totally unknown territory. You need a basic faith in human nature. You have to see the good in each person, respect that good, and you will get respect in return. Concerns about water, food, shelter and physical security then vanish. All those flow toward you as part of the relationships you build along life's way. My love for and friendships with these people are born of moving with them, eating with them, sharing a few drops of water from a desert well, sleeping under the stars at night, and always laughter. India has always been the destination of my explorations and travels.

On my first visit to India I flew with stops in Alaska, Japan, China, Taiwan, Cambodia and Thailand. When I landed in Phnom Penh I was told I could not enter because there were no diplomatic relations between America and Cambodia. By the time the argument was over the plane had deserted me in the small airport and they had no choice but to allow me to stay. Adversaries became friends. The one or two taxis had left the airstrip and the soldiers got me a ride on an elephant up to the edge

of town, from where I took a cycle rickshaw at dusk into the strange city with red dust roads. They were of a grand scale, lined with white stones and swept neatly into order every morning. I spent ten days in the country and was the only American who was not in jail. There were fourteen imprisoned US Air Force pilots, who had been shot down for illegally intruding into Cambodian airspace while flying out of South Vietnam toward the North. There were Viet Cong soldiers strolling about the streets off duty; young men of my age enjoying a respite from their battle against capitalism and from the invading American forces. That was an encounter with a society and a culture that was soon after lost to the ravages of the Khmer Rouge terror. It was my privilege to move freely amongst these beautiful, ancient people, sharing with them their simple and dignified lives before their culture was destroyed, vanishing into the pages of history books.

My youth is now an antique land drifting away into memories. It is a secret place that only a few can enter through our dreams and recollections. But the passions, dreams and hopes of those I met way back then persist as a force within me today.

Now in India I am an architect, living a dream with my life partner, my fellow architects and my wonderful patrons, working in India House at Balewadi on the edge of Pune, an emerging metropolis in this chaotically changing world. Change is all around me. Yesterday is becoming an antique land of memories and we are thinking of tomorrow in terms of bits and pieces, in the form of designed little enclaves that hopefully will act as models for a better future. All of us, young architects and old, students and teachers face huge challenges.

India, like America, is a land of individuals. Like America, its composite parts are diverse. India is composed of a billion initiatives of a billion people. These are sometimes in conflict and sometimes in alignment. Living in India is a huge challenge demanding understanding, patience and perseverance. This vast complexity of personalities, multiple visions, values and ways of doing things is a continuous source of inspiration and motivation. The very name of the country sparks my imagination and my curiosity even after living more than half a century amongst its wonderful people. The search here is to find the common thread that weaves all of these strands into one cloth. That is the fun of it all. What is amazing is that within all of this diversity there are so many threads that tie everything together into a stable pattern; always in flux, always in transition, always changing, yet dependable and unified.

Somehow I have always felt most at home in India; more so even than in America or Europe. I love the chaos, the dynamic synergy and the way things settle into their own unique order. I love the warmth of the people, expressed through smiles and laughter. I love the variety of characters and the complexity of the society. I love the smell of rain on parched earth, wafted on the breeze from the distant mountains as the monsoon approaches. I love the sounds of insects at dusk and the songs of birds at dawn, when the sun peeks over the horizon onto verdant fields.

I love the early morning chatter of the boys who run India House, as they prepare for yet another day. I have loved people and people have loved me, making India my natural abode. But most of all I love the inherent curiosity of Indians. They never leave you alone until they have queried every aspect of your private and family life. Probably I am just the *avatar* of another traveler from a previous life here in what is now an antique land. My soul knows this place! Perhaps I was a migratory bird living winters on the shores of Sri Lanka and summers in the high Himalayas! I find a harmony between the soil and myself, enjoying friendships with people of many communities, religions, castes, and ethnic groups from across India. It is surely my destined home. The passion

behind my plans and designs for India, Sri Lanka and Bhutan emanate from being a part of this harmony and vibrating within it. As an architect I am merely a hand following its innate force, giving materiality to India's deep well of wisdom.

The search here is to find the common thread that weaves all of these strands into one cloth.

As I have said to you several times in these letters, there is only one kind of good luck in life and that is to have good teachers. I have had more than my share of them, dating way back to the 1960s when I studied under Jose Lluís Sert, and when Walter Gropius strolled through the studios at Harvard, where I was a young student. I can go back further as a teenager studying under unknown masters like Harry Merritt and Robert Tucker, or I can go back to my childhood of green lawns, flowering trees, rolling hills, clear streams filling lucid lakes, adorned with water lilies, turtles, alligators and blessed with yellow butterflies hovering above.

Within one's memory lies a vast treasure house of images, puzzles, nostalgia, possibilities, constructs, values, ideas and thoughts of the future. It is the unique ability of the human race that we live primarily in the past, using the present to forge images and scenarios of the future. Unlike other animals we lack the peace of living only in the present, intuitively resolving the challenges of the moment that our instincts are prepared for. We are restless by nature, concerned with multiple future scenarios, both good and bad. Nothing in the future seems certain to us, which makes us curious to know what tomorrow will bring. Every sunset forebodes a hundred sunrises. The setting sun makes us contemplate what will be our contribution for a better tomorrow.

Having said all of this, having shared my stray thoughts and emotions, I feel in the end life is all about being a perpetual student, finding good teachers and walking starry-eyed under the continuous spell of curiosity.

Glossary

Akhada: wrestling arena

Ashram: spiritual hermitage; a stage of life

Beedi: a thin, South Asian cigarette filled with tobacco flakes and wrapped in a leaf tied with a string at one end.

Bindaas: being fun, frank, fearless and valuing freedom in all its forms.

Bollywood: informal term popularly used for the Hindi-language film industry based in Mumbai

Brahmacharya: one of the four stages of life (*ashramas*) in traditional Indian civilization as outlined in the *Manusmruti*. It refers to an educational period of 14–20 years which starts before the age of puberty. During this time the traditional vedic sciences are studied along with the religious texts contained within the *Vedas* and *Upanishads*. This stage of life was characterized by the practice of strict celibacy. One who is in this stage is called a **Brahmachari**.

Busti: a hutment or small inhabitation of people.

Chawls: buildings which are often 4 to 5 stories with about 10 to 20 tenements, sharing sanitary facilities.

Chela: religious student or disciple.

Chortens: a Tibetan Buddhist shrine or monument.

Dal: a dried legume (as lentils, beans, or peas); an Indian dish made of simmered and usually pureed and spiced legumes.

Dilli haat: a combination craft bazaar and food plaza located in the heart of Delhi.

Gharana: In Hindustani music, a *gharānā* is a system of social organization linking musicians or dancers by lineage or by apprenticeship, and by adherence to a particular musical style. A *gharana* also indicates a comprehensive musicological ideology.

Gopuram or Gopura: a monumental tower, usually ornate, at the entrance of any temple, especially in Southern India.

Jaali: a perforated stone or latticed screen, usually with an ornamental pattern constructed through the use of calligraphy and geometry.

Kuchcha: something raw, unfinished.

Kund: well or water reservoir built as part of a temple complex in Indian temples.

Mandala: a Hindu or Buddhist graphic symbol of the universe/ cosmos; specifically: a circle enclosing a square with a deity on each side that is used chiefly as an aid to meditation.

Mani walls: stone structures with intricately carved stone tablets, mostly with inscriptions.

Mumbaikar: a person from the city of Mumbai.

Nalla or Nallah: an arm of the sea, a stream, a watercourse, or a steep narrow valley.

Otta: a platform in front of the house for informal seating.

Rishi: a Hindu saint or seer.

Serai: a traditional travelers' halting place, usually within a protective wall in which sleeping niches are built. There are trees in the central open space with sitting/sleeping platforms.

Shamiana: popular Indian tent shelter, which is commonly used for outdoor parties, marriages, feasts, etc.

Shishya: student

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